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IGCSE 0448 & O Level 2059

PAKISTAN STUDIES

(GEOGRAPHY)

A Complete Revision Guide
for

Environment of Pakistan
According to Latest 2023-2025 Syllabus

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List of Contents

Chapter 1	The Natural Topography.....	4
Chapter 2	Climate of Pakistan.....	11
Chapter 3	Water Resources.....	21
Chapter 4	Forests.....	31
Chapter 5	Minerals.....	39
Chapter 6	Fishing.....	45
Chapter 7	Agriculture.....	50
Chapter 8	Power Resources.....	66
Chapter 9	Secondary and Tertiary Industries.....	77
Chapter 10	Trade.....	94
Chapter 11	Transport and Communication.....	100
Chapter 12	Population.....	111
	High Grade Responses.....	123

CHAPTER 1: THE NATURAL TOPOGRAPHY

Northern and North Western Mountains

Ranges	Avg. Height	Highest Peak
❖ Himalayas	❖ 4000m	❖ Nanga Parbat 8126m
❖ Karakoram	❖ 6000m	❖ K-2 8610m
❖ Hindu Kush	❖ 5000m	❖ Tirich Mir 7690m

Relief Topography	Passes	Economic Activities	Drainage Pattern	Environmental Issues	Glaciers
❖ Snow capped peaks ❖ Sharp peaks ❖ Passes ❖ Glaciers ❖ V-shaped valleys ❖ Steep slopes ❖ Forests ❖ Fast flowing rivers	❖ Khunjerab ❖ Shandur ❖ Babusar ❖ Lowari	❖ Nomadic lifestyle ❖ Cattle breeding ❖ Terraced farming ❖ Cottage industry ❖ Honey making ❖ Fishing ❖ Sugar refining	❖ Gorges ❖ Rapids ❖ Springs ❖ Streams ❖ Lakes ❖ Fast flowing rivers ❖ Water falls	❖ Deforestation ❖ Floods ❖ Earthquake ❖ Land sliding ❖ Siltation ❖ Soil erosion	❖ Baltoro ❖ Batura

Western Mountains

Ranges	Passes	Relief	Rivers	Economic Activities
❖ Waziristan Hills ❖ Safed Koh ❖ Suleman Range ❖ Kirthar Range	❖ Khyber ❖ Kurram ❖ Tochi ❖ Bolan ❖ Gomal	❖ Height is less than 4000m ❖ Mineral deposits ❖ Bare of vegetations ❖ Passes ❖ Snow – capped peaks ❖ Valleys	❖ Kabul ❖ Kurram ❖ Tochi ❖ Gomal	❖ Inaccessible mineral deposits ❖ nomadic lifestyle ❖ sheep and goat rearing ❖ lack of infrastructure ❖ lack of trade towns only Kohat and Peshawar ❖ lack of cultivation because of less rainfall ❖ No irrigational system due to rugged landscape.



Balochistan Plateau

Balochistan Plateau	Rivers
❖ Basins – depressions – no way out to sea ❖ Inland drainage basins forming lakes called as Hamuns ❖ Example Hamun-e-Mashkel ❖ Salt Lakes ❖ Saltpans – salt appeared on the surface after evaporation	❖ Hab ❖ Hingol ❖ Dasht ❖ Niehing ❖ Zhob

Economic Potential	Ranges
❖ Mineral oil – gas – coal ❖ Saindak: coal & copper mine ❖ Gwadar and other ports for trade ❖ Fishing (pasni-omara-jiwani-sonmiani) ❖ Livestock farming ❖ Fruits and vegetables	❖ Ras Koh ❖ Central Brahui range ❖ Hala range ❖ Toba Kakar range ❖ Makran Coast range ❖ Siahan range

Q: If Balochistan would develop, it can feed whole of Pakistan. Give arguments in favour or against it. [5]

Potwar Plateau (badland topography)

Topography/Relief	Economic Activities
❖ Irregular depressions ❖ Dissected and eroded land because of running water ❖ Ridges ❖ Ravines ❖ Troughs ❖ Soan river forming gullies	❖ Agriculture (barani or rain fed land) rugged landscape does not allow canal irrigation. ❖ Minerals: oil and gas ❖ Limestone salt and gypsum ❖ Attock oil refinery ❖ Cattle breeding Trade as system of roads, railways and airports developed

Indus Plain

Relief	Rivers	Economic Activities
❖ Active flood plain ❖ Old flood plain ❖ Alluvial terraces ❖ Piedmont plain ❖ Tidal delta ❖ Deserts	❖ Indus ❖ Jhelum ❖ Chenab ❖ Ravi ❖ Sutlej	❖ Trade ❖ Industry ❖ Agriculture ❖ Fishing ❖ Livestock farming ❖ Tertiary (services)



Deserts	Features
❖ Thal Desert	❖ Sand storms
❖ Thar Desert	❖ Rolling sand dunes
❖ Cholistan	❖ Weathering of rocks because of wind
❖ Nara	❖ Lack of vegetation
❖ Tharparkar	❖ Bare rocks
❖ Kharan	❖ Severe temperature at day time
❖ Desert	❖ Cool nights
	❖ Drought resistant species of plants
	❖ Scanty rainfall
	❖ Water table is extremely low

(a) Study Fig. 1 which is a map of northern Pakistan.

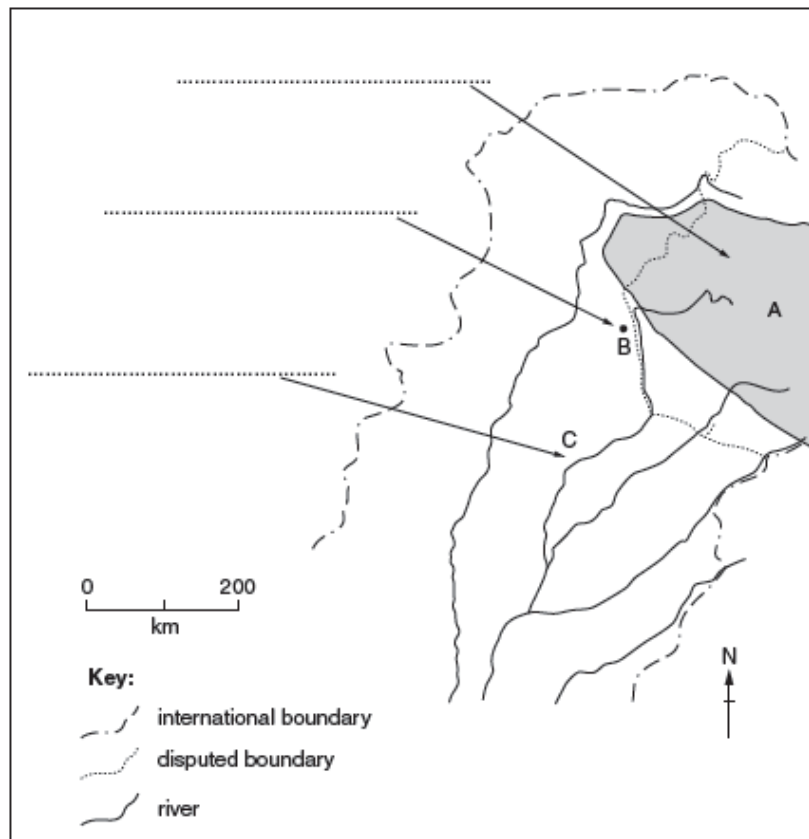


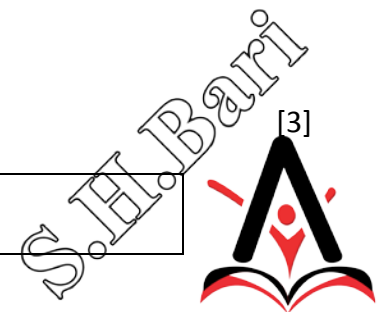
Fig. 1

(i) On the map name the following:

- Mountain range A
- City B
- River C

Answer: Study Fig. 1 which is a map of northern Pakistan.

A: Himalaya(s) B: Murree C: Jhelum



Study Fig. 9 which is a map of southern Pakistan.

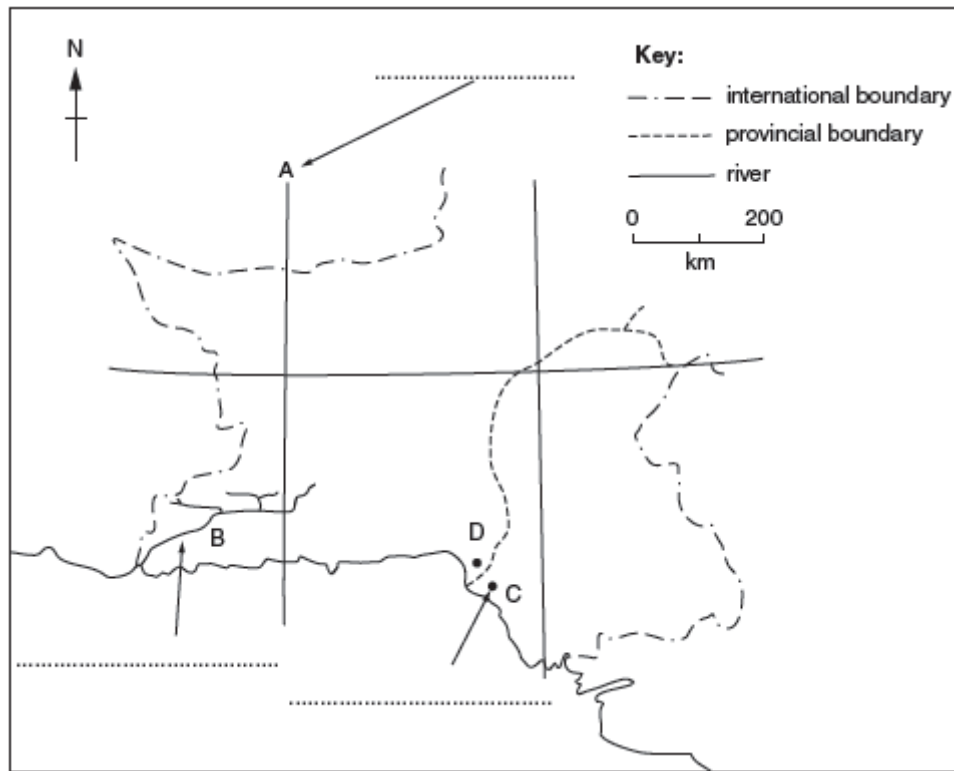


Fig. 9

(i) On the map name the following:

- Line of longitude A
- River B
- City C

[3]

Answer:

Study Fig. 9 which is a map of southern Pakistan.

On the map name the following: Line of longitude A; River B; City C

[3]

A: – 64E

B: – Dasht

C: – Karachi

Study Fig. 5, which is a map of southern Pakistan.

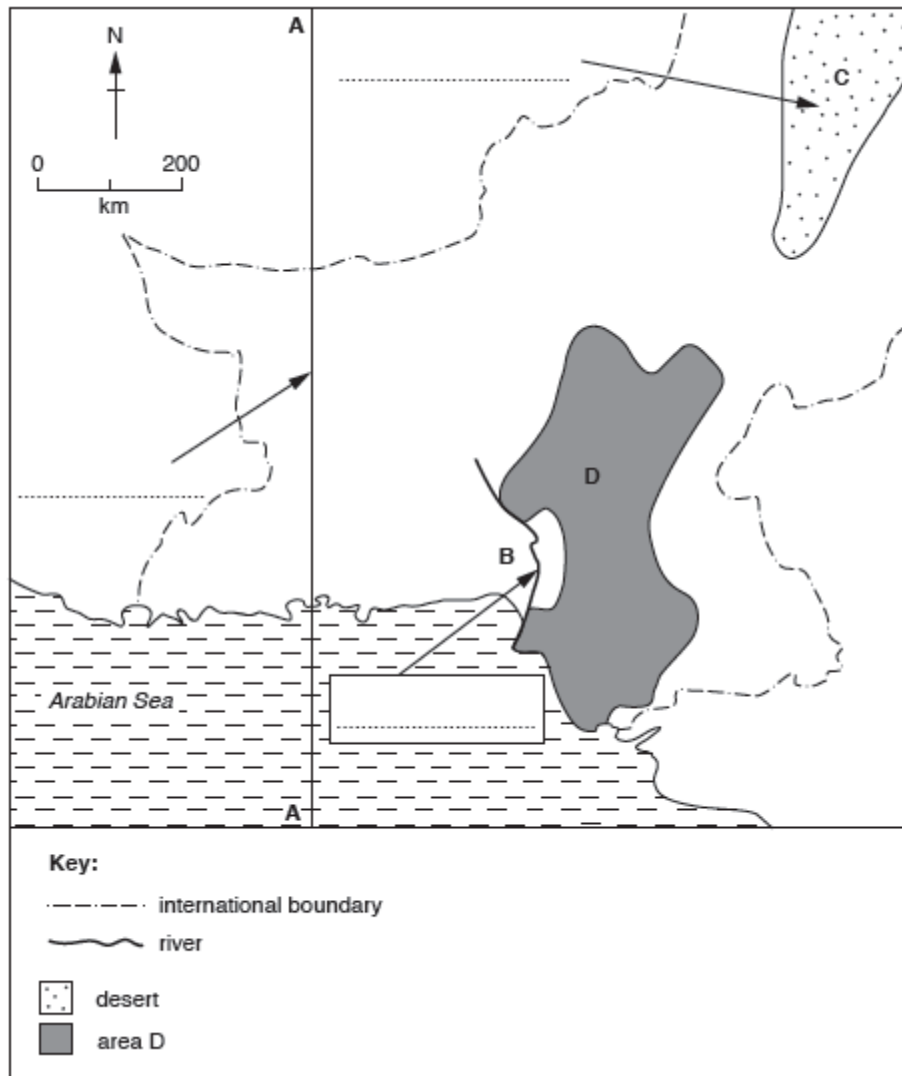


Fig. 5

On the map name the following: Line of longitude A–A; River B; Desert C.

[3]

Answer:

A 64°E

B Hab

C Thal

3 @ 1 mark

Describe the Figures.

Fig. 1



Fig. 1.2



Study Fig. 1.1, an outline map of Pakistan.

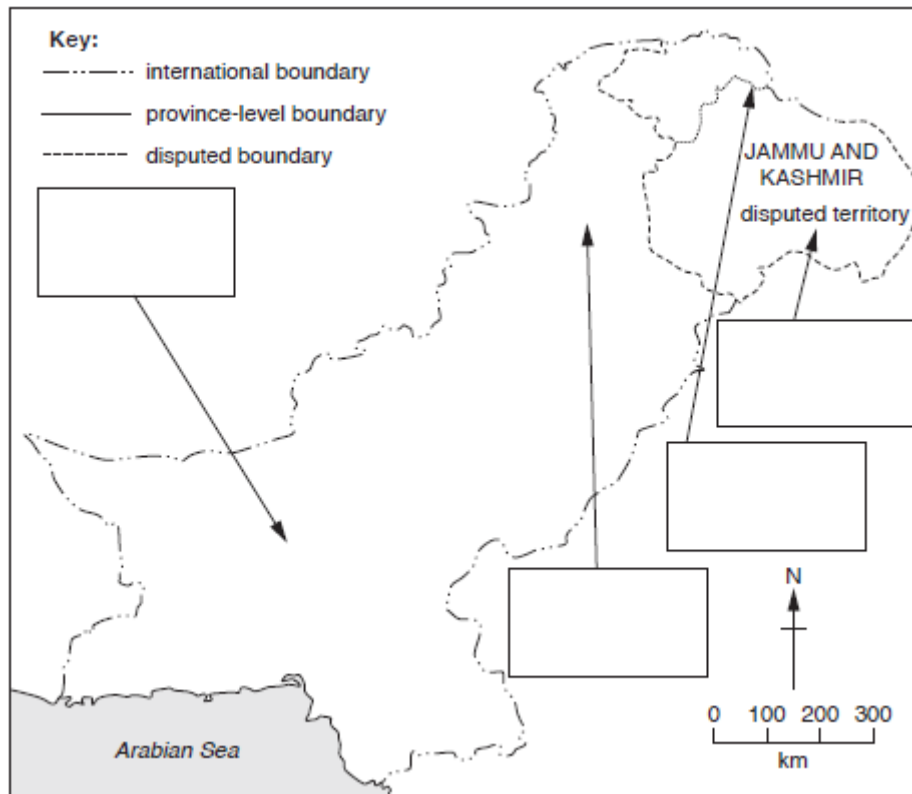
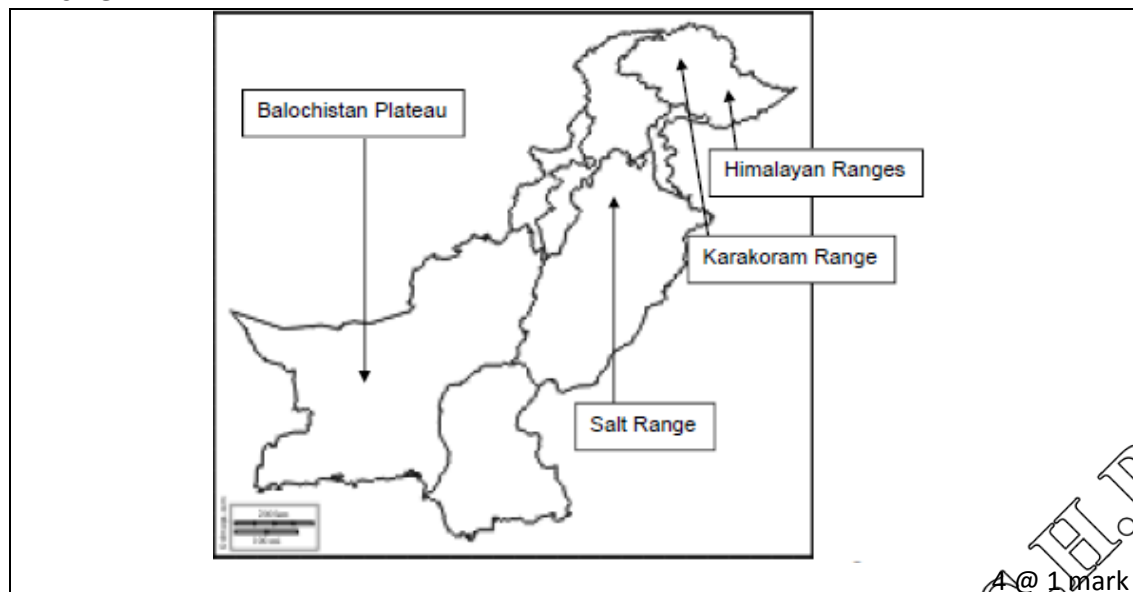


Fig. 1.1

On the map name the following landforms in the boxes provided: Balochistan Plateau; Himalayan Ranges; Karakoram Range; Salt Range.

Answer:



Chapter 2: CLIMATE OF PAKISTAN

Climate

It is the atmospheric condition of an area for a longer period of time e.g. 20 to 30 years.

Weather

It is the day to day atmospheric condition of an area for a short period of time.

Climatic Zones

Highland Zone

- ❖ Extremely cold long winters and snowy
- ❖ Summers are short and cool or mild
- ❖ Precipitation is in the form of snow as well as rainfall
- ❖ The temperature vary moving from north to west
- ❖ Rainfall (relief rainfall, monsoon rainfall, thunderstorm and western depression)

Low Land Zone

- ❖ Cool winters generally short
- ❖ Hot summers severe temperature generally long
- ❖ Monsoon rainfall (upper Indus plain receives more rainfall and lower receives less)
- ❖ Semi arid and arid climate
- ❖ Western depressions but less rainfall
- ❖ Convectional current
- ❖ Continental effect

Coastal Zone

- ❖ The climate is dominated by sea breezes throughout the year.
- ❖ Maritime influence: In summer the land heats up while the ocean remains cool. In winter, the land loses heat quickly and becomes cold while ocean is warmer.
- ❖ The humidity level in the area is high.
- ❖ Monsoon rainfall, Western Depression, Tropical cyclones.
- ❖ Temperature is generally mild.



Arid Zone

- ❖ Hot dusty winds
- ❖ Scanty rainfall
- ❖ Aridity....dry spells
- ❖ Hot summers which are long
- ❖ Cool nights
- ❖ Sand storms are common

Life style of People Living in Different Climatic Zone

Highland Zone

Due to extreme cold, the people live a hard life. The temperature usually falls below freezing point farming is impossible in winter. Mostly cottage industries are common. People live a nomadic life. Transhumance is practiced as they move to low lands and return back in spring season. People are mostly herdsman in summer as green valleys are there. Agriculture is practiced on terraced fields. Life becomes normal in summer and is difficult due to snowfall in winter.

Lowland Zone

The Punjab plains are fertile areas for cultivation. Indus plain is densely populated. People are greatly engaged in primary, secondary and tertiary activities. (see economic activities on Indus Plain in chap 1)

Coastal Zone

Climate is moderate primary activities as fishing. Boat making and repairing are common. Trade is a major occupation. People usually wear cotton clothes. (Same as Indus Plain for urban area as Karachi but different for rural areas)

Arid Zone

Life is difficult due to extreme heat, scarcity of water leads to many problems. Agriculture can only be practiced where water is available. Karez system helps to grow orchards of date palms, coconut, peaches, apples, apricots and melons. Crops as millet, barley and wheat can be grown. People have livestock and mostly live a nomadic life moving from one place to another with their animals in search of food, water and shelter.



Sources of Rainfall

Monsoon Winds

When summer sun heats up in tropical continents, air becomes warm and starts rising overland. It creates a low pressure which attracts cool moist air from sea. The main monsoon winds blow from the Bay of Bengal, after crossing Bangladesh and North India, the tail end of the wind enters northern Pakistan. The winds start to blow in June and cause heavy rainfall in northern Punjab, in July onwards.

Western Depression

These winds originate in the Mediterranean sea, It travels across Afghanistan and Iran then reaches the western areas. These winds cause rainfalls in areas from December to March.

Convectional Currents

At the start of summer, hot air rises up which also contains a high moisture content. When it reaches to the higher layers of the atmosphere, condensation takes place and this cause rainfall. Only northern and north western areas receive rainfall form convectional currents e.g. Rawalpindi, Lahore, Peshawar.

Relief Rainfall

It is related to height of land, it occurs where moist air moves up. This chilling causes heavy condensation and precipitation on the Windward side. The Leeward slope does not receive rain.

Tropical Cyclones

It may bring a few hours of very heavy rainfall with destructive winds to the coastal areas. They originate over the Arabic Sea quite often but only rarely they reach the coast of Pakistan. This source of rainfall is totally unreliable.

Floods

Causes

- ❖ Melting of snow from mountains in summer
- ❖ Heavy rainfall especially monsoon
- ❖ Lack of high embankments of rivers to control water
- ❖ No reservoirs to store and control the flow of water.
- ❖ Deforestation in the foothills of the mountains



- ❖ India opens the syphons flowing water in three rivers

Benefits

- ❖ More alluvium flow can result in fertility of soil.
- ❖ More water can be used for producing HEP.
- ❖ Water can help in the irrigation for cultivation.
- ❖ More fish can flow in rivers to the delta of the sea.
- ❖ High level of reproduction, breeding of fish in fresh water areas.
- ❖ Recharge the ground water supply.

Damages

- ❖ Residential areas will cut off from shops and services.
- ❖ Destruction of crops.
- ❖ Houses can be damaged.
- ❖ Epidemics break out causing people to be sick.
- ❖ The roads and railway tracks be damaged and cut off.
- ❖ Threat to human life
- ❖ Threat to animals

How to control the damages caused by flood?

- ❖ A forestation programs should be started to grow trees in foothills of the mountains.
- ❖ Building and raising the embankments.
- ❖ Extra channels and enlarging the channels to divert the flow of flood water.
- ❖ Building of more reservoirs, barrages and dams to store water to produce electricity and provide water for cultivation.
- ❖ Flood warning before time
- ❖ Evacuation of flood threat zones.
- ❖ Rehabilitation program be planned for flood victims and flood struck areas.
- ❖ Prevent, building in areas where flood can destroy the life.

Storms

These cause very strong winds which are common in April to June in summer and September to October in post monsoon season. Strong winds from 60 – 80 km per hour blow. Stormy rainfall or hail can occur.



Effects (Storms-Cyclones)

- ❖ Destruction of crops.
- ❖ Electricity supply is destructed.
- ❖ Communication system is damaged. Telephone lines, internet connections can be destroyed.
- ❖ Transportation is difficult.
- ❖ Trees can fall
- ❖ Accidents can occur
- ❖ Bill boards / sign boards can fall.
- ❖ Destruction of fruits as pears, apples etc

Droughts

It is the lack of rainfall resulting in massive scale aridity leading to starvation and death.

Causes**Natural Factors**

- ❖ Unreliability of rainfall by monsoon
- ❖ Long dry spells
- ❖ Hot winds
- ❖ Sandy soils lacking moisture
- ❖ Arid lands
- ❖ Global climate is changing

Human Factors

- ❖ Mismanagement of water resources due to seepage of unlined canals
- ❖ Dispute on water distribution among provinces
- ❖ overgrazing of land leading to soil erosion reduces the soil fertility

Effects

- ❖ Animals cannot get fodder
- ❖ Failure of crops
- ❖ Shortage of food
- ❖ Malnutrition>starvation>famine>Human beings can die
- ❖ Cattle can die
- ❖ Social set up destroyed
- ❖ Migration



How to manage drought?

- ❖ Desalination of sea water
- ❖ Water filtration plants
- ❖ Efficient management of water resources
- ❖ Artificial supply of water
- ❖ Thrifty usage
- ❖ Awareness among people
- ❖ Drought resistant species

Types of Drought**❖ Seasonal Drought**

It occurs in areas with good rain but dry season e.g., Potwar Plateau or Southern Punjab. It can be in any one season.

❖ Invisible Drought

Results due to water deficiency in soil which reduce crop yields. It is not forever.

❖ Permanent Drought

It is a situation when crop cultivation is not possible without irrigation e.g., Thal, Thar, Kharan.

❖ Unpredictable Drought

It is the result of abnormally low rainfall and occurs in areas having humid climates.

Factors affecting Temperature / Factors responsible for Climatic Variation

- ❖ Continental effect
- ❖ Maritime influence
- ❖ Latitudinal Effect: Areas closer to the equator receiving direct sunrays. Southern part of Pakistan is mostly under latitudinal effect.
- ❖ Away from equator
- ❖ High altitude: Air is denser at sea level because it absorbs water vapours, dust particles and solar radiation. Air is least dense at high altitude because less solar radiation is absorbed at that level. There is drop of 6°C for each 1000 meter altitude
- ❖ All types of rainfall
- ❖ Angle of sun (low in winter and high in summer)
- ❖ During summer, the northern hemisphere is tilted towards the sun and therefore the land absorbs more heat because of the angle of the sun and it faces the sun for longer duration (as longer days and shorter nights).
- ❖ During winter, the northern hemisphere is tilted away from the sun i.e. low angle of the sun so the earth absorbs less heat and faces the sun for shorter duration (shorter days and longer nights)



- ❖ Cloud cover also affects temperature to some extent. In day time it reduces the amount of incoming heat to outer spaces, resulting in a drop in day temperature. At night cloud cover traps the outgoing heat, temperature does not drop as low as temperature on a clear night.

(b) (i) Study Fig. 2 which shows rainfall data for two cities on the River Indus.

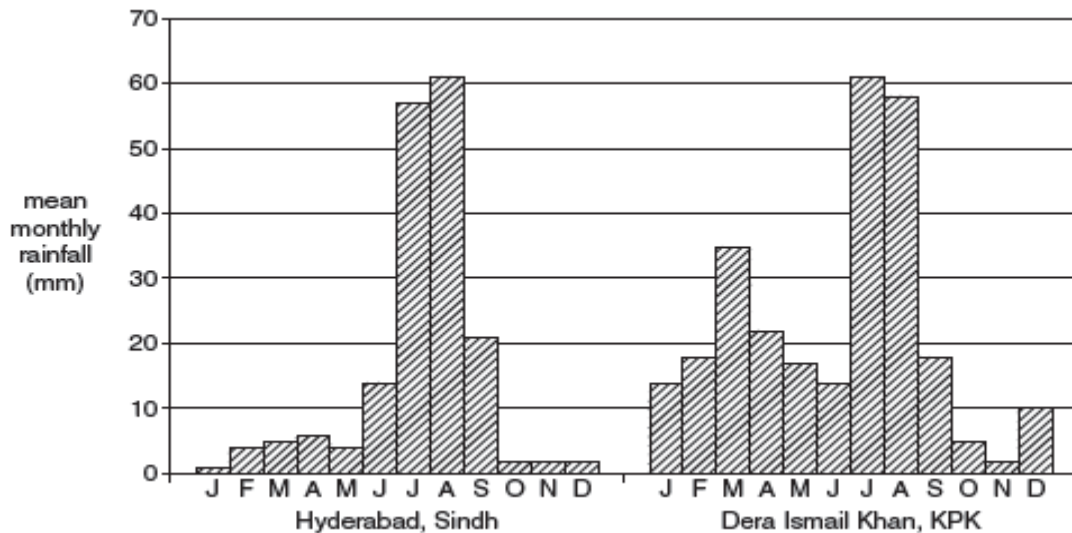


Fig. 2

A. Compare the amount and pattern of monthly rainfall in Hyderabad with that of Dera Ismail Khan.

Answer:

Compare the amount and pattern of monthly rainfall in Hyderabad with that of Dera Ismail Khan.

Amount

- Both high Jul and Aug
- Both identical Jun/Nov
- Both low Oct/Nov

For Dera Ismail Khan (accept converses for Hyderabad)

- Greater total
- 274 mm as opposed to 179 mm
- Higher in all months except Aug and Sep/any named month / lower in Aug/Sep
- A pair of stats to illustrate for any month (e.g. May H – 4 mm, DIK – 17 mm)
- Max 1
- Tolerances: ± 1 mm

Pattern

- Both maximum Jul–Aug
- For Dera Ismail Khan (accept converses for Hyderabad)*
- Has double maximum Jul–Aug and Mar (H – one maximum)
- Has more evenly distributed rainfall over the year (H – more variable)



B. Give three reasons for any similarities or differences in the two patterns of rainfall.

Answer:

Give three reasons for any similarities or differences in the two patterns of rainfall.

Both experience monsoon rainfall [Jul–Sep]

Dera Ismail Khan experiences rainfall from western depressions [Dec–Mar]

Dera Ismail Khan experiences some thunderstorm rainfall [Apr–Jun]

Accept converses for Hyderabad

Study Fig. 1 which shows the distribution of monthly rainfall in Karachi.

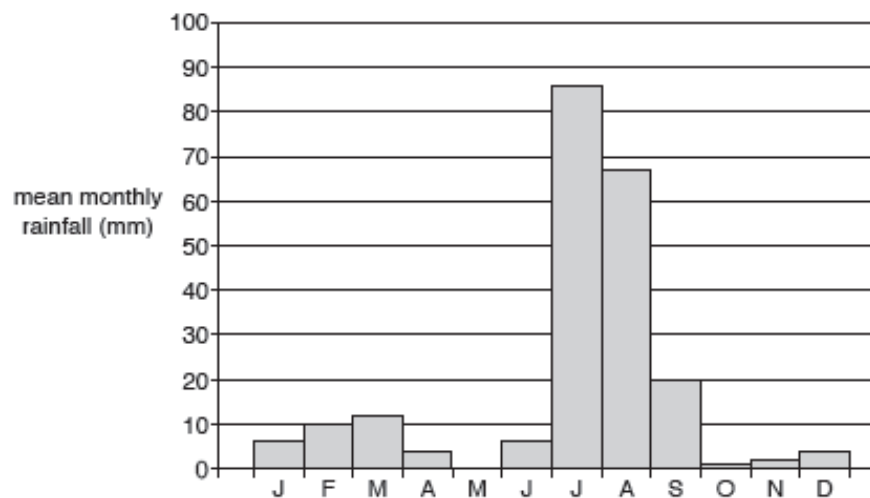


Fig. 1

A For how many months does Karachi experience less than 10 mm rainfall?

B Estimate the total rainfall in Karachi for the period July to September.

Answer:

Study Fig. 1 which shows the distribution of monthly rainfall in Karachi.

A 7 (may simply list the 7 months)

B 173 mm Tolerance 171–175 mm

Study Fig. 7, which shows climate data for Lahore, Punjab. Lahore has a monsoon climate.

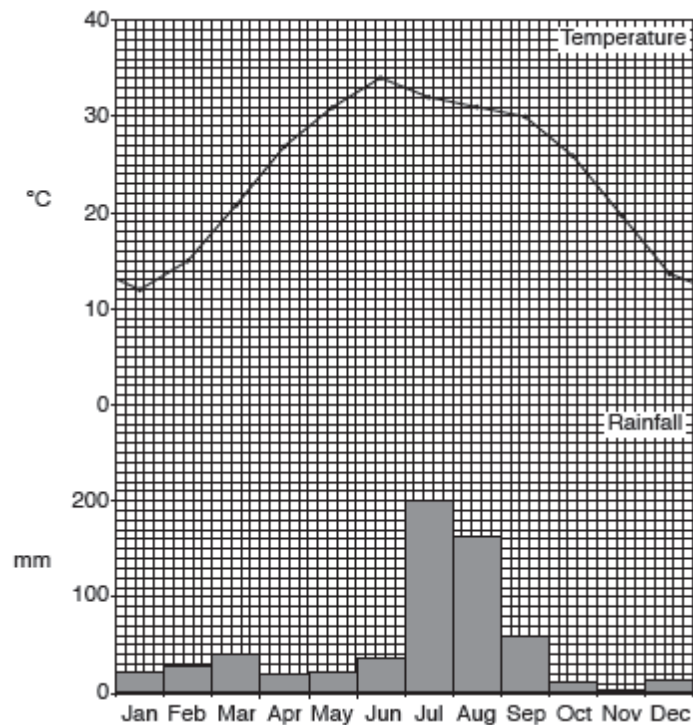


Fig. 7

Rivers are an important resource for human settlement and economic activity, but flooding is an increasing problem which can hinder development. Read the following two views about ways to manage flooding in Pakistan:



Which view do you agree with more? Give reasons to support your answer and refer to examples you have studied. You should consider View A and View B in your answer.

Answer:

Levels marking
No valid response 0

Level 1

Simple point referring to any view (1)
Simple points referring to any view (2)

Level 2

Developed point referring to one view only (3)
Developed points referring to both views (4)

Level 3

Developed points referring to both views with evaluation or relevant example (5)

Developed points referring to both views with evaluation and relevant example (6)

Content Guide

Answers are likely to refer to:

Building more flood management schemes

- So many people are affected;
- Many people killed / injured;
- Homes / farms / businesses destroyed;
- Cost of clear up and losses on economy;
- Floods hinder development as constantly having to rebuild / replace infrastructure;
- Loss of days at school affects literacy rates / skills base;
- Loss of days at work has an impact on revenue and tax collected;
- Tax changes to cover cost of damage;

Etc.

Allowing rivers to flood naturally

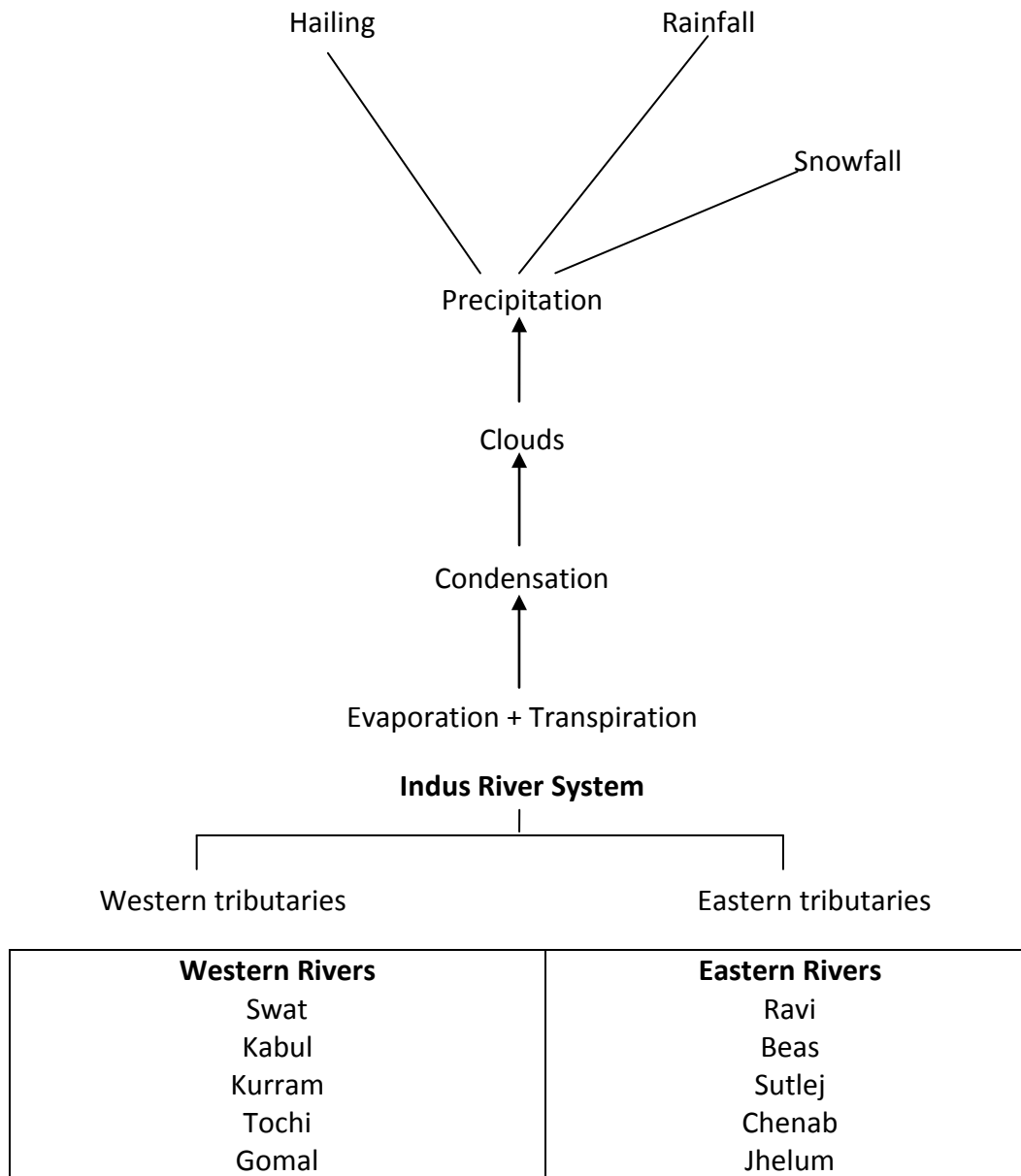
- Rivers provide nutrients to soil, so good for farming;
- Should not build on flood plain;
- Cost of management / flood protection is high;
- Flood management needs to be maintained and updated, incurring ongoing costs;
- Money is better spent on developing infrastructure, e.g. services, transport and industry;
- Flood plains can be used for fish farming;
- Flood plains can be used to channel water for irrigation;

Etc.



Chapter 3: DEVELOPMENT OF WATER RESOURCES

Hydrological Cycle



Balochistan River System

River Zhob, Khandhar and Kalacchi drain into river Indus. The rivers Loralai, Chakar, Bolan and Mula flow into Kathci Sibbi Plain. Rivers Hub, Porali, Hingol and Mashkel drain into Arabian Sea.

Uses of Water

Industrial

- ❖ Tanning industry for washing leather
- ❖ Food processing industry for juices, squashes beverages etc
- ❖ Chemical industries or making acids, liquid bleach and solutions
- ❖ In textile industries for washing bleaching, bluing, dying printing
- ❖ In iron and steel industry for cooling down of furnaces
- ❖ Thermal power stations for making steam
- ❖ Mineral water industry as raw material
- ❖ For pharmaceuticals e .g, syrups, drips, injections

Domestic

- ❖ Cooking
- ❖ Sanitation
- ❖ Washing
- ❖ Drinking
- ❖ Heating

Factors responsible for development of irrigation system

- ❖ Unreliability of rainfall
- ❖ Long dry spells
- ❖ High rate of evapo – transpiration
- ❖ Variability of amount of rainfall
- ❖ To store water
- ❖ To provide water where there is no rainfall

Conventional Methods of Irrigation

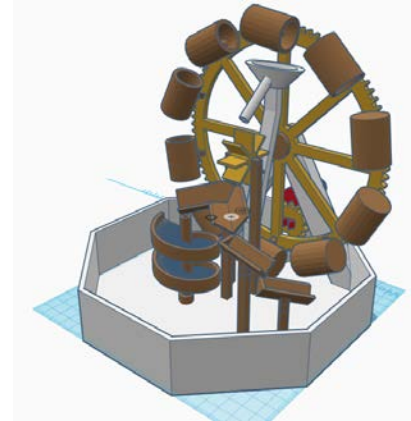
Shaduf

Bucket is attached to the pole on one side and weight on other side to get water out from well, river etc. A small area can be irrigated. It is not used now.



Persian Wheel

It is designed traditionally powered by a blind folded bullock. It turns a horizontal wooden wheel geared to a vertical wheel at the distant end of the shaft. A chain is attached with earthen pots; the water is spilled in the channels to flow from fields.

**Charsa**

Animal power is used to pull water from wells. It is not used now.

Tanks

These are made in fields to store water.

Karez

It is a horizontal underground water channels in the foothills that brings ground water to surface. These are only found in Balochistan to stop the evapo-transpiration because of high temperature.

Inundation Canals

Canals have water in rainy season only.

Modern Methods**Perennial Canals**

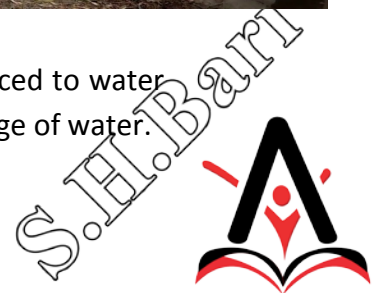
These canals are taken out from dams and barrages having water all the year round.

Tube wells

They are installed in ground. They have diesel or electrically operated pumps that can raise water flowing in channels to the field.

**Sprinklers**

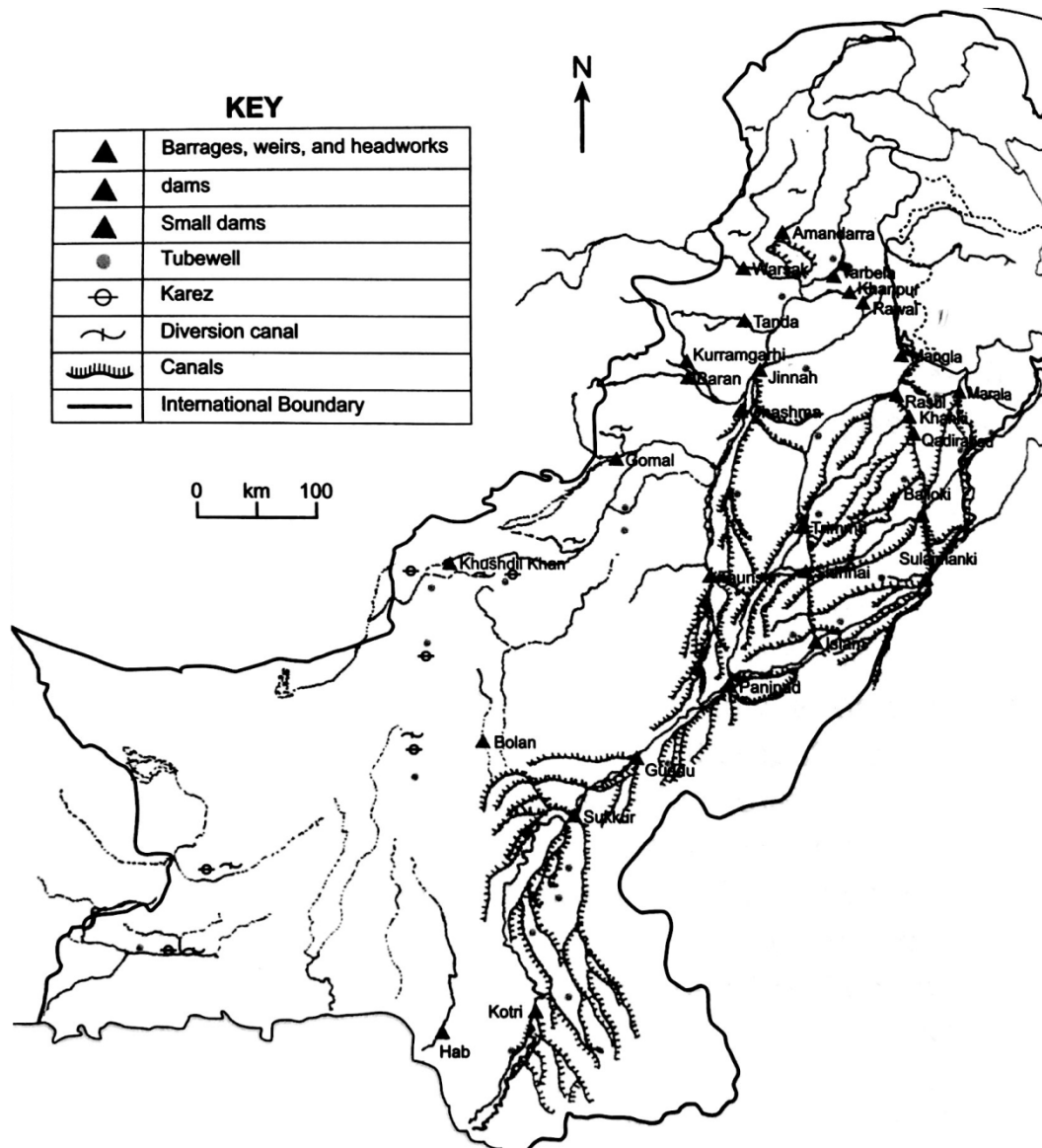
The sprinklers are connected to public water supply pipes. They are placed to water fields or orchards but they are an expensive method. There is less wastage of water.



Tankers

They collect water from lakes and ponds to provide it to fields and houses in case of emergency.

Q: Name any three systems of irrigation other than perennial canals and describe how they work? (6)



Conventional methods of irrigation system

Advantages	Disadvantages
• Less expensive • Less maintenance cost • No fuel / power required • Environmental friendly • Employment	• Time consuming • Labour intensive • Less water • Low area of cultivation • Low output • Cannot meet the growing demands of population

Modern method of irrigation system

Advantages	Disadvantages
❖ More water available ❖ More land irrigated ❖ More output ❖ Not labor intensive ❖ Less time consuming ❖ Technology used ❖ Can meet the growing demand of population	❖ Unemployment ❖ Air pollution ❖ Noise pollution ❖ High installation cost ❖ High maintenance cost ❖ Power failure will stop the work and supply ❖ Wastage of water

Indus Water Treaty (1960) Arbitration of World Bank

Pakistan got authority of three rivers – Chenab, Jhelum and Indus and three rivers were given to India as Ravi, Beas and Sutlej.

Under the treaty the following dams, barrages, and link canals have been completed.

Dams

Mangla, Tarbela and many more.

Barrage

Chashma, Marala, Rasul, Qadirabad.

Link Canals

- ❖ Marala Ravi
- ❖ Ravi – Badian – Dipalpur
- ❖ Rasul – Qadirabad
- ❖ Qadirabad Balloki
- ❖ Balloki – Sulaiman 1,2
- ❖ Chashma Jhelum



- ❖ Trimmu Sidhani
- ❖ Taunsa Punjnad
- ❖ Sindhani Mailsi Bahawal

Dams

Suitable place to build Dam

- ❖ Mountainous area as high altitude available to make water falls to move turbines
- ❖ Steep slopes and Natural basins help to make reservoirs easily
- ❖ Availability of water by rainfall and melting of snow
- ❖ Low temperature so less evaporation during summers
- ❖ Low evacuation cost because of less population density

Importance

- ❖ Produce HEP (Hydro Electric Power)
- ❖ Storage of water in reservoirs for domestic and industrial usage
- ❖ Store water for irrigation purpose through canals
- ❖ Controls flood and save from its devastations
- ❖ Attract tourists
- ❖ Fishing in reservoirs

Prepare

- ❖ Difference between small dam & large dam
- ❖ Difference between barrage & dam

Lined Canals

Canals having cemented embankments are called as lined canals. There is no seepage of water.

Unlined Canals

Canals having no cemented embankments are called as unlined canals.

Water Table

The level of water under the ground is called as water table.



Water Logging and Salinity

Excess of water in land causes water logging. It is due to high water table.

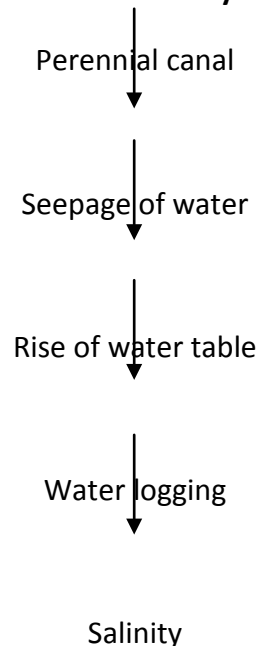
When water dries up and white salty patches appear on the surface. It is called as salinity.

How to control water logging and salinity?

- ❖ Lining of canals to stop seepage of water
- ❖ Canal closure on temporary basis
- ❖ Installing tube wells to pump out water to lowers the water level
- ❖ Planting of eucalyptus trees
- ❖ Treating the land with chemical or limestone

SCRAP (Salinity Control and Reclamation Project) is working for treatment of water logged and saline areas

How perennial canals destroy the farmland?



Siltation

Causes

- ❖ Slit eroded from the mountains with the help of flow of river.
- ❖ Eroded material from narrow and deep valley causes siltation.
- ❖ Due to deforestation, fast flow of silt accumulates.



Effects

- ❖ Weakens the foundation of dams.
- ❖ Blockage of canals.
- ❖ Reduction in storing capacity of water in dams.
- ❖ It can result in floods.
- ❖ Fluctuation of electricity due to silt in turbines.

Solutions/ How to control?

- ❖ Afforestation programs.
- ❖ Cemented embankments.
- ❖ Silt traps should be installed before flow of water in dams.
- ❖ Raising the height of dam to increase the capacity of reservoir.
- ❖ Operating the water level at low pace when flood flows.
- ❖ Stop deforestation

Q: Name a dam and describe its location.

Dams	Constructed on Rivers
Tarbela	River Indus
Mangla	River Jhelum
Warsak	River Kabul

Barrages: They are long structures used for irrigation and flood control. They can be made on flood areas. The cost of their construction is low and can be constructed in plain areas.

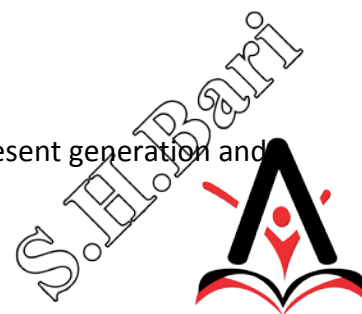
Famous Barrages

Marala, Chashma, Jinnah, Taunsa, Sukkur, Kotri, Guddu, Qadirabad, Rasul.

Effect of Climatic Changes

- ❖ Shortage of water supply
- ❖ Depletion of ozone layer
- ❖ Speedy melting of glaciers is expected
- ❖ Heavy flooding
- ❖ Flood in low lying areas due to rising seas levels
- ❖ Droughts in arid and semi arid area

Sustainability: Sustainable development is which meets the needs of present generation and available for future generations to meet their needs.



Study Fig. 5, which shows a diagram of the karez system of irrigation.

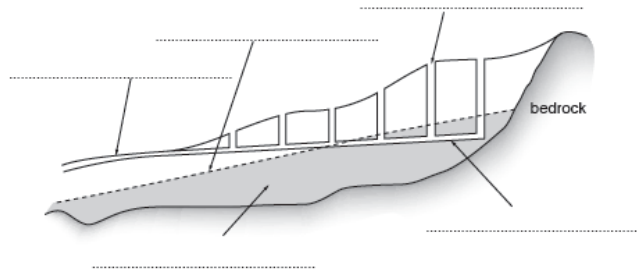


Fig. 5

Choose three terms from the list below and use them to label the diagram in three of the spaces provided.

Maintenance shaft aquifer canal tunnel water table [3]

Read the following two views about water shortages in Pakistan:

A

The best way to prevent water shortages in Pakistan is to build more dams and other infrastructure projects. These will store or supply more water.

B

The best way to prevent water shortages in Pakistan is to educate people about different methods of saving water. These methods could be carried out in agriculture, industry, and in homes.

Study Fig. 2.1, a map showing the major rivers of Pakistan.

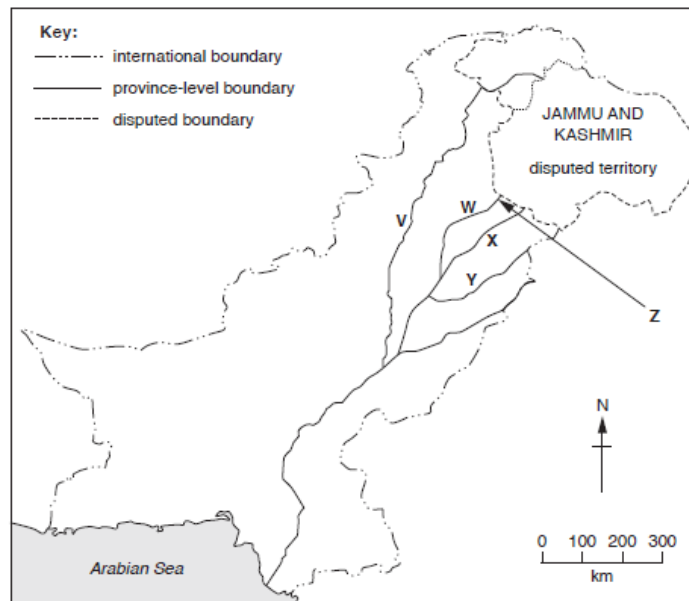


Fig. 2.1

Name rivers V, W, X and Y.

Answer:

V = Indus

W = Jhelum

X = Chenab

Y = Ravi

4 @ 1 mark

Read the following two views about providing solutions to the challenges of water supply in Pakistan:

A

Small dams are the best way to solve water shortage problems and assist development.

B

Large dams are the best way to solve water shortage problems and assist development.

Chapter 4: FORESTS

A large area continuously covered with trees is called as forest. Pakistan has 4.8% forests.

Productive Forests

One third of the area of Pakistan is covered with these forests. They are natural forests having high density of trees and canopy is closed they have great commercial value. Timber is usually extracted.

Protection Forests

These are man-made forests. They are linear plantations mostly in parks and along roads and railway lines. They prevent soil from erosion.

Importance of Forests

- ❖ Industrial usage: Timber, herbs for pharmaceuticals, wood for paper etc
- ❖ Fuel wood
- ❖ Prevent soil erosion
- ❖ Natural habitat for animals
- ❖ Reduces air pollution
- ❖ Increase soil fertility
- ❖ Fruits are available
- ❖ Provide science beauty and attract tourists
- ❖ Source of earning (wood and wood based products)
- ❖ Control floods
- ❖ Brings rainfall
- ❖ Regulates the temperature
- ❖ Provide job opportunity

Deforestation

Causes

- ❖ Industrial purposes (furniture, paper, pharmaceutical)
- ❖ Fuel wood particularly in northern areas
- ❖ Urbanization
- ❖ Farming
- ❖ Due to over grazing by animals
- ❖ Building of roads/railways.
- ❖ Mining



Effects

- ❖ Soil erosion
- ❖ Loss of natural habitat to animals
- ❖ Extinction of species
- ❖ Results in flood
- ❖ Results in siltation in dams reducing the storage capacity of water in dams
- ❖ Increase surface run off
- ❖ Air pollution
- ❖ Less oxygen
- ❖ Increases the temperature
- ❖ Less generation of HEP in dams due to less water because of siltation

How to control

- ❖ Afforestation programs such as Rachna Doab Afforestation Project.
- ❖ Develop more irrigational forests
- ❖ Trees cut should be substituted i.e. if a tree is cut down it must be replaced.
- ❖ Supply of natural gas for fuel as an alternate instead of wood.
- ❖ Increase the awareness among people
- ❖ Strict forest laws to be imposed to stop illegal cutting.
- ❖ Selective cutting methods to be used
- ❖ A tree should be cut with the same ratio it grows.

Learn these terms

- ❖ Reclamation: To convert a barren land to a cultivate land with the help of irrigation and fertilizers etc.
- ❖ A forestation: To replant a deforested area.
- ❖ Terracing: Crops are grown on the steps of mountains; small embankments are made on the side.
- ❖ Contour Ploughing: Contours are made; trees and plants are grown parallel to each other.
- ❖ Strip Farming: A small crop or plant can be grown under the shade of a large tree or plant.

How mangroves survived in Salt water?

- ❖ Filters on root tips
- ❖ Internal tissues exhibit a high tolerance to salt
- ❖ Excess salt is carried to leaves and excreted onto their surfaces

Threats to Mangroves

- ❖ The cutting of mangroves for fodder
- ❖ Firewood



- ❖ Making tourists resorts
- ❖ The removal of sand from beaches
- ❖ Reclamation of beaches for constructing buildings and roads
- ❖ Chemical waste of industries dumped into Mangrove areas.
- ❖ Construction of barrages and dams on River Indus stops the supply of nutrients.

Importance of Mangroves

- ❖ Breeding grounds for fish, shrimps, crabs.
- ❖ Increase in fisheries will increase the annual catch leading to more foreign exchange.
- ❖ Fallen leaves of Mangroves give rise to a fertile habitat in the estuaries.
- ❖ The losses by tropical cyclones are reduced as Mangroves act as a barrier.
- ❖ Mangrove forests produce honey from the blossoms of *Avicenna Marina* species.



Types of Forests

Types	Areas	Importance
Alpine forests	Northern areas (Chitral, Dir, Kohistan)	Used as fuel wood only.
Coniferous forests	Northern areas. NWFP (Abbottabad, Mansehra, Kohistan, Shangla, Swat) Rawalpindi, Islamabad, Murree. Balochistan Mountains. (Quetta & Kalat divisions)	Important sources of timber for making furniture. Environmental protection. Help in checking floods. Good breeding centers for birds and wildlife. Attract tourists and promote tourism industry. Add to the scenic beauty of the area.
Tropical Thorn forests (Rakh)	Punjab Plains. Southern and western Balochistan. Sindh Plains.	They are used as firewood.
Sub Tropical Scrub Forest	Hills and foothills of Lower Himalayas. Suleman and Kirther Ranges. Western Mountains (Peshawar, Waziristan, Kohat, Mardan)	Watershed protection. Supplying firewood. Grazing purposes.
Riveration or Bela Forest	River Indus and its tributaries.	Provide Shishum and Babul, two valuable species, which are used for making furniture, agricultural implements.
Mangrove Forests	Coastal areas of Sindh. Coastal areas of Balochistan Deltaic regions	Supply of firewood. Coastal communities use these forests for timber. Breeding grounds for fish shrimps. Camels and livestock feed on the leaves of mangrove. Fallen leaves provide nutrition for marine life. Protect coastline from erosion, storm, damage and wave action. Act as barrier against intensity of earthquakes and tsunamis by absorbing aback waves.
Irrigated Forests	Changa Manga near Lahore. Chichawatni in Sahiwal district	Important source of timber, firewood. When planted in linear form they provide shade.

Q: How can a forest are described as Irrigated? (4)

Q: Why is there a need for more irrigated plantation in Pakistan? (4)



Describe the Figures.

Photograph A for Question 1



Photograph B for Question 1



Study Fig. 2 which is a map showing different forest types in Pakistan.

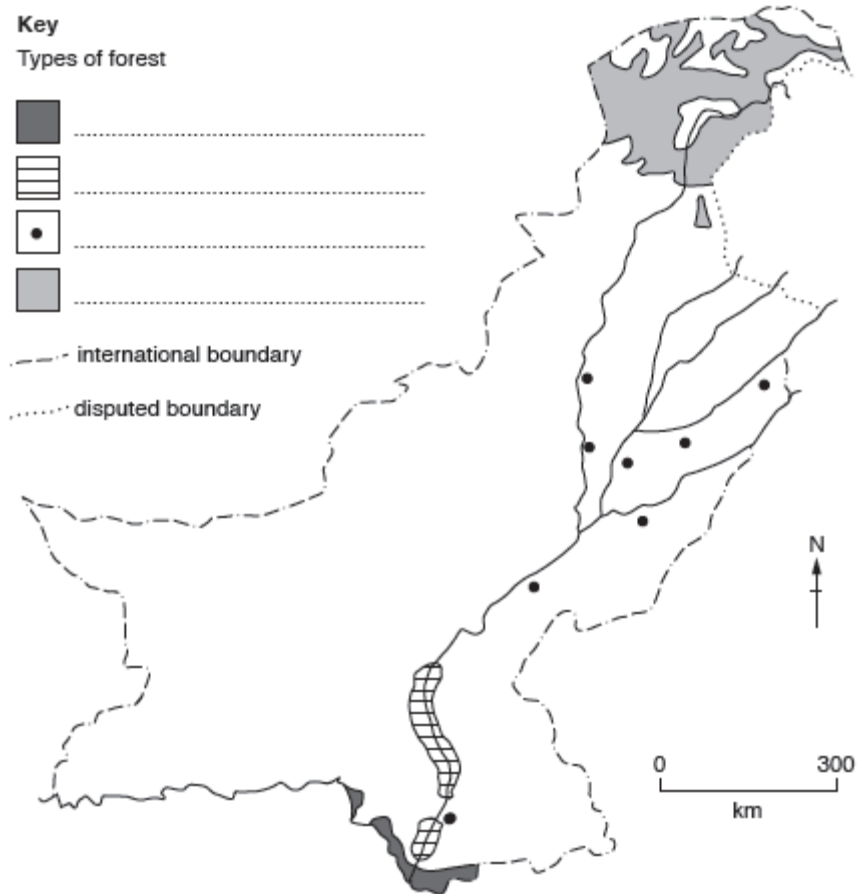


Fig. 2

(i) In the key, name the types of forest shown on the map.

[2]

Answer:

Study Fig. 2 which is a map showing different forest types in Pakistan.

(Top to bottom) Mangrove, Riverain / Bela, Irrigated, Coniferous / Alpine

(Mark as one or two correct 1 mark; three or four correct 2 marks)

Study Fig. 2.1, a map showing forest types in Pakistan.

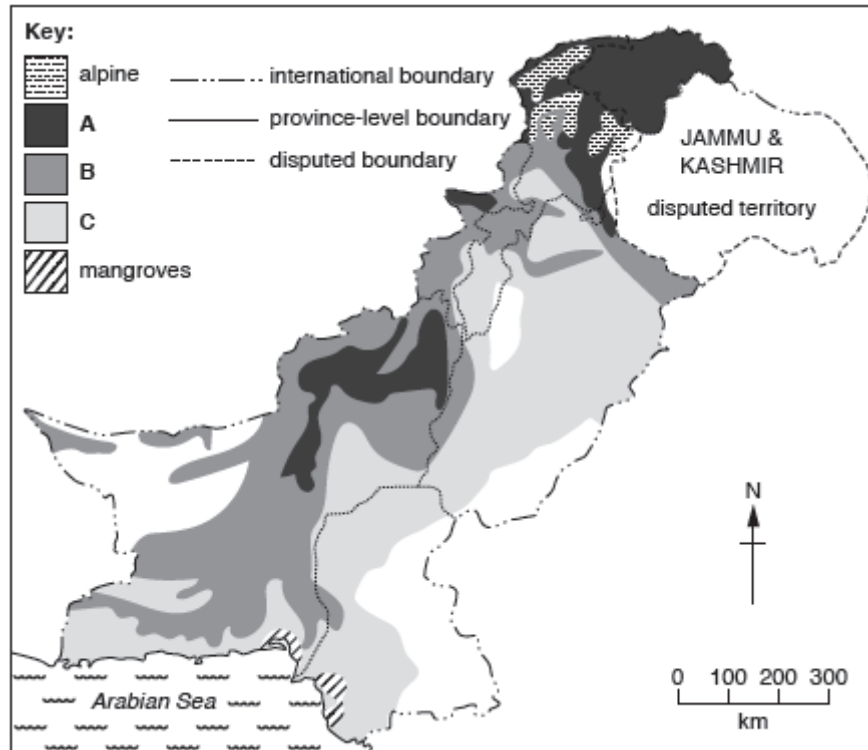


Fig. 2.1

Name the forest types shown at A, B and C:

Answer:

- Forest type A – Coniferous;
- Forest type B – Subtropical scrub / subtropical dry / subtropical thorn / dry thorn scrub;
- Forest type C – Tropical thorn / rakh.

3 @ 1 mark

Using Fig. 2.1, name two areas in Pakistan where mangroves grow.

Answer:

- Coastal areas of Sindh / Indus Delta;
- Coastal areas of Balochistan / Sonmiani Bay;
- Along the coastline of / near the Arabian Sea.

@ 1 mark

Study Fig. 2.2, which shows a coniferous tree.



Fig. 2.2

Explain how this type of tree has adapted to the climatic conditions it grows in. You should develop your answer.

Answer:

Ideas showing how the tree has adapted to the climate such as:

- Evergreen – no need to renew leaves / short growing season;
- Compact conical shape – stabilises the tree in windy conditions / releases snow / prevents snow accumulation;
- Needles instead of leaves – reduces moisture loss;
- Tall / straight trunk – in order to grow straight towards sunlight;
- Cones – protect seeds during cold months;
- Downward pointing branches – allows snow to easily fall off;
- Shallow root system – as soils are thin / subsoil is frozen for most of the year;
- Long / wide spreading roots – helps to anchor the tree against strong winds;
- Thick bark – protects from cold winds etc.

Note: One mark for identification of appropriate idea and a further mark for development (in parentheses).

Note: Max. 2 marks if no development.

2 @ 2 marks

Around 5 per cent of the land area of Pakistan is forested. To meet sustainable development targets the amount of land covered by forest needs to increase to 25 per cent by 2030. Read the following two views about possible uses of land in Pakistan:

There are more important uses of land for the future development of Pakistan than increased forest cover.



Chapter 5: MINERALS

Types of Mining

Open Cast Mining

If a seam of mineral is near the surface, digging through excavators takes place for getting the mineral.

Adit Mining

Horizontal tunnels are dug to enter and then extraction takes place.

Shaft Mining

Vertical shafts are dug deep, than horizontal digging takes place to extract the minerals.

Quarrying

Open excavation method, when a seam of the rock is observed on the surface especially soft rock, limestone. It can be extracted with the help of power shovels, hammer, wedges and spades.

How are minerals formed? (Consult book)

Problems of Mining industry

- ❖ Lack of technology/machinery
- ❖ Poor infrastructure
- ❖ Lack of expertise
- ❖ Lack of skilled labor
- ❖ Less capital
- ❖ Lack of interest by government for the sector
- ❖ Institutional mismanagement
- ❖ Inaccessible mineral deposits areas due to lack of infrastructure as roads, electricity etc.

Solutions

- ❖ Government should provide capital
- ❖ Machinery should be imported
- ❖ Government should subsidize the machinery
- ❖ Government should provide education for developing expertise and skilled labor
- ❖ Government should be willing to promote it
- ❖ Develop infrastructure to explore minerals
- ❖ Attract private companies for investment through various scheme



Major Organization Working for Mineral Exploration

- ❖ Geological Survey of Pakistan: For the investigation and mapping of mineral deposits.
- ❖ Pakistan Mineral Corporation: For the exploration and marketing of minerals
- ❖ Resource Development Corporation: To investigate and develop copper mines at Saindak
- ❖ Gemstone Corporation of Pakistan: To develop gemstone resource

Effects of mining on Environment

- ❖ Noise pollution because of drilling and blasting
- ❖ Air pollution from dust and smoke
- ❖ Deformation of landscape. Depressions are caused.
- ❖ Destruction of natural habitat
- ❖ Water pollution if mining waste will not be properly disposed
- ❖ Deforestation as vegetation is cut down

Solutions

- ❖ Mining should be on scientific lines to minimize the damage
- ❖ Waste of mining should be properly disposed
- ❖ Land should formally leveled after mining
- ❖ Plantation in the mining area for better environment(oxygen)

Effects of mining on minors

- ❖ Poor health condition
- ❖ Explosions
- ❖ Poisonous gasses can kill them
- ❖ Lungs and skin diseases
- ❖ Lack of pure drinking water
- ❖ Lack of precautionary measures as masks
- ❖ Many poisonous gases can result in suffocation and death of the miners.
- ❖ Rock blasting can block the miners inside the mine.

Solutions

- ❖ Proper health facilities
- ❖ Safety precautions as emergency exists
- ❖ Pure drinking water should be available
- ❖ Improved working hours
- ❖ Recreational activities
- ❖ Proper protective clothing
- ❖ Safety masks to save from poisonous gases



Difference b/w Metallic and Non-metallic Minerals

Metals	Non-metals
Economically more valuable	Economically less valuable except for the power resources as oil gas and coal.
Hard, tough and shiny	Softer, rough and may not shine.
They can change shape without breaking.	They break away when shape is changed.
Metals are more reactive with water and acids.	Non – metals minerals are less reactive with water and acid
Many are good thermal and electrical conductors.	Poor thermal and electrical conductors.
They can be stretched and compressed	They cannot be stretched or compressed.

Q: Differentiate between metallic and non – metallic minerals?

[4]

Sustainable Development

- ❖ The government must introduce concept of sustainable development when making policies.
- ❖ The application of science and technology to enhance the industry's competitiveness and environmental protection.
- ❖ Discharge of toxic substances and the release of heat which is harmful to environment should be checked.
- ❖ Mining should be done in a way that it does not effects ecosystem.

Q: Describe some measure for the sustainable development of mining?

[4]

Non Metallic Minerals

Rock Salt

The rocks are white or pink in color. The salt is overlain by gypsum and clay. Rock salt is used for cooking and preservative purposes and for the manufacture of soda ash, bicarbonate of soda, caustic soda and other sodas for laundries, textiles, and tanning.

Brine

Used in the chemical and fertilizer industry.

Limestone

Limestone is major sedimentary deposit and is widespread in Pakistan. It is the main raw material for cement. It is also used in the manufacture of bleaching powder, glass, soap, paints and lime. It is used to treat sugarcane waste to produce alcohol fuel. It is painted on barks of trees to counter pests and termite attacks. Also used to treat salinity.



Coal

Pakistan has low-quality coal. Coal is mainly used in brick kilns, some is used to make coke and a small percentage is used for power generation. It is planned to build a thermal power station to use to coal from a new coalfield in Thar District.

Natural Gas

Domestic, industrial and transportation uses.

Mineral Oil (Petroleum)

It is used as power source, as a lubricant for machines, and as motor fuel.

Gypsum

Found in grey, white and pink color. It is used in the manufacture of paints and fertilizers. White gypsum is used for making cement and Plaster of Paris. Spread on saline soil to help land reclamation for farming.

Marble

Found in bands of white, grey, yellow and brown. It is used in buildings and for making chips for flooring and decorative pieces.

Clays

Clays are fine-grained minerals. In Pakistan the most important industrial clays are China Clay, Fire Clay and Fuller's Earth.

- ❖ China Clay is used in the ceramic industry, for a special type of cement has other industrial uses.
- ❖ Fire Clay, fine clay capable of enduring high temperature to make fire bricks and insulating bricks. It is also used to make pottery and chemicals.
- ❖ Fuller's Earth is used to manufacture steel, oil drilling machine and in the process of oil refining.

Magnesium

It has a high percentage of magnesia(about 50%). It is used in the manufacture of cement, fertilizer, paper pulp, chemicals and pharmaceuticals.

Sulphur

Sulphur is used in chemical industries to manufacture sulphuric acid, paints, explosive material, dyes and fertilizers.



Name	Uses
1. Chromite	Chromite gives hardness and electrical resistance to steel. It is used for bridges and railway carriages. It is also used as a lining in metallurgical furnaces and for making engineering tools and stainless steel etc.
2. Iron Ore	Steel making, construction and the transport industry.
3. Copper	Making electrical wires and other electrical appliances, especially switches that carry current, also used in making alloys, water pipes and tanks.
4. Manganese	Used in making dry batteries, paints. It is a vital alloy in steel making, flares and flash bulbs.
5. Bauxite	Aluminum is mainly obtained from bauxite and is a valuable metal. Uses: utensils, tins, cans etc. and many other products.
6. Celestite	Found in the cavities of sedimentary rocks. Uses; tracer bullets, fireworks, ceramics, paints and plastics.

Q: Name any two minerals and describe their uses?

[n]

Study Fig. 3 which is a map showing the locations where three different non-metallic minerals are extracted in Pakistan.

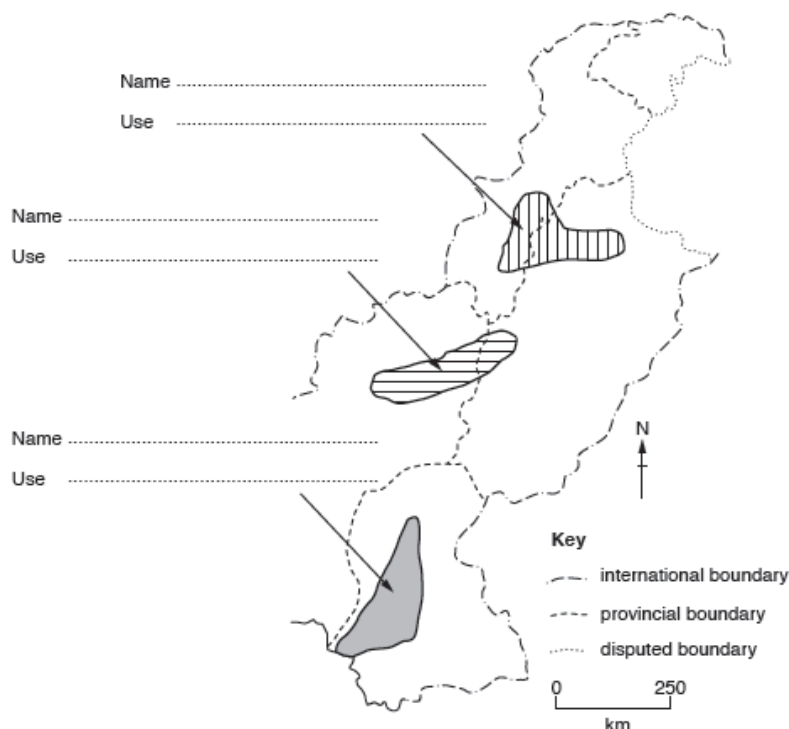


Fig. 3

(i) For any two locations, state the name of the mineral extracted and a use for this mineral. Write your answers in the spaces provided on Fig. 3.

You should choose from the following list:

Gypsum limestone rocksalt

Answer:

Study Fig. 3 which is a map showing the locations where three different non-metallic minerals are extracted in Pakistan.

You should choose from the following list: gypsum limestone rocksalt

Location

NW – rocksalt / limestone / gypsum

Central – limestone / gypsum

S – limestone

Mark any two correct

Uses

Rocksalt: cooking / preservation / soda (used in laundries / textiles / tanning) flavouring food

Gypsum: paints / fertilisers / boards / cement / to treat saline soil / plaster of paris

Limestone: for building / cement / bleach / glass / soap / paints / to treat saline soil / bleaching powder / paper

Study Fig. 4 which is a diagram of a coal mine.

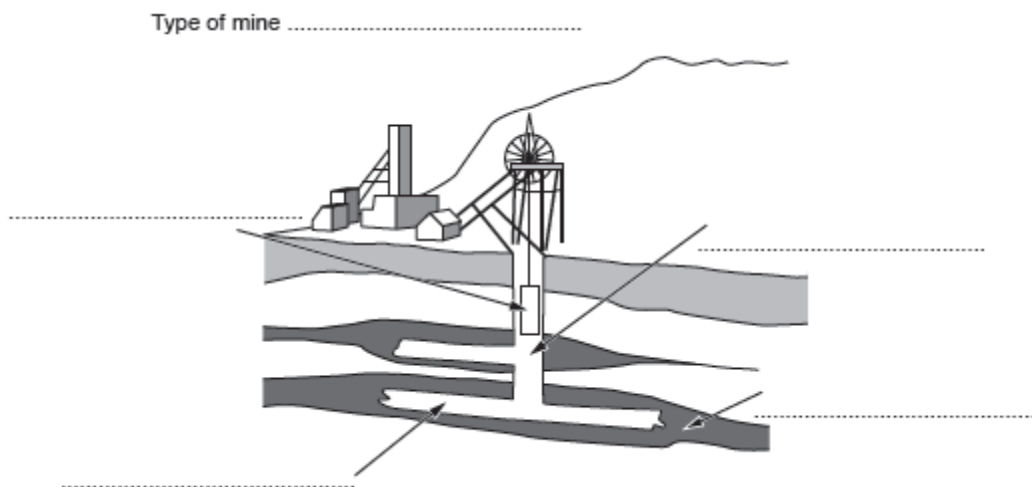


Fig. 4

Choose two terms from the list below and use them to label the diagram in any two of the spaces provided.

Adit cage shaft open-cast seam tunnel

Chapter 6: FISHING INDUSTRY

Fish farming is called 'Aqua Culture'.

Uses of Fish

- ❖ Fish is a delicious food.
- ❖ It is nutritious white meat.
- ❖ It is preferred as a low cholesterol diet.
- ❖ Fish waste is used to make fertilizer and poultry feed.
- ❖ Oil extracted from fish provides a source of vitamin A and D.
- ❖ It adds 0.9% GDP.
- ❖ Pakistan earns 6% of its total foreign exchange.

Subsistence Fishing

To catch fish for self consumption is called as subsistence fishing. Traditional methods are used by fisherman.

Commercial Fishing

It is on a large scale to earn huge profits. Modern techniques are used as mechanized boats and refrigeration facilities.

Inland Fishing

It is practiced in rivers, lakes and reservoirs of dams. Small ponds are also made for fish breeding and catching. 32% people are engaged in Inland fishing.

Main fishing centers are:

- ❖ Manchar lake in Dadu district.
- ❖ Kairi, Keenjhar lake, north of Thatta.
- ❖ Haleji lake, west of Thatta.
- ❖ Reservoirs of Mangla and Terbela Dams.
- ❖ River Indus at Sukkur, Kotri and Thatta.

Types of Fish

Mahasheer, Pella, Thalla, Rahu, Tront, Grass silver carp.

Marine Fishing

It is practiced in seas and ports. 68% are engaged in marine fishing.

Sindh: Karachi-Kemari

Balochistan: Gawadar-Pasni-Ormara—Jiwani-Sonmiani



Types of Fish

Sharks, Croakers, Skates, Drums, Cat Fish, Rays

Fish Farming

The man made rectangular ponds are made. They have concrete or cemented base to prevent water losses through leakage. Side of farms is edged with solidified mud. Trees are planted around the fish farms to provide oxygen and to minimize water losses.

How to develop Fishing Industry?

- ❖ Processing facilities must be increased for development of value added products as canned or frozen fish.
- ❖ The introduction of new fishing methods.
- ❖ Weather information is required in marine fishing particularly. through meteorological departments.
- ❖ New development projects by Marine Fisheries Department
- ❖ Providing loans to fishermen
- ❖ Modern boats and nets should be provided.
- ❖ Increasing storage facilities for fisheries.

Benefits of Developing Fish Farming

- ❖ Greater return to fishermen.
- ❖ Use of technology and up grading of skills will increase the fish production and revenue as well.
- ❖ Increased employment.
- ❖ Reduces burden on crop and livestock in rural areas
- ❖ Foreign exchange will come as export will increase.

Problems of the Fishing Industry

- ❖ Water Pollution by spillage of oil from ships
- ❖ Water Pollution by domestic and industrial waste is causing harm for fishing.
- ❖ Numerous chemicals have been found in marine life that can disturb the users if use in excess. (cadmium, aluminium and nickel)
- ❖ Over fishing i.e. catching fish throughout the year even in breeding season.
- ❖ Cutting of Mangroves
- ❖ Lack of storage and refrigerator facilities
- ❖ Lack of modern handling system
- ❖ Catching baby fish
- ❖ Use of illegal nets



Solutions/Sustainable Fisheries

- ❖ Banning the use of illegal nets.
- ❖ Not allow foreign deep sea trawlers to operate in the Arabian Sea area under Pakistan control.
- ❖ Strict enforcement of laws against cutting of Mangroves forests
- ❖ Laws to control water pollution
- ❖ Destruction of mangroves should be controlled
- ❖ Stop catching fish during breeding season
- ❖ Overfishing should be stopped.

Fish Marketing

- ❖ The catch from marine areas is supplied to local fish markets through wholesale dealers.
- ❖ Street hawkers buy the catch and provide door to door service.
- ❖ Frozen or processed fish is available at large departmental stores in cities.
- ❖ Most of the fish catch is sold to other countries.

The per capita consumption is 1.6kg annum in Pakistan. In European countries per capita consumption is 20kg per annum. Pakistan exports fish to Japan, USA, UK and France. Pakistan exports shrimps in a large quantity to Japan.

Development of Fishing in Sindh Coast

- ❖ It has numerous creeks, sheltered harbours and a wide shallow sea beyond.
- ❖ The Indus delta is rich in fish food which is brought down to the delta by River Indus.
- ❖ It has advantages of a wider continental shelf than Makran Coast.
- ❖ It has better export and processing facilities. In Karachi, Government has developed Korangi fish harbor where storage and packing facilities are available for fishermen.



Study Fig. 2.1, which is a map of the southern coastline of Pakistan.

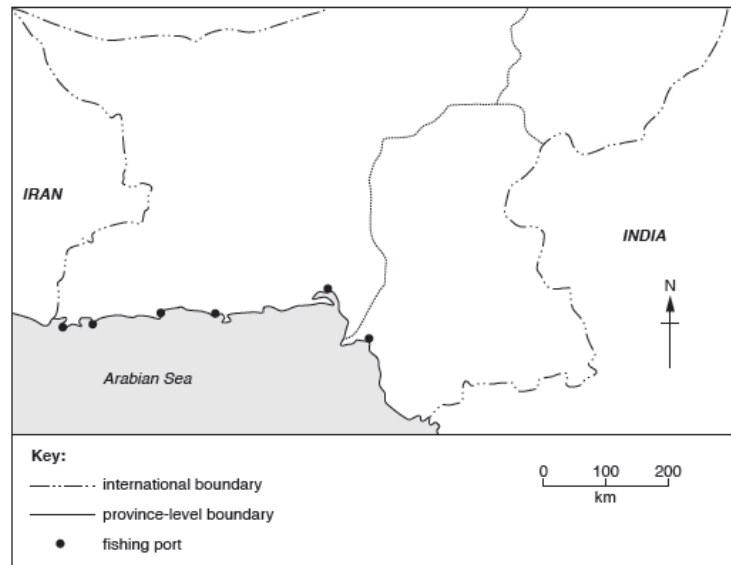


Fig. 2.1

Name two fishing ports on this coastline.

Answer:

- Jiwani;
- Gwadar;
- Pasni;
- Ormara;
- Sonmiani;
- Karachi / Keamari.

2 @ 1 mark

State two uses of fish

Answer:

- Fish waste for fertiliser;
- Money / income / export / selling;
- Fish waste for poultry feed / livestock fodder;
- Oil / oil extracted (to provide a source of nutrition – vitamins A & D) / medicines;
- Food supply / white meat / consumption / eating / healthy food.

2 @ 1 mark

Fishing is an important and growing economic activity in Pakistan. Read the following two views about ways to develop the fishing industry in Pakistan sustainably:



A

Developing inland fish farming has greater benefits for the people and economy and is more sustainable.

B

Developing marine fishing has greater benefits for the people and economy and is more sustainable.

Which view do you agree with more? Give reasons to support your answer and refer to examples you have studied. You should consider View A and View B in your answer.

Answer:

Levels marking

No valid response 0

Level 1

Simple point referring to any view (1)

Simple points referring to any view (2)

Level 2

Developed point referring to one view only (3)

Developed points referring to both views (4)

Level 3

Developed points referring to both views with evaluation or relevant example (5)

Developed points referring to both views with evaluation and relevant example (6)

Content Guide

Answers are likely to refer to:

Developing inland fish farming

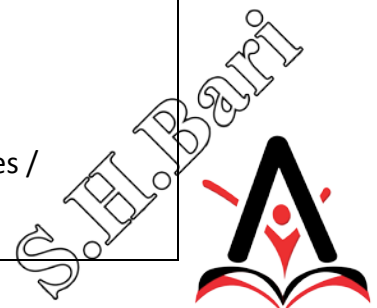
- Greater return to fishermen / fishing industry;
- Increased employment;
- Use of technology and upgrading of skills will ensure increased fish production;
- Favourable impact on foreign exchange earnings;
- Reduces burden on crop and livestock;

Etc.

Developing the marine fishing industry

- Improved facilities are being developed along the coast for preservation and storage;
- Almost all communities along the coast of Balochistan depend on fishing for their livelihood;
- Infrastructure facilities provided at Gwadar, Ormara and Pasni;
- Ice factories and cold storage developed at Gwadar;
- One fish harbour has been constructed at Pasni;
- Could set up fish processing for further employment opportunities / potential to catch a lot of fish;

Etc.



Chapter 7: AGRICULTURE

Inputs for Agriculture

Natural Factors (Physical)

Land – Soil – Rainfall - Temperature

Human Factors

Small Scale/Subsistence	Large Scale / Cash Crop Farming
Less Capital	Large Capital
Labour	Machinery (tractor, harvester, thresher)
Conventional irrigational methods	Modern irrigational methods
Natural manure	Chemical Fertilizers
Desi yielding variety of seeds	High yielding variety of seeds
Traditional inherited knowledge	Modern knowledge and skills

Processes

Ploughing, sowing/ transplanting, fertilizing, irrigating, weeding, threshing and harvesting.

Small Scale Subsistence Farming

It is practiced on small scale. A farmer works with his family. Traditional methods are used. The output is for self-consumption.

Large Scale / Cash Crop Farming

Cultivation is done on a large farm. Modern methods are used. It produces large outputs and profits.

Rabi Crop

Crops sown in Oct – Dec and harvested in May – June e.g., wheat, barley, oil, seeds.

Kharif Crop

Crops sown in May/ June and harvested in Oct- Nov e.g., cotton, sugarcane, rice.



Barani Lands

These are rain fed regions which only depend on rain. Output of these areas is low. e.g. Potwar plateau

Q: Discuss about difference of conventional and modern methods of irrigation ?

Wheat (Rabi Crops)

Geographical Requirements	Seeds	Areas	Cultivation:	Importance
Moderate rainfall – 1000 mm. A little rain before the harvest swells the grain and ensures a better yield.	Maxipak (HYV)	Punjab Multan, Gujrat, Sialkot, Faisalabad, Okara, Rahim Yar Khan, Bahawalpur, Jhang, Gujranwala Sindh Sanghar, Nawabshah, Thar, Khairpur, Sukkur, Larkana. KPK Kohat, Peshawar, Bunair, Mardan, Malakand, Bannu. It is not grown in Balochistan.	It is sows in Oct-Dec. Seeds are sown in the ground. First irrigation is done one month after sowing and second a month before harvesting. Chaff is separated from the grain.	Wheat is a staple food. Its demand is increasing with the increasing population. Low grade by- products of flour are used as feed for livestock. Wheat, when grown in surplus is exported to earn foreign exchange.
Temperature: 10°C- 20°C at the time of growing and warm 25°C-30°C for ripening				
Irrigation bridges the rainfall gap. It is also grown in Barani/Rainfed lands. (Potwar and KPK areas)				
Soil – Stiff loamy or clayey soil				
Flat land				

Rice (Kharif Crop)

Geographical Requirements	Seeds	Areas	Cultivation:	Importance
Rainfall Over 2000mm is ideal. Heavy shower. (Monsoon)	HYV: Irri Pak Basmati	Punjab Sialkot, Gujranwala, Sheikhupura, Kasur, Okara,	Seeds are sown in nurseries. When it is 9 inches high, it is transplanted into the prepared field. Then it is	It is used with wheat as a major food product. Rice is a major



20°C to 30°C with no cold season. A warm dry period for harvesting.	Super	Gujrat Sindh Along the river Indus It is not grown in KPK and Balochistan.	flooded with 30- 37 cm of water. Water is kept for growing. It is drained before harvesting. Threshing is done in Oct – Nov.	export of Pakistan. Rice husks are used for making cardboard or covering roofs of houses after mixing it with mud and water.
Irrigation bridges the rainfall gap mainly.				
Soil Loamy and clayey				
Flat land				

Cotton: King of Fiber (Kharif Crop)

Geographical Requirements	Seeds	Areas	Cultivation	Importance
Rainfall – 1000 mm. Irrigation usually bridges the gap. Rain at picking spoils the bolls.	Old varieties: Pak Upland, Desi HYV: Nayab 78, B-557, 149-F	Punjab: Bahawalpur, Rahimyar Khan, D.G. Khan, Leiah, Kasur, Faisalabad, Vehari, Multan Sindh Nawabshah, Sanghar, Hyderabad, Sukkur, Ghotki, Badin, Mirpurkhas. It is not growing in KPK Balochistan in Lasbela.	Seeds are sown at a distance of 30cm-45cm. One month later the land is irrigated. Second irrigation is done after further two months. The land is dried before harvesting in Oct-Nov. Plant has a height 130 – 135 cm.	It is used for making clothes and industrial fabrics. Backbone of textile industry. It is widely used textile fibre in Pakistan. It is used for exports. Leaf Curl virus is a major disease for cotton.
Temperature - 25°C to 35°C. Mild night temperatures are better for development of bolls. For harvesting sunny days and dry weather is best. It is sensitive to frost.				
Flat land				
Irrigation bridges the rainfall gap				
Soil: Medium loam				



Sugar-cane: (Kharif Crop)

Geographical Requirements	Seeds	Areas	Cultivation	Importance
Rainfall – 1520mm of rain	HYV: JN 0 88 Thatta – 10	Punjab Okara, Sialkot, Faisalabad, Sargodha, Gujrat, Khnewal, Vehari, Bahawalpur, Rahimyar Khan Sindh Khairpur, Naushero Firoz, Ghotki, Badin, Thar, Thatta KPK Charsada, Swabi, Peshawar	30cm stalks are planted in April-May. Distance is 30cm between the stalks, a good plant yield size is 6 to 7.3 feet. Crop can be rationed and harvested for 2 – 3 consecutive years. A ratoon is left when it is cut. It is cut by manual labour.	It is used to make sugar, brown sugar and gur. Molasses is used to make chemicals. It is used for citric acid, cattle feed, baker's yeast, synthetic rubber. Bagasse is a fibre to make paper, chipboard, animal feed. It is also used for fuel in sugar mills. Boilers are used to produce steam for generation of power to run machinery.
Temperature: 25°C - 35°C. Tolerant to frost for a short period.				
Irrigation bridges the rainfall gap				
Soil: Loamy, Clayey. Fertilizers containing nitrogen, phosphorous and potash are needed.				
Flat land				

Sugar cane is immediately transported as if it is delayed sugar content is reduced and starts losing weight. It is bulky and expensive to transport.

Maize, Pulses, Millets, Fruits, Oil seeds, Tobacco, Fruits and Vegetables: Consult book
Livestock Farming (Pastoral)

Livestock

- ❖ Cows
- ❖ Buffaloes – Nilibar, Kundi and Ravi are famous examples. Found in Punjab and Sindh.
- ❖ Sheep and goats
- ❖ Poultry
- ❖ Camels

Types of Livestock Farming**Nomadic Livestock Farming**

The farmer moves from one place to another with his animals in search of fodder, water and shelter.



Transhumance

Animals are kept on pastures high up in mountains in summer. They are brought down to the low land during winter season. When the season is feasible for animals, the farmer returns back to their highlands.

Settled Livestock Farming

It is mostly practiced in the villagers of Punjab and Sindh. Proper farms are made where fodder rooms are present. Milking, breeding takes place. Excess milk is converted to ghee, butter and cheese. Hens are also kept to get eggs.

Q: Why is settled livestock farming common in many areas of Punjab and Sindh? (4)

Q: Why is it difficult to rear buffaloes in Balochistan? (4)

Q: Why are many settled livestock farms are near to urban areas? (4)

Importance of Livestock Farming

- ❖ Cows and buffaloes are a source of milk and meat.
- ❖ Sheep and goats are reared for meat, wool, and skin.
- ❖ Hens provide meat and eggs.
- ❖ Bullocks are used for animal power, especially draught power in Persian wheel.
- ❖ Ox for the ploughing of fields.
- ❖ Bullocks to carry the bulk on the cart.
- ❖ Camels, mules are used for transport.
- ❖ The hides, skins and wool are a source of agro based industries.
- ❖ Cow dung or other animal waste is helpful for natural manure.
- ❖ Biogas is produced by cow dung.

Problem Faced by Livestock Farming

- ❖ Lack of grazing grounds led to over grazing
- ❖ Soil erosion
- ❖ Few veterinary hospitals and vaccination facilities
- ❖ Unhygienic conditions in animal husbandry lead to unhealthy animals
- ❖ Primitive breeding methods result in low quality animals
- ❖ Inadequate facilities for storage of meat
- ❖ Gap in price of livestock products in rural and urban areas
- ❖ High prices of animal feed. Difficult to keep in cities

How to Development

- ❖ Selective breeding and cross breeding for better quality animals
- ❖ Programs to fatten cows for milk and meat



- ❖ Control of disease and better animal husbandry practices
- ❖ Improvement in vaccination facilities
- ❖ Cultivation of fodder crops
- ❖ Improvement in livestock research

Factors Affecting Agriculture

Physicals / Natural	Human / Economic
Topography / Land	Irrigational facilities
Soil	Fertilizers
Water or Rainfall	Mechanization
Temperature	High yielding variety of seeds
Pests and Diseases	Marketing of Agriculture Products
	Size of Farms
	Pesticides
	Knowledge

Q: To what extent mechanization in agriculture is useful? (6)

Problems of Small Size of Farms

- ❖ Machinery cannot be used
- ❖ Small and fragmented holdings are difficult to supervise
- ❖ Loans cannot be obtained
- ❖ No research work can take place.
- ❖ More capital is not present. No machinery can be used.
- ❖ Output is small so profits are very low.

Government Measures to Increase Agricultural Output

- ❖ Develop plant protection program.
- ❖ Distribution of improved seeds.
- ❖ Provide financial support.
- ❖ One window operation to provide official service and advice.
- ❖ Loans for farm machinery on easy installments.
- ❖ Irrigational water to be provided.
- ❖ Increase the use of fertilizer.
- ❖ Awareness for modern methods and technology.



Land Reforms

These were made by the rulers in 1959, 1972, and 1977.

Aims

- ❖ Breaking the hold of land lords
- ❖ Increasing the output
- ❖ Equitable distribution of land
- ❖ Protection of rights of tenants
- ❖ Consolidation of holdings

It could not be properly achieved as the landlords did not show their actual holdings but the land which was taken under control of government was given to the tenants and the production increased.

Effects of Farming

- ❖ Use of sprays can contaminate subsoil water.
- ❖ Pesticides weakened the human immune system and can spread diseases.
- ❖ Chemical of fertilizers when is drained out of field can cause water pollution.
- ❖ Seepage of water from canals causes, water logging and salinity.
- ❖ Water from dams in canals can deposit silt in fields making them less reproductive.
- ❖ Deforestation for making farming areas.



Sustainable Development

- ❖ The problem of water logging and salinity must be addressed promptly to protect the cultivable land through various schemes.
- ❖ Government and Private institutions must develop new, highly productive and environmentally sustainable production technologies and systems.
- ❖ Soil management through afforestation projects is another measure to maintain the fertility of soil by improving its organic contents. Over cropping or multi cropping should not be allowed. To avoid soil erosion, forests should not be cut.
- ❖ Organic farming which stimulates bacteria in the soil act as natural nutrients for the fertilization of soil. It improves the fertility and productivity.
- ❖ Irrigational technologies must be used to reclaim the defects.
- ❖ Avoid poor farming methods

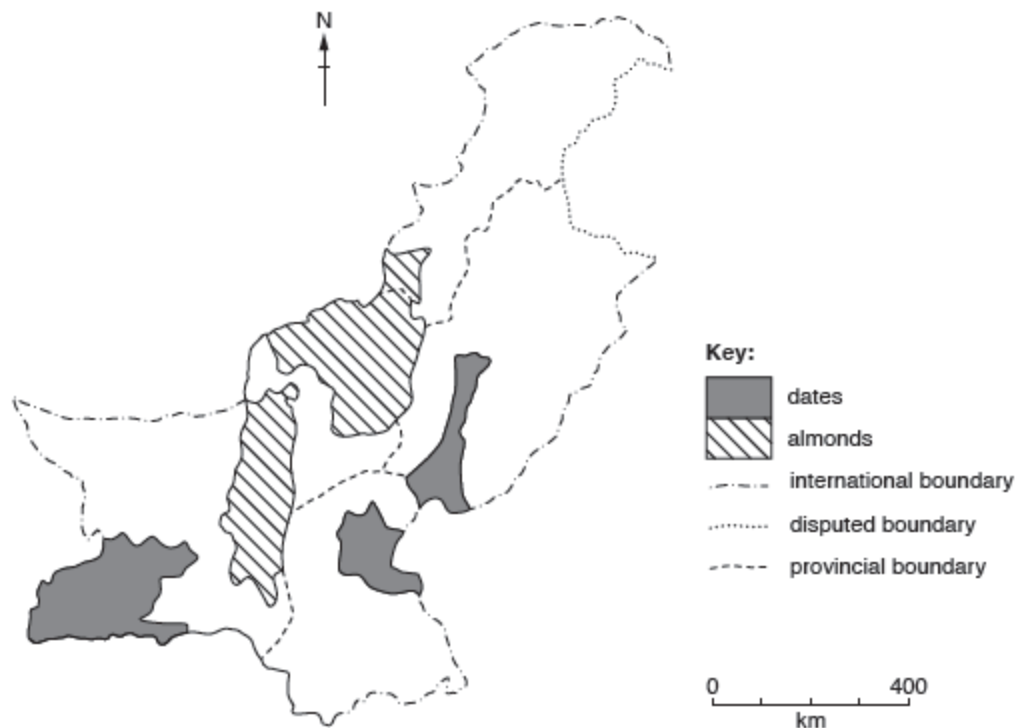


Fig. 6

Describe the distribution of the areas where almonds are grown.

Answer:

Study Fig. 6 which shows date and almond growing regions in Pakistan.

Describe the distribution of the areas where almonds are grown.

Central Balochistan / Khuzdar/Kalat/Mastung

N/NE Balochistan / Pishin/Zhob/Qila Saifullah/Loralai/Kohlu/Barkhan/Musa Khel

S/SW KPK/FATA / S Waziristan

Near boundary of Balochistan and KPK/Waziristan/FATA

Upper/lower = 0

Describe the Figures

Photograph A for Question 1



Photograph B for Question 1



Chemical fertilisers to help increase agricultural production are one of Pakistan's main imports. These imports are expensive. Read the following two views:

A

Pakistan should manufacture more of its own chemical fertilisers to reduce the need for importing them.

B

Pakistan should rely less on chemical fertilisers and reduce the need for importing them by using natural alternatives.

Which view do you agree with more? Give reasons to support your answer and explain why it is important to reduce imports of chemical fertilisers.

Answer:

L3	5–6 marks	6 – <i>Developed points addressing both views and the importance of reducing imports of chemical fertilisers. Evaluation gives clear support to one view</i> 5 – <i>Developed points addressing both views and the importance of reducing imports of chemical fertilisers. No evaluation</i>
L2	3–4 marks	4 – <i>Two developed point(s) addressing any view</i> 3 – <i>Developed point addressing any view</i>
L1	1–2 marks	2 – <i>Two simple point(s) addressing any view</i> 1 – <i>Simple point addressing one view</i> 0 – <i>No valid response</i>

Read the following two views:

A

Pakistan should plant more cash crops on its land to generate export earnings.

B

More land should be used to grow crops to feed the growing population of Pakistan.

Which view do you agree with more? Give reasons to support your answer and refer to examples you have studied.

Answer:

L3	5–6 marks	6 – <i>Developed points explaining both views. Evaluation gives clear support to one view. At least one reference to an appropriate example</i> 5 – <i>Developed points explaining both views. Evaluation gives clear support to one view</i>
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L2	3–4 marks	4 – <i>Developed point(s) explaining both views. No evaluation</i> 3 – <i>Developed point(s) explaining one view</i>
L1	1–2 marks	2 – <i>Simple point(s) addressing both views</i> 1 – <i>Simple point(s) addressing one view</i> 0 – <i>No valid response</i>

Photograph A for Question 1



Read the following two views about increasing food supply in Pakistan:

A

Development of the livestock sector is the best way to increase the amount of food available in Pakistan.

B

Increasing crop production and cultivable area is the best way of ensuring there is enough food in Pakistan.

Which view do you agree with more? Give reasons to support your answer and refer to examples you have studied. You should consider **both** View A and View B in your answer.

Answer:**Levels marking**Level 1

(1)

Simple point addressing any view (1).

Simple points addressing any view (2).

Level 2

(3)

Developed point(s) explaining one view (3).

Developed point(s) explaining both views (4).

No evaluation.

Level 3

(5)

Developed points explaining both views. Evaluation giving clear support to one view or a named example (5).

Developed points explaining both views. Evaluation giving clear support to one view and a named example (6).

Content Guide:

Answers are likely to refer to:

For livestock

- Large multi-national farms
- Bigger source of protein
- Source of milk/ghee/meat
- Sheep/goats can survive on marginal land

Against livestock

- Insufficient land for fodder crops
- Inadequate storage facilities
- Lack of grazing land
- Overgrazing
- Lack of funds
- Unhygienic husbandry

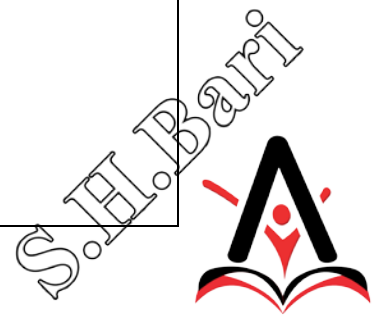
For food crops

- More land can grow food for people
- Well-developed irrigation
- Multi-cropping
- Access to fertilisers/pesticides, etc.

Against food crops

- Mismanagement
- Overuse
- Of water/seepage from canals
- Over-cultivation

ETC.



Describe the Figures

Photograph B for Question 3



Fig. 3.1 for Question 3



Describe the Figures**Fig. 3.2 for Question 3**

Study Fig. 1.2 (Insert). Describe the main features of the desert area shown in the photograph.

Answer:

- Sand / sandy;
- Sand dunes / ridges / hills / hilly;
- Large area / expanse / plain / plains;
- Sparse / scant vegetation / not much greenery / few trees / lack of trees;
- Small bushes / thorny bushes / scrub / rakh / shrubs;
- Barren / bare / dry;
- Oasis.

3 @ 1 mark

Study Fig. 1.3 (Insert). State two features of the climate typical of the environment shown in the photograph.

Answer:

- Cold / cool / low temperatures / freezing temperatures;
- Relief rainfall;
- Snow (capped peaks) / blizzards;
- Ice / hail;
- Windy;
- Dry;
- Sunny / bright / clear skies OR few sunny days / cloudy.

2 @ 1 mark

Describe the Figures

Fig. 1.1 for Question 1



Fig. 1.3 for Question 1



Describe the Figures**Fig. 1.4 for Question 1**

Study Figs. 1.1, 1.2, 1.3 and 1.4 (Insert), photographs showing different crops growing in Pakistan.

Identify the crops shown in each photograph

Fig. 1.1

Fig. 1.3

Fig. 1.4

Answer:

Fig. 1.1 = Rice

Fig. 1.3 = Cotton

Fig. 1.4 = Wheat

4 @ 1 mark

Chapter 8: POWER RESOURCES

Renewable Resources: which can be utilized again e.g., water, wind, sunlight.

Non-Renewable Resources: which cannot be used again e.g., coal, oil, gas, uranium.

Coal

Types of Coal

- ❖ Anthracite – Found very deep, best quality coal with most carbon content.
- ❖ Bituminous – hard coal which can burn easily.
- ❖ Lignite – Lower quality and with high moisture and ash content. It has low heating value.
- ❖ Peat – Least quality coal having low carbon contents.

Coal is transported outside mine on the back of miners, donkey backs or trolleys, it is then loaded on trucks, trolleys and railway carriages to be taken to the industries. It is a preferred source of energy as it is cheap and Pakistan is rich in its reserves.

Uses of Coal

- ❖ Thermal power generation
- ❖ Heating purposes
- ❖ Brick kilns
- ❖ Making cement
- ❖ Steam engines

Areas

- ❖ Balochistan: Three coal mines at Shahrig Sor-range and Degas.
- ❖ Sindh: Thar coal fields.
- ❖ Punjab: Salt Range and Kalabagh areas. Bituminous and lignite coal is found.

Mineral Oil

It is also known as 'black gold'. It is trapped in dome shaped anticlines between two layers of non-porous rocks gas is present above and water below. Once the drilling site is selected, a derrick or drilling rig is setup. The derrick is a steel structure that holds the drilling pipes and other equipment. The oil is pumped up and flown by the pipelines to the refinery.

Uses of Oil

- ❖ It is used as a lubricant for machines
- ❖ By products as paraffin, wax, plastics, synthetic rubber, detergents, insecticides, pharmaceutical products, chemical products.



- ❖ Furnace oil.
- ❖ Diesel, petrol, kerosene oil for vehicles and burning.
- ❖ Generating electricity

Transportation of Oil

It is transported by

- ❖ Roads
- ❖ Railways
- ❖ Pipelines

Pier

It is an oil handling system. Oil is transport in oil ship tankers from different Middle East countries to Karachi ports.

Refineries

- ❖ Pak Arab Refinery Corporation Limited (PARCO)
- ❖ Attock Refinery
- ❖ Pakistan Refinery
- ❖ National Refinery

Oil Pipelines (prepare maps)

- ❖ Mid-country Mehmud Kot oil pipeline
- ❖ WOPP (White Oil Pipeline Project)

Natural Gas

It is an important fossil fuel made up of methane, ethane, propane and butane. It is pumped and transported mostly by pipelines. Natural gas was discovered firstly at Sui in 1952. Oil and Gas Development Corporation established in 1961 is working for extraction and exploration. It is a cheap fuel.

It is mostly transported by the pipelines.

- ❖ SNGPL – Sui Northern Gas Pipeline Limited
- ❖ SSGPL – Sui Southern Gas Pipeline Limited.

Uses of Gas

- ❖ Cooking
- ❖ Heating
- ❖ Cheap fuel used in fertilizer and cement industry



- ❖ To make electricity
- ❖ Used as fuel in car CNG(compressed natural gas).

Thermal Energy

Electricity which is produced by the use of fossil fuels as oil, gas and coal is thermal energy. The construction of thermal power projects is low but its running cost is very high producing very expensive electricity.

Nuclear Energy

The nuclear power stations are based on atomic fission. In atomic fission, energy is released when atoms split into their parts. They use heat energy to make steam which turns the turbines to make electricity. Uranium is used for the purpose which is found in a large quantity in Pakistan.

KANUPP (Karachi Nuclear Power Project) in 1971 started working, capacity is 137 MW. CHASNUPP (Chasma Nuclear Power Project) in 1999 at Chashma with the help of China

Advantages	Disadvantages
Nuclear energy can solve energy problems as other sources are becoming rare	The reactors produce rays which are dangerous as they result in cancer
Mutli-purpose project as provide fuel for nuclear weapons.	There are problems of reprocessing and storing nuclear waste
Running cost is cheap.	Nuclear power plant is very expensive to be built
Environment friendly as no air pollution	Output is low.

Renewable Energy Resources

Hydro Electric Power (HEP)

These stations use the force of flowing water to move turbines. A shaft is going into the generator. Water helps to move the turbine and shaft spins rapidly inside a magnetic field in generator which generates current it is then regulated in the transformer which is then sent through transmission lines at required voltage. Major HEP stations are Tarbela, Warsak and Mangla.

Suitable place to build DAM

- ❖ Mountainous area as high altitude available to make water falls to move turbines
- ❖ Steep slopes and Natural basins help to make reservoirs easily



- ❖ Availability of water by rainfall and melting of snow
- ❖ Low temperature so less evaporation during summers
- ❖ Low evacuation cost because of less population density

Importance

- ❖ Produce HEP (Hydro Electric Power)
- ❖ Storage of water in reservoirs for domestic and industrial usage
- ❖ Store water for irrigation purpose through canals
- ❖ Controls flood and save from its devastations
- ❖ Attract tourists
- ❖ Fishing in reservoirs

WAPDA - Water and Power Development Authority

KESC – Karachi Electricity Supply Corporation

IPP – Independent Private Producers are helping in power generation.

Advantages	Disadvantages
Environmental friendly.no air pollution.	The initial/installation cost of HEP is high.
Water is used in it which is a renewable resources.	Maintenance cost is high.
Running cost is very low.	Evacuation is required in nearby areas.
Cheap electricity	Deforestation
Mega projects producing thousands of MW.	Burden on weak economy.

Solar Power

It is produced by collecting sunrays in photovoltaic cells. Solar panels are used.

Advantages	Disadvantages
Environmental friendly. No air pollution.	The initial/installation cost of is high.
It is a renewable resource.	Maintenance cost is high.
Running cost is very low.	Not suitable for cloudy areas.
Cheap electricity	Burden on weak economy.
Safe and limitless supply as 250-300 sunny days.	



Wind Power

It is the process by which the wind is used to generate electricity. A wind turbine has a generator which converts the mechanical power into electricity. Large scale wind farms consists of many turbines

Advantages	Disadvantages
Environmental friendly. May attract small scale industries in rural areas providing job opportunities.	The initial/installation cost is high. 7500 turbines are needed to produce the same amount of electricity as one nuclear power station.
It is a renewable resources.	Maintenance cost is high.
Running cost is very low.	It will disturb the Eco system.
Cheap electricity	Wind Turbines can automatically stop working when there is a storm.
Wind power can help reduce energy problem in Pakistan	Burden on weak economy.
Wind farms are an attraction for tourism.	

Geothermal Power

It is the energy derived from the heat of Earth. The holes are drilled into the land and steam is used to derive the turbines to produce electricity. Pakistan has a potential of commercially exploitable sources of geothermal energy in Himalaya region.

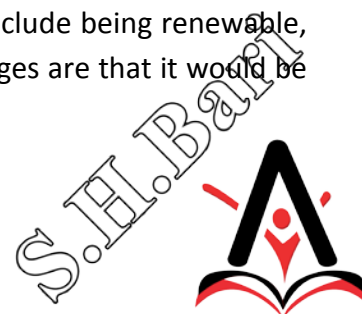
It has advantages as it is renewable, provides constant supply and is pollution free. It has some disadvantages as high cost of construction of power station. It has the threat from eruptions and earthquakes.

Wave Power

Wave motion can be used to compress air to drive a turbine to generate electricity. Makran coastal area has strong wave energy which could help generate electricity. The coastal cities as Gwadar, Pasni, Ormara, Gidani etc are being developed. It is expensive and produces limited supply.

Tidal Power

It is a renewable method of producing electricity by movement of tides. Tides are generated due to gravitational pull of the moon. The advantages of Tidal power include being renewable, providing a constant supply and environmental friendly. The disadvantages are that it would be very expensive to build.



Biomass

It refers to organic material that can be used as fuel to generate electricity.

Biogas

It is the cheapest source of energy. It will increase air pollution because methane is a greenhouse gas. If cow dung is used in Biogas, we will be short of natural manure in fields.

Load Shedding

When the demand is more than the supply of power, load shedding occurs. It results in great economic damages.

Causes

- ❖ Many power plants are not working to their full capacity.
- ❖ During winter the HEP decreases due to decrease in flow of water in rivers.
- ❖ Long transmission lines cause loss of electricity.
- ❖ Siltation in reservoirs results in low storage capacity of water in dams.
- ❖ Power theft on a large scale.
- ❖ Waste of electricity due to line losses.
- ❖ High price of fossil fuels increases cost of electricity generation.

National Grid System

It is an electricity distribution system which distributes the electricity to the whole country.

Rural Electrification

To provide electricity to rural areas is rural electrification. It can help

- ❖ Economic advantages as tube wells can be used.
- ❖ Small scale industries can be developed.
- ❖ More employment opportunities will be provided.
- ❖ Rural urban migration will be controlled.
- ❖ The standard of living will increase.
- ❖ Awareness among the people by media.

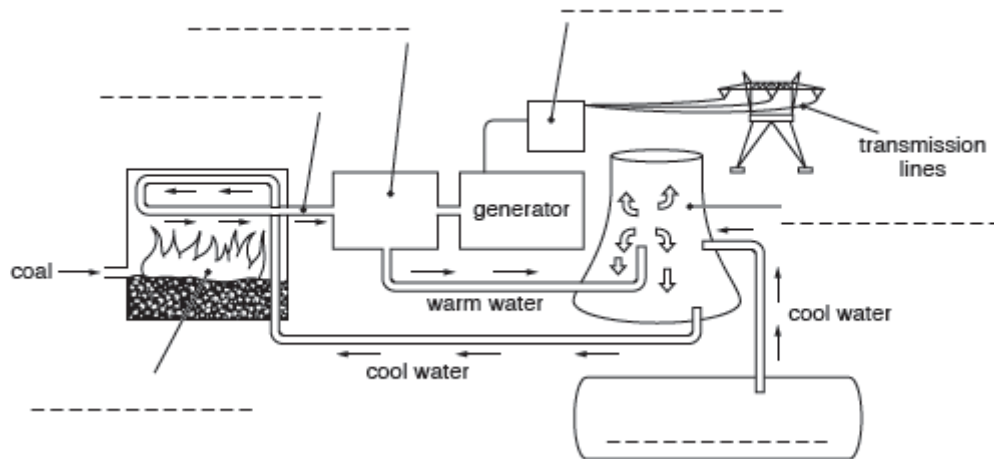
Sustainable Development of Power Resources

- ❖ Use of advanced technology can help to develop renewable resources.
- ❖ Public awareness programs.
- ❖ Avoid non productive uses.
- ❖ Replacement of faulty and damaged transmission lines as well as outdated machinery.



- ❖ Strict laws for power theft.
- ❖ Preservation and conservation of non renewable resources.

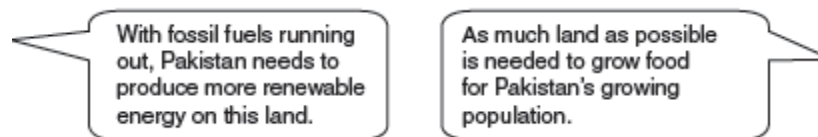
(c) (i) Fig. 5 is a diagram of a thermal power station.



Choose three terms from the list below and use them to label the diagram in three of the spaces provided.

Reservoir transformer boiler turbine cooling tower steam [3]

Read the following two views about this proposal:



Study Fig. 7 which is a diagram of an HEP (Hydel) power station.

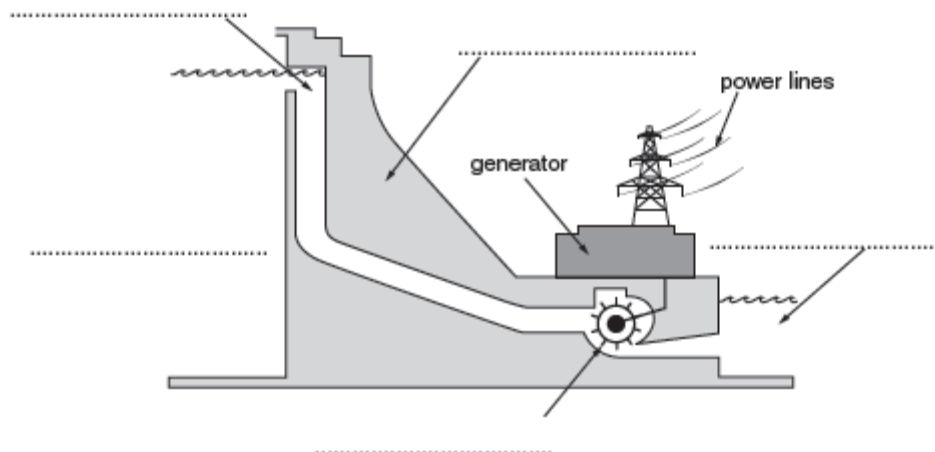


Fig. 7

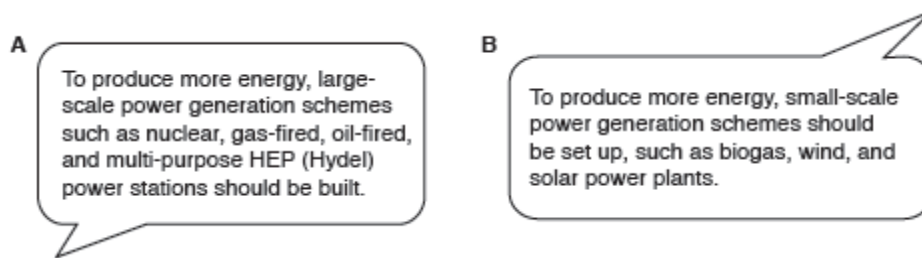
A. On the diagram place an arrow or arrows to indicate the direction of movement of water through the power station.

B. Choose two terms from the list below and use them to label the diagram in two of the spaces provided.

Outflow turbine reservoir dam

[3]

Read the following two views about solutions to Pakistan's shortage of energy:



Which view do you agree with more? Give reasons to support your answer and refer to examples you have studied. You should consider both View A and View B in your answer.

Answer:

Levels marking

Level 1

(1)

Simple point addressing any view (1).

Simple points addressing any view (2).

Level 2

(3)

Developed point(s) explaining one view (3).

Developed point(s) explaining both views (4).

No evaluation.

Level 3

(5)

Developed points explaining both views. Evaluation giving clear support to one view or a named example (5).

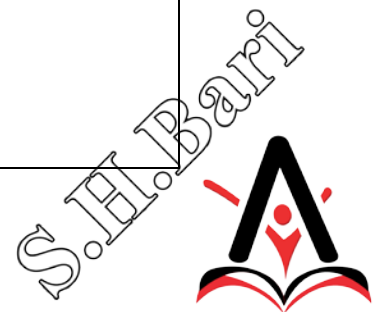
Developed points explaining both views. Evaluation giving clear support to one view and a named example (6).

Content Guide:

Answers are likely to refer to:

For large-scale

- Funding available from China



- Provide very large amounts of power from small amount of uranium
- Large coal reserves

Against large-scale

- Large sums of money/loans needed
- Problems with disposing of/reprocessing/storing waste
- Danger of insurgency threat/accident
- Danger of flooding
- Coal extracted in Pakistan is poor quality for power generation
- Oil expensive to import
- Large coal reserves not exploited
- Fossil fuel reserves are depleting
- Political issues between provinces with the construction of multipurpose dams over division of water

For small-scale

- Lower cost to maintain
- Renewable resources do not deplete
- Renewable resources do not pollute the environment
- Biogas – cheap source of energy
- Wind – available land in Balochistan highlands
- Solar – many parts of Pakistan experience 250–300 sunny days per year

Against small-scale

- Do not contribute/only small amount to national grid
- Renewables only generate small amount of energy
- Wind/solar have high construction cost
- Wind is considered unsightly/harms wildlife
- Not constantly producing energy
- Biogas decreases availability of manure for organic fertiliser

ETC.



Fig. 1.2 for Question 1



Study Fig. 3.1, a map showing gas producing regions and the natural gas pipeline network in Pakistan.

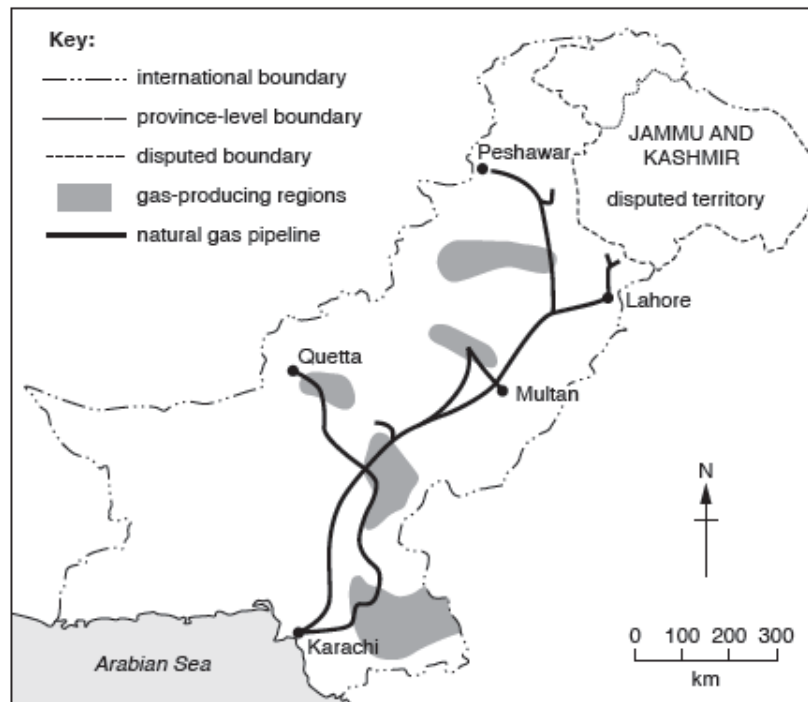


Fig. 3.1

Name three gas producing regions in Pakistan.

- 1
- 2
- 3

Answer:

- Sui / Eastern Balochistan / Pirkoh / Uch / Zin / Loti;
- Lower Sindh / Tharparker / Thora / Tando Adam / Nazari / South Mzari Deep / Pasakhi;
- Northern Sindh / Mari;
- Northern gas region / Potwar Plateau / Meyal / Dhurnal / Adhi / Pindori /
- Balkassar / Fimkassar.

3 @ 1 mark



Chapter 9: SECONDARY & TERTIARY INDUSTRIES

Factors / Inputs Required for establishing an Industry

- ❖ **Capital:** Money and machinery.
- ❖ **Land:** Flat land is required.
- ❖ **Raw Material:** It is the basic commodity to manufacture a product.
- ❖ **Power supply** is required.
- ❖ **Skilled Labour** is required to work on machines. (**unskilled labour** as well)
- ❖ **Enterprise:** The management having skills.
- ❖ **Water supply:** Basic need of every industry.
- ❖ **Marketing:** To advertise a product.
- ❖ **Transport and Communication:** Means of transport is through land (road, rail) air and sea. Communication devices are required to market.

Classification of Industries

Primary Industry It is related with direct extracted material e.g. agriculture, fishing, mining and quarrying.

Secondary Industry (Manufacturing)

- ❖ Small scale industry
- ❖ Medium scale industry
- ❖ Large scale industry

Tertiary Industry These provide services e.g. banking, insurance, transport, hotels, teaching and medicine.

Cotton Textile Industry

Types	Areas	Importance	Problems
Ginning Spinning Weaving Ready-made	Karachi Faisalabad Hyderabad Peshawar	It provides job opportunities to a large number of people. It contributes approximately 7% to GDP.	Shortage of raw material due to leaf curl virus Recession in international market. Competition on international level from Egypt, Hong Kong, Thailand, Taiwan etc. Lack of capital Lack of modern machinery Power failure Alternate supply increase the cost



Sugar Industry

Sugar cane starts losing its sugar content as soon as it is harvested. It needs to be crushed immediately. It is also bulky so it is expensive to transport. In mill sugarcane is crushed to get the juice which is then converted into sugar.

Areas

The sugar mills are mostly found near the sugar growing areas as:

Sarghoda, Faisalabad, Pattoki, Samundri, Kamalia, Rahimyar Khan and Bahawalpur in Punjab, Nawabshah, Badin, Thatta, Mirpurkhas, Dadu in Sindh. Peshawar, Mardan, Charsadda in KPK.

By-products of Sugarcane

Molasses is used to make chemicals. It is used for citric acid, cattle feed, baker's yeast, synthetic rubber.

Bagasse is a fibre to make paper, chipboard, animal feed. It is also used for fuel in sugar mills. Boilers are used to produce steam for generation of power to run machinery.

Fertilizer Industry**Raw Material**

Raw materials used are phosphorous, sulphur, potash, phosphate, gypsum, and nitrogenous compounds make up the fertilizer. Cheap fuel gas is used.

We need to have more fertilizers from the time of green revolution. As the population is increasing, need for crop increase. The soil is deficient in nutrients, Therefore, it required the chemicals in the fertilizer. If we prepare such fertilizers in Pakistan to meet the needs for agriculture we can save precious foreign exchange for paying import bills.



Areas

The areas where fertilizer factories are found are Dandkhel and Haripur in NWFP, Faisalabad, Jaranwala, Sheikhpura, Multan in Punjab. Machighot, Sukkur and Dadu in Sindh.

Cement Industry**Raw Material**

Cement needs raw material as gypsum and limestone. It is easier to make cement as we have the raw materials available in a large quantity. Cheap fuel gas is used.

Areas

Karachi, Thatta, Hyderabad, Khairpur, Rohri in Sindh. Nowshera, Kohat in NWFP. Jauharabad, Daud Khel, Dandot, Attock and Rawalpindi in Punjab.

The demand of cement has increased because:

- ❖ Population increase resulted in more construction.
- ❖ More industries established.
- ❖ New roads, bridges and underpasses are made.
- ❖ Lining of canals

Steel Industry

Pakistan steel mill was established in 1973 at Pipri in Karachi on Ghoro Creek near Port Bin Qasim. It was financed by USSR. It provides raw material to engineering and construction industries. The products of steel mills are coke, pig iron / hot metal, rolled and cast billets, galvanized products and raw steel.(see exercise of chap 9 for pictures) Heavy mechanical complex Ltd. (HMC) was established at Taxila in 1979 with Chinese assistance. The Heavy Forge factory (HFF) is also established at Taxila. The HMC is processing equipment of HEP, thermal power plant, petro chemical plants, boilers, cranes, construction machinery, material handling equipment, steel structures and railway equipment.

Why is Steel mill located at Pipri in Karachi?

- ❖ Port Qasim has a natural harbor that facilitates imports and exports
- ❖ Cheap flat land was available in Karachi.
- ❖ Iron ore, manganese etc can be imported easily.
- ❖ Limestone and clay are brought from Thatta
- ❖ Water is brought from Haleji Lake near Thatta
- ❖ Thermal and nuclear power station is present to produce electricity.
- ❖ Cheap labour is available due to more population.
- ❖ Transport, by road rail and sea is also available.



Processes involved in Industries

Industry	Process
Textile	Ginning, spinning, weaving, dying, printing
Woolen	Weaving, knitting, stitching
Leather	Tanning, stitching, printing
Iron/ Steel	Heating, molding, packing
Cement	Grinding, mixing, processing

Industrial Estates

These are the areas reserved for industries only.

Special Industrial Zones

These are present where infrastructure is not provided. Government provides them with roads, gas supply, electricity telephones etc. The investors will be allowed to develop their residential, commercial and social facilities in the vicinity of industrial zone. Government then offers incentives as tax holiday, foreign exchange control and relaxation on import of machinery. Stability of govt. is required for local and foreign investor's attraction.

Industrialization	Nationalization	Denationalization	Privatization
When in 1947, private sector was reluctant to invest in capital intensive industries. Pakistan Industrial Development Corporation (PIDC) was established to do so. It helped in establishing large scale industries, then later they transferred many projects to the private sector during 1960s. The development was quite encouraging. It was the Era of industrialization.	In 1972, 10 basic industries were transferred to public sector which gave heavy blow to private sector. The industrialization's pace become slow.	In 1977, government reversed the policy. The martial law govt. gave assurance that no further industry will be nationalized, but the previously nationalized were given back to owners, those who could not buy it, were auctioned.	State Owned Enterprises (SOE) has been promoted in 1991. The objectives were to create better opportunities for private sector for expansion and to modernize to improve productivity, and profitability to facilitate economic activities for the private sector, to achieve more rapid industrialization and to reduce the burden on government economic resources. So far 103 (SOE) have been privatized.



Difference between Formal and Informal sector

Formal Sector	Informal Sector
It is employed by an institution. It is capital intensive with few workers generally mechanized. It has regular working hours and certain fixed wages. Work is located in offices or factories. It is legal or registered. Normally male workers. Quality of good is high.	It is self employed, labor intensive, using mainly hand tools, very few modern machines are used, irregular working hours and uncertain wages, often low standard in quality of goods, work is done at home, often women and children work in them.

Cottage and Craft / Small Scale Industries

Advantages

- ❖ They provide employment opportunities to the people.
- ❖ Industrial waste can be used.
- ❖ The demands of local market can be fulfilled.
- ❖ Due to exports, foreign exchange is earned.
- ❖ Women of rural areas can be employed.
- ❖ It reduces the rural urban irrigation.
- ❖ It reduces regional disparity.
- ❖ Local raw material can be used.
- ❖ Less capital and less sophisticated technology involved.
- ❖ Standard of living improves especially in rural areas.

Problems

- ❖ Economies of scale are not available so production cost is higher.
- ❖ They cannot compete in the open market if the same goods are manufactured on a large scale.
- ❖ Profits are limited as they are spent on daily expenses of owners.
- ❖ They do not have extra capital for the expansion of industrial units.
- ❖ There is lack of standardization and quality control.
- ❖ Production methods and machinery are outdated.
- ❖ Old, traditional methods are used by the craftsmen in which training is limited.
- ❖ Due to shortage of electricity, the production is reduced. Commercial rate of electricity have decreased the growth.



Q: Suggest some solutions of the problems faced by cottage industries? (5)**Famous small scale industries are:**

Carpets, Textiles, Embroidery, Jewelers, Ceramics, Wood work, Metal work, Cutlery industry, Fan industry.

Sports Goods

It is located in Sialkot. It is mostly exported. Raw material used is leather, rubber, stitching of thread, the sewing is done mainly in small workshops manually and final stage of production is done in factories. However, issue of child labor in stitching footballs had attracted worldwide attention. In most cases embargo was laid on the Pakistan's sport goods exports, but the sports goods meet the international standards.

Surgical Instruments

These are manufactured in Sialkot. The industry has transformed from a small level to a large scale 95% of surgical instruments made in Sialkot are exported. Pakistan have a meager share of 1% of world exports of medical and dental instruments. It has been reported that due to lack of education and increased poverty, children are forced to work in such industries.

Brick Kiln Industry

It is made with raw clay preferably mix with 25 to 30% of sand to reduce the shrinkage. The clay is first grind and mixed with water to the desired consistency. The clay is then processed into steel moulds. The shaped brick is then fired (baked) at 900 - 1000°C.

Brick kilns are found near Sheikhpura, Kasur, Batapur area of Lahore, Gujranwala and Islamabad as well. Child labour is a major issue in this industry.

Effects of Industrial Pollution

- ❖ Dumping of industrial waste causes land pollution.
- ❖ Sea port pollution due to nearby industries.
- ❖ Industrial waste dumped in water resources is harmful for the irrigation of crops including food crops.
- ❖ The turning of machinery causes noise pollution resulting in deafness and irritability.
- ❖ Serious health hazards in the form of various diseases.
- ❖ Threat to marine life – contaminated seafood.
- ❖ Threat to mangroves, reduces the fish production.
- ❖ Contamination of sub soil water affects food crops and suppliers of drinking water from wells.



How to Control?

- ❖ Government should facilitate the import of machinery for treating industrial waste.
- ❖ Afforestation programs must be started to grow trees to reduce air pollution.
- ❖ Industries must be away from residential areas.
- ❖ Strict laws should be made and implemented to check industrial pollution.
- ❖ Companies should not be allowed to dump their industrial waste into Arabian Sea.
- ❖ It is necessary to create awareness among people about all types of pollution and its damages.

Tourism (Tertiary Industry)

It is related to the services. Tourism means the whole business of providing accommodation and recreation facilities for those people who are travelling and visiting or staying in a place for a relatively limited period of time. It is mostly for pleasure.

Factors hindering Development of Tourism in Pakistan

- ❖ Lack of capital
- ❖ Security issues (terrorism)
- ❖ Lack of political stability to implement long term plans.
- ❖ Lack of infrastructure such as roads
- ❖ Lack of standard accommodation
- ❖ Lack of quality and safe food
- ❖ Lack of safe drinking water
- ❖ Poor Management of tourist attractions.

Advantages of Developing Tourism

- ❖ Foreign exchange earnings
- ❖ It creates employment e.g., in hotels, entertainment and as guides.
- ❖ It increases the sale of local cottage industries.
- ❖ Locals can also use tourist facilities.
- ❖ Profit earned by tourism can be used to improve local housing schools, hospitals, electricity and water suppliers.
- ❖ It develops a good and soft image of Pakistan in international world.
- ❖ It is a sustainable industry. The tourist attractions will continue to exist.

Disadvantages of Tourism

- ❖ It provides seasonal employment.
- ❖ The building of hotels means deforestation which leads to
 - land sliding
 - soil erosion
 - siltation
- ❖ Domestic tourists may not be able to afford tourist facilities.



- ❖ During tourist season, the prices of basic necessities rise beyond the affordability of local people.
- ❖ The natural habitat may be affected because of deforestation.
- ❖ Tourism often destroys local culture and traditions. New social problems of crime,
- ❖ Less Marketing and publicity at international and local level.

Types of Tourists coming to Pakistan

- ❖ Visitors on business come to Pakistan to invest or to work.
- ❖ People who live abroad come to visit their families.
- ❖ Local tourists who want to enjoy the sceneries and climate.
- ❖ International tourists who only come to gain pleasure and knowledge.
- ❖ Foreign delegates and diplomats visit in Pakistan.
- ❖ Staff of multinational companies such as foreign banks and organizations as Banks etc.

Five Star Hotels of Pakistan Are: Marriot, Pearl Continental, Avari etc.

PTDC – Pakistan Tourism Development Corporation also has motels or rest houses in different mountains areas and historic cultural places as Kaghan, Ayubia, Ziarat, Chitral, Gilgit, Hunza, Taxila, Moenjodaro etc.



Great Potential to develop Tourism in Pakistan?

Natural Attractions	
Kaghan Valley	Fishing is chief sport. Brown trout and Mahasheer are stocked in pure silvery waters. They Kunhar river trout is famous. Fish license is issued by fisheries department at Naran or by Trout Hatchery at Shinu. Some go to Saif-o-Maluk lake which is a lovely spot.
Swat Valley	Beautiful lakes, orchards, flowery slopes are found.
Gilgit Valley	Favourite sport here is polo, a beautiful valley with mountains and greenery.
Skardu	It is the capital of Baltistan. It is the starting point for expedition to k-2.
Hunza Valley	It is a beautiful rugged valley mostly Ismailies (Agha Khans) live there, local language is Burukshahi.
Chitral	Famous Kalash tribe lives there, Shandur Polo festival is famous which is held in July every year.
Cultural Attractions	
Archaeological areas as Harappa Taxila (Indus civilization, Gandhara civilization) in Punjab.	
Moenjodaro (Indus civilization) in Sindh	
Historical Areas	
Forts as Baltit, Khyber Pass, Khunjerab.	
Mosques (Badshahi, Mahabat Khan, Faisal.	
Tombs of Shah Jehan, Nurjehan, Jehangir	
Shalimar garden – Lahore Fort	
Allama Iqbal's Tomb, QUAID-E-AZAM'S Tomb.	
Salt mine in Khewra.	



Sustainable Industry

- ❖ It is to provide raw material to agro-based industries.
- ❖ Energy crisis destructs the industry but if renewable or non-renewable resources help to increase energy.
- ❖ Skill enhancing working conditions for the development of its human resources. It is possible through in service training programs for industrial labor force.
- ❖ Safe working conditions, economically security and a professional environment for the industrial labor force in order to maximize production.
- ❖ Efficient use of technology in industrial processes to reduce or utilize the waste for generation of electricity or for other purposes.
- ❖ Conservation of environment by applying recycling techniques.
- ❖ Safe disposal of toxic Industrial waste that may pose health threats.
- ❖ Sustainable development of tertiary sector such as transport, banking, insurance and services to serve for financial and administrative needs for Industrial sector.
- ❖ Political stability on a sustainable basis to ensure a favorable investment climate for the growth of industry.
- ❖ Exploration of new market opportunities and maintenance of the existing ones through constant market research and quality control techniques.

Study Figs 10 and 11 which give information about visitors into Pakistan in 1999.

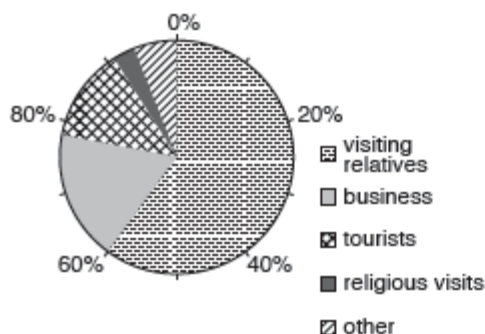


Fig. 10

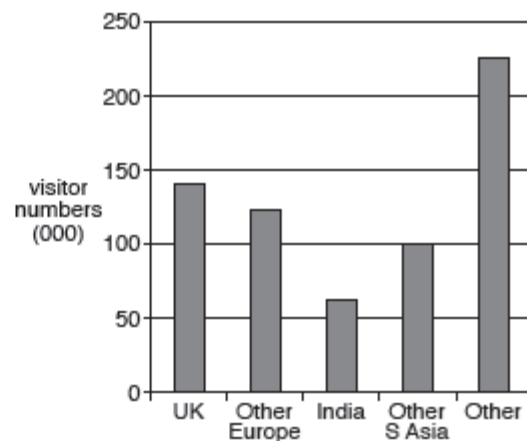


Fig. 11

A. What percentage of visitors into Pakistan was classed as tourists?

B. What was the total number of visitors into Pakistan?

C. Suggest **one** reason why more people visited relatives compared to visiting Pakistan as tourists.

Answer:

Study Figs 10 and 11 which give information about visitors into Pakistan in 1999.

- A** What percentage of visitors into Pakistan was classed as tourists?
13% (accept 12–14%)
- B** What was the total number of visitors into Pakistan?
655 000 (accept 640 000–670 000)
- C** Suggest one reason why more people visited relatives compared to visiting Pakistan as tourists.

[3]

Lower cost
Large/extended families spread out in different countries
Work abroad
Few tourist attractions
Terrorism/lack of security/political instability
Poor transport infrastructure

(a) Study Fig. 9, which is a map giving information about cotton textile industries in Pakistan.

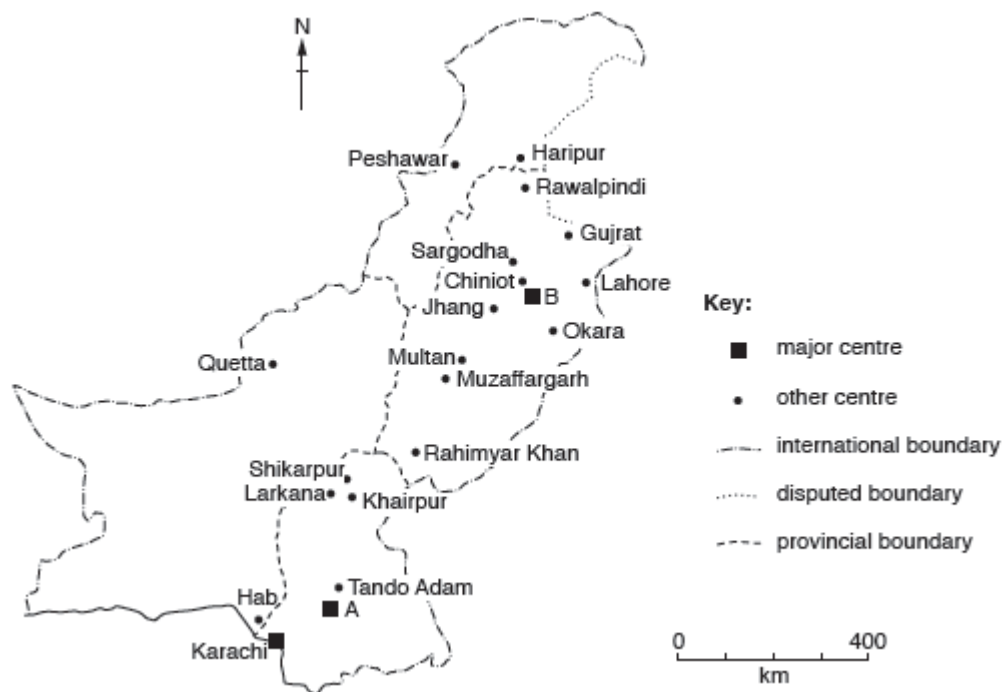


Fig. 9

Read the following two views:



Study Fig. 6 which shows the distribution in Pakistan of selected cottage industries and the engineering industry.

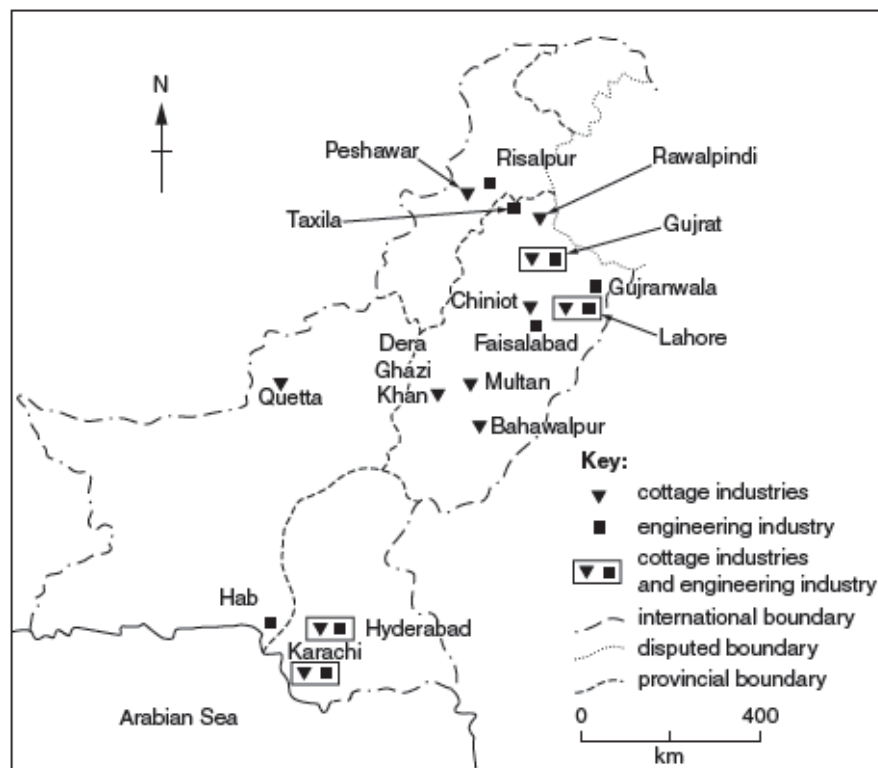


Fig. 6

A. Give an example of a cottage industry.

Answer:

Study Fig. 6 which shows the distribution in Pakistan of selected cottage industries and the engineering industry.

A: Give an example of a cottage industry.

One of: furniture / pottery / woodwork / metalwork / jewellery / hand woven rugs/carpets/shawls / handicrafts / embroidery/sewing / hand-made leather goods/shoes / candles

[1]

B. Compare the distribution of cottage industries and engineering industry as shown in Fig. 6.

Answer:

Compare the distribution of cottage industries and engineering as shown in Fig 6.

[3]

Similar

Both in all four provinces / spread throughout provinces

Both concentrated in Punjab

Both present in Karachi/Hyderabad/Lahore/Gujrat

Both have two locations in Sindh/one location in Balochistan/one in KPK / equal in number in S/B/KPK

Different

Cottage has more locations in Punjab/use of stats to exemplify e.g. cottage 7 locations in Punjab whereas engineering 5 locations

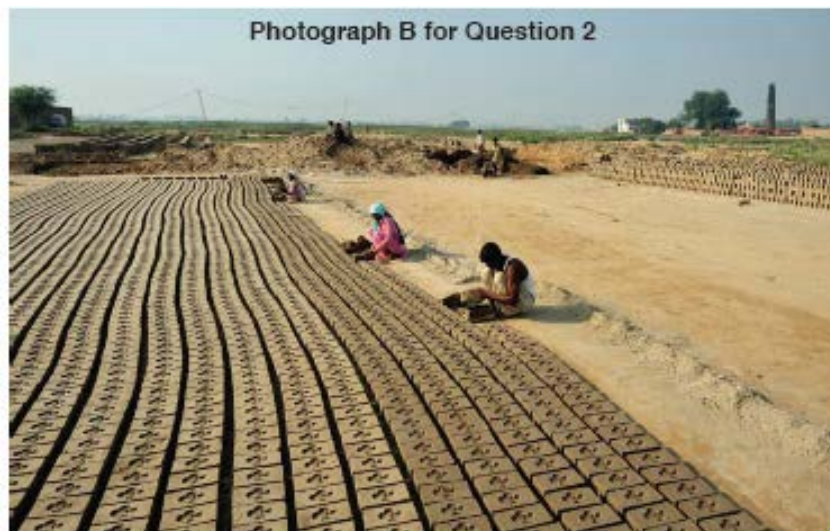
Cottage present only in

Quetta/Peshawar/Rawalpindi/Chiniot/Multan/DGK/Bahawalpur/S Punjab

Engineering present only in Hab/Faisalabad/Gujranwala/Taxila/Risalpur

Engineering more clustered / cottage more spread out

Photograph B for Question 2



Study Fig. 4.1, a map showing the location of selected textile industries in Pakistan.

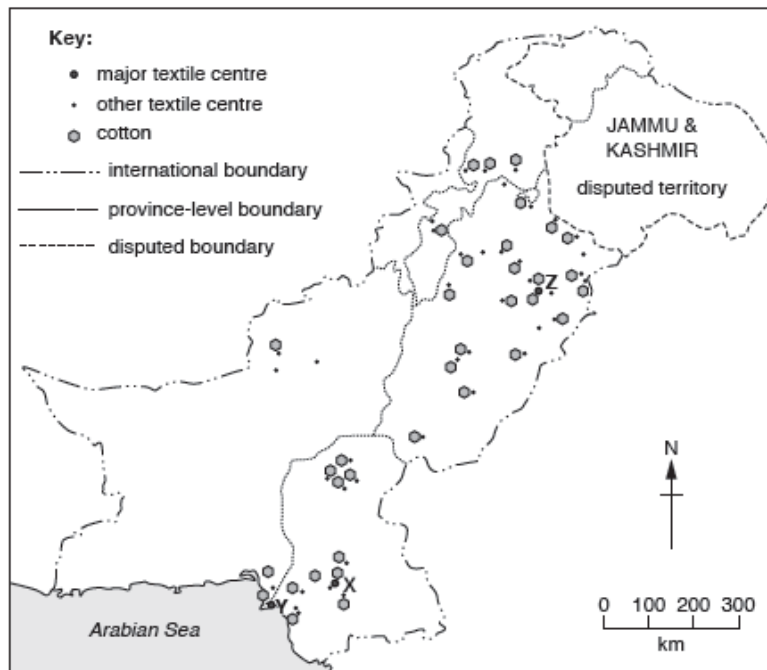


Fig. 4.1

Name the three major textile centres labelled X, Y and Z on Fig. 4.1.

Answer:

- X = Hyderabad;
- Y = Karachi;
- Z = Faisalabad.

3 @ 1 mark

Describe the location of the major textile centre labelled Y on Fig. 4.1.

Answer:

Karachi is:

- In south of Pakistan;
- Next to the coast / near sea / Arabian Sea;
- In the province Sindh;
- Close to border with Balochistan;
- At centre of three cotton areas;
- Distance from named feature, e.g. within 100 km of other textile centres / cotton areas;
- Direction from named feature, e.g. west / southwest of X.

@ 1 mark

Study Fig. 4.2, a diagram showing three of the processes involved in the spinning of cotton. Describe processes A, B and C in the boxes on Fig. 4.2.

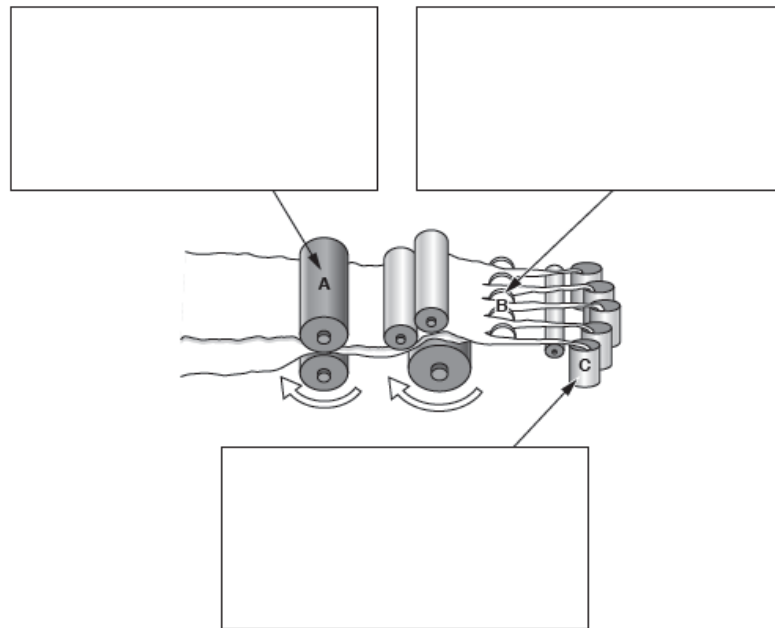


Fig. 4.2

Answer:

- A: Opening / bales of cotton laid down uniformly in layer / rollers used to flatten cotton fibres / smooth out folds / straighten fibres / spread out fibres;
- B: Carding / sorts fibres to produce a continuous web or slivers / teasing wires produce loose bundles of fibres (web / slivers) / the fibres (web / slivers) are divided into threads;
- C: Drawing / fibres are straightened / the combined sliver (threads) is collected in moving cylinders where fast and slow rollers further divide slivers / slivers stretched / twisted / pulled out further.

3 @ 1 mark

Describe the Figures

Fig. 4.2 for Question 4



Fig. 4.3 for Question 4



Fig. 4.4 for Question 4

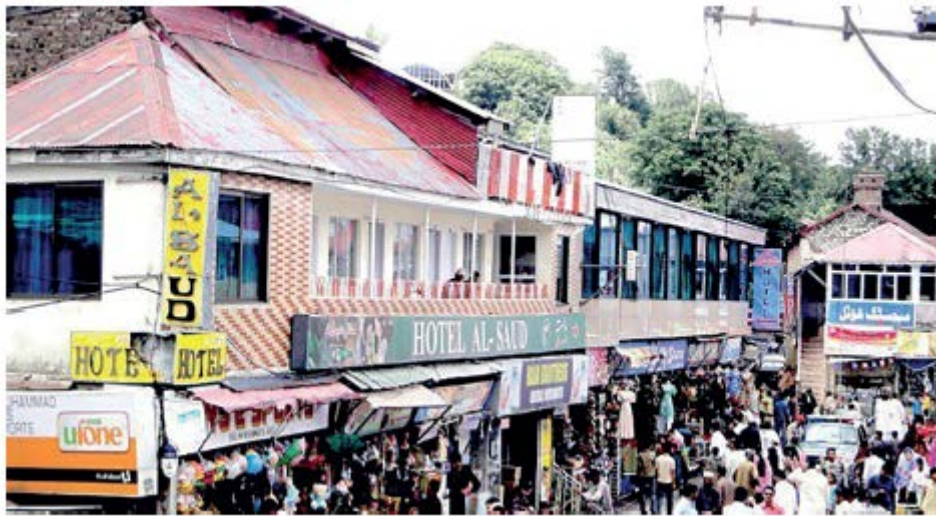


Fig. 4.5 for Question 4



Fig. 4.6 for Question 4



Chapter 10: TRADE

Trade is the exchange of goods and services among countries.

Exports: Goods we sell to other countries are exports.

Imports: Goods we buy from other countries.

Foreign Exchange: Money earned on exports is called foreign exchange

Balance of Trade: The difference between the exports and imports of goods is the balance of trade.

$BOT = \text{Export of goods} - \text{Import of goods}$

Balance of Payments:

The difference between the value of imports and exports is the balance of payments.

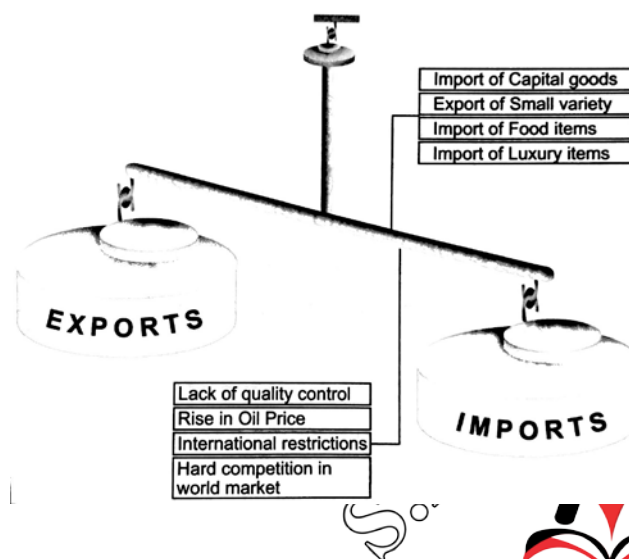
$\text{Balance of payments} = \text{Value of exports} - \text{Value of imports}$

Major Exports	Major Imports
Raw cotton Cotton yarn Readymade garments Fruits and vegetables Fish Leather Carpets and rugs Sports good Surgical instruments Leather products	Petroleum and its products Machinery Fertilizer and other chemicals. Cosmetics Edible oil Tea Automobiles
Countries	Countries
UK, Europe, USA, Middle East Countries, China, Hong Kong and Japan	UK, Europe, USA, Middle East, Malaysia, Japan, Sri Lanka

Negative Balance of Payment

Reasons

- ❖ Lack of standardization
- ❖ Low quality
- ❖ Import of expensive oil
- ❖ Pakistan is heavily dependent on the exports of few items like cotton.
- ❖ Lack of access to the major world markets
- ❖ Lack of industrialization
- ❖ Failed to develop value added products
- ❖ Political instability resulted in lack of long



term plans to boost trade.

- ❖ Child Labour is a major issue
- ❖ USA imposed embargo on carpets and rugs, sports goods, surgical instruments on October 20, 1996 on the issue of child labor.

How Does the Negative Balance of Trade Affect the Economy?

- ❖ Development projects have to be curtailed.
- ❖ Imbalance of trade has to be filled by taking loans.
- ❖ In case of non - payment of loans, an economic or trade embargo may be imposed.
- ❖ In critical situations to repay the loans, assets of the country may have to be sold.
- ❖ Increase in taxation for payments will result in inflation. Higher taxation limits the purchasing power of consumer resulting in lower demand and less production.
- ❖ Business and commercial activity slows down.

How can we correct the Balance of Payments

- ❖ By increasing exports
- ❖ To export value added products
- ❖ Develop the cottage industries
- ❖ Strict quality control for high standards
- ❖ Reduce taxes to provide incentives for export.
- ❖ Export processing zones should be setup in different parts of the country to promote export goods.
- ❖ Impose taxes to restrict imports(trade barriers)

Primary Goods: The raw material or natural products are the primary goods as crops.

Capital Goods: The goods which helps to manufacture other goods.

Consumer Goods: The goods which are ready to use.

Exports Processing Zone

EPZA (Exports Processing Zone Authority) in Pakistan was established in 1980 with a mandate to plan, develop and operate. Export processing zones in Pakistan.

Objectives:

- ❖ To boost industrialization.
- ❖ To create job opportunities.
- ❖ To transfer hi-tech from the developed world to the developing countries.



- ❖ To increase the country's exports by creating facilities for local and foreign investors to set up export oriented units.

Incentives Offered to Investors Are

- ❖ 100% ownership rights.
- ❖ No minimum or maximum limit for investment.
- ❖ Duty free imports of machinery or equipment and material.
- ❖ No sales tax in input goods and services including electricity and gas bills.
- ❖ Exemption from import duties and freedom from national import restriction.
- ❖ The domestic market to be available on same conditions as for imports from other countries.
- ❖ EPZ's are set up in Karachi, Sialkot, Risalpur, Lahore, Faisalabad and Gwadar.

Infrastructure for EPZ's

- ❖ It shall be near the sea port to facilitate export and import of goods and import of required machinery.
- ❖ Consistent government policies for stability.
- ❖ Adequate road, rail transport for marketing of finished goods.
- ❖ Adequate air travel facilities are provided.
- ❖ Efficient links for transport of raw material.

Importance of Gwadar as EPZ

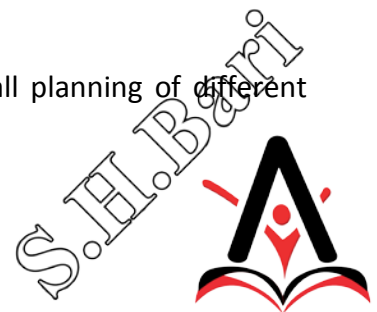
Gwadar's location between Karachi and UAE is helpful for central Asian Republic Foreign investment could be done in Gwadar and Ormara. It could be a Regional Trade Hub with infrastructure development. A deep water port can be a great help. Major development of River Dasht, Mirani Dam, Desalination Plants and Power Generation plants at Pasni can be a great source of progress.

Export Promotion Bureau:

It was formed in Pakistan for organizing and regulating export activities. It helps to create awareness among manufacturing service sector. It helps to assist entrepreneurs to secure entries in the international market. The TDAP (Trade Development Authority of Pakistan has replaced EPB).

Reasons

It has a vast domain apart from restricted EPB. It is involved in overall planning of different



sectors of economy. It helps to link with international trade requirements. It will help to succeed in international trade in goods and service resulting in massive change from the creation of WTO (World Trade Organization).

World Trade Organization:

It came into being in 1995. It is the successor to the General Agreement on Tariffs and Trade (GATT). It is an international organization to allow free trade between its members Countries by reducing or abolishing certain restrictions imposed by government of other countries on exports and imports. After December 2004, Pakistan became one of the signatories of WTO along with 146 countries.

Opportunities Challenges of WTO's Membership for Pakistan from 2005

- ❖ Textile industry requires modernization by replacement of industrial machinery. Import of updated High Tech is required.
- ❖ The service sector will make quite a few adjustments as it is already competing with international players.
- ❖ Pakistan has to reduce to import duties.
- ❖ The small scale and medium sized industries had to face the competition due to imported goods because of high cost of production.
- ❖ A strategy calls for measures to make our industry comply with international quality standards.
- ❖ We need to modernize the production process training and education.
- ❖ Proper infrastructure is also required and flood of information between government and exporters can help to face the challenge.



WTO; A Step Towards Globalisation of International Trade



Exchange Rates

It refers to the price of one currency in term of another currency e.g. 1 US dollar = 101 Rs. Exchange rates are important in determining the cost of imports and prices of exports. An exchange rate is said to depreciate when one unit of that currency buys fewer units of another country's currency e.g. Pakistan's exchange rate against US dollar was \$1 = Rs. 101 last month but then changes to \$1 = 98, the dollar is said to have been depreciated since now it can buy lower value of the rupee. Currency depreciation makes imports expensive in terms of domestic currency and exports cheaper in terms of foreign currency. Appreciation of the exchange rate takes place when one unit of a currency can buy a greater value of another currency. Appreciation of the exchange rate makes imports cheaper and exports more expensive.

Trade Barriers**Advantages**

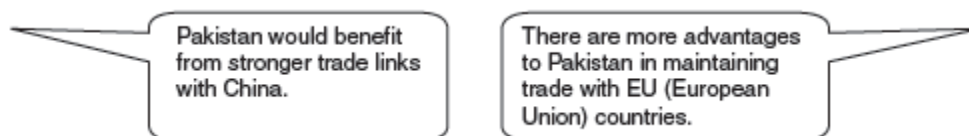
- ❖ Protect local industries and create employment opportunities.
- ❖ Creates domestic demand that leads to greater exploitation of local resources.
- ❖ Give rise to greater self sufficiency thereby reducing foreign dependency.
- ❖ Improve the balance of payments position.

Disadvantages

- ❖ Local industries become complacent due to lack of international competition and thus loose efficiency.
- ❖ Consumer choice is limited to domestically produced goods.
- ❖ Even those goods which the country produces inefficiently and at high cost would need to be produced.

Trading Blocs

Trading blocs refers to regional groupings of international economies to allow for greater economic co-operation and facilitation of trade. Trading blocs involve lower or zero trade restrictions between members and strong trade barriers against non-members. Pakistan is a member of SAARC (South Asian Association of Regional Co-operation) and ECO (Economic co-operation Organization) but they have not been so effective due to differences in political situations. European Union (EU) is a strong trading bloc.



Study Fig. 3.2, a world map. Shade and label two countries that are major importers of products from Pakistan.

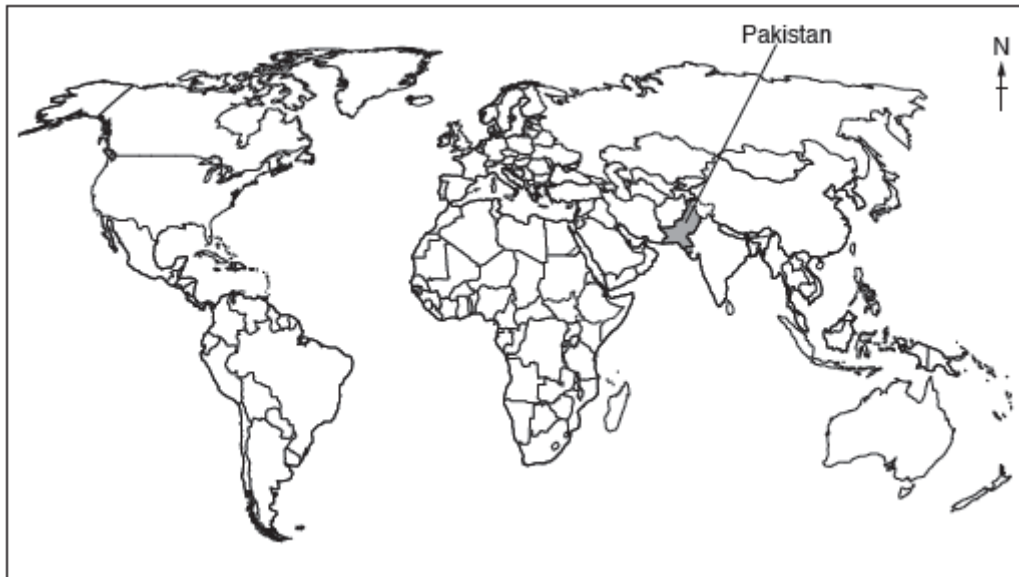


Fig. 3.2

Chapter 11: TRANSPORT & COMMUNICATION

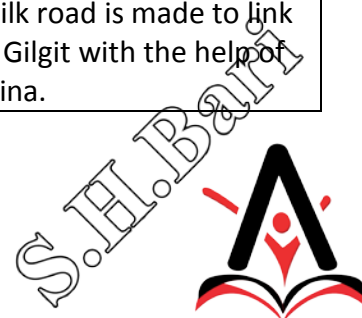
Four means of Transport

Road – Railway – Airways - Sea Route

Roads

In Pakistan, road transport carries 82% of total passenger traffic and 54% of freight traffic. Roads are easy to be built in plain areas but it is difficult and expensive to build roads in the difficult terrain as mountains.

Roads	Routes
Grand Trunk Road (N5 National Highway)	It is stretched from Karachi to Lahore is moves from Lahore to Peshawar. It is often called G.T road. It is 1260 kms long.
Indus Highway	It is called Super highway the link from Karachi to Kotri is a new dual road, from Kotri it moves to wards Dadu, Shikarpur, Larkana, Dera Ghazi Khan and Dera Ismail Khan, then it leaves Indus and reaches Peshawar via Kohat.
RCD Highway	The road crosses Balochistan entering from Karachi, after through Lesbela, Khuzdar, Quetta, Nushki and Nokkundi in Pakistan, it links Iran through Zahidan and then moves to Turkey. It is 600km long in Pakistan.
Sukkur-Quetta	It moves from Sukkur to Shikarpura on the Indus Highly and then to Quetta passing through Jacobabad and Sibi.
Karachi-Gawadar	It connects Karachi with Gawadar.
Lahore – Quetta	It passes from Lahore to Faisalabad then links Dera Ismail Khan, the road later passes through Derajat in NWFP crossing Sulaiman range into Balochistan come to Zhob linking Quetta.
Karakoram Highway	It is a trade route, the silk road is made to link china, It connects from Gilgit with the help of Khunjerab pass links china.



Benefits of Motorways

Motorways help the road transport to be quicker and more efficient. Pakistan railway has a multiguage system. The estates could be established on its route. It would promote industrial growth by supplying raw material to the industries and manufactured goods to the markets. Industrial expansion would open up employment opportunities for the people. The motorways could further be connected to Afghanistan and the Central Asian Republics which will increase foreign trade. May new settlements could be established.

Famous Motorways Are

- ❖ Islamabad – Lahore Motorway 339km.
- ❖ Islamabad to Peshawar Motorway M-I 154km.
- ❖ Pindi Bhattian – Faisalabad Motorway M-3 (52km).
- ❖ Karachi – Hyderabad Motorway M-9 35km.

Railways

Railway is easily built in plains. It is convenient for longer routes. It carries bulky, heavy goods. In high mountains there are no railway tacks e.g, Gilgit, Murree, Chitral etc. The foothills of the mountains like Kohat, Bannu and Peshawar had rail tracks as they are found in the lower areas. It is a very expensive to build the rail tracks rather than roads. The railway engines, sleepers and wagons are expensive. The tracks are difficult to build and maintain. Railway is preferred for long distance as it is economical.

Major railway routes are

- ❖ Karachi to Lahore
- ❖ Rawalpindi to Peshawar
- ❖ Quetta to Chaman
- ❖ Kotri to Larkana
- ❖ Faisalabad to Khanewal
- ❖ Rawalpindi – Kohat
- ❖ Sibi – Kohat
- ❖ Faisalabad, Sargodha, Khushab
- ❖ Peshawar to Karachi via Rawalpindi and Lahore
- ❖ Karachi to Quetta via Kotri, Larkan and Islamabad.



Railway needs specific stations. These are long term returns on investment. The railway network in Punjab and Sindh is dense because they are the flat lands and demand is present.

Famous trains are Karakoram Express, Karachi express

Problems of Railway

- ❖ Lack of investments
- ❖ Poor reservation system
- ❖ Overstaffing
- ❖ Corruption
- ❖ Worn out rails and sleepers,
- ❖ Poor time management
- ❖ Presence of single tracks in many areas
- ❖ Out-dated locomotives

Dry Ports

The inland cities which are far from seaport have dry ports to promote foreign trade. These speed up export and import procedure. Famous dry ports are Quetta, Lahore, Multan and Peshawar. Railway is the key for transportations of goods.

Importance

- ❖ It helps to reduce the work load at Karachi Port and Port Bin Qasim.
- ❖ It helps to collect revenue.
- ❖ They help to stimulate foreign trade activities in those cities which are far away from the seaport.
- ❖ They help to reduce the pressure on exports and imports by providing easy transportation from different cities to seaport directly.
- ❖ Custom clearance at different dry ports saves ports in Karachi from burden.
- ❖ Less time consuming
- ❖ Traders don't need to go all the way to Karachi saves their money as well.

Airways

Internal Factors for Development of Air Transport

- ❖ Air transport is faster than rail and road.
- ❖ Industrialists and businessmen prefer to travel by air because of less time.



- ❖ It is effective mode of transport for high value, light weight and perishable goods.
- ❖ People who can afford the higher cost prefer to go by air.
- ❖ In mountainous regions where there is difficult access by road, air routes make these areas accessible.

External Factors

- ❖ Transport has increased to Middle East because of large local work force working there.
- ❖ Trade of perishable items like fruits and vegetables to Middle East also increased.
- ❖ Improvements in transport have turned the world into a global village with more passengers.
- ❖ Migration has also increased.
- ❖ Tourism has also increased.

Air Travel Companies

- ❖ PIA (Pakistan International Airlines)
- ❖ Air Blue
- ❖ Shaheen Airlines

Requirements of an International Airport

- ❖ Open vast land, not far from the city.
- ❖ Cargo services for trade must be available.
- ❖ 20 planes can stand at one time.
- ❖ Multiple long runways.
- ❖ Access to the transport system.
- ❖ Demand for air travel.
- ❖ A huge international standard waiting area having good food available.

Examples are

- ❖ Quaid-e-Azam International Airport – Karachi
- ❖ Allama Iqbal International Airport – Lahore
- ❖ Benazir International Air port-Islamabad

Requirements of Regional Airport

- ❖ A huge area
- ❖ Civil aviation, cargo system
- ❖ 5 planes can land at one time
- ❖ Normal runway
- ❖ Access to transport system

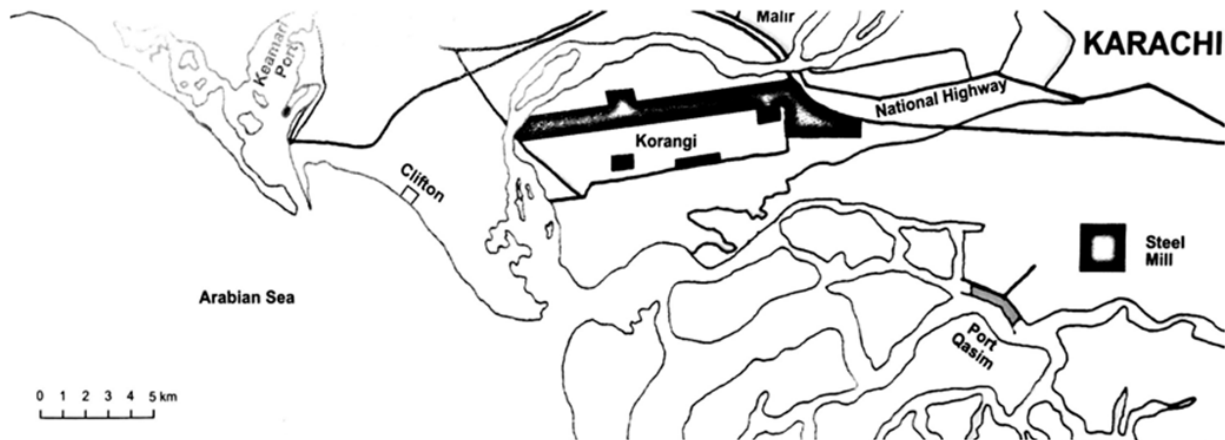


❖ Small waiting area

Regional Airports are found in many cities as Sargodha, Chitral, Gilgit, Zhob, Jacobabad, Skardu, Kohat, Mianwali, Nawabshah, Sukkur etc.

Water Transport

Seaports in Pakistan provide access to the cities in the world. Trade links are easy to develop.



Major Sea Ports

- ❖ Karachi(Keamari)
- ❖ Port Bin Qasim
- ❖ Gwadar

Importance of Port Bin Qasim

It is the second deep seaport and was built in 1980. It has modern machinery to relieve the pressure at Karachi port and for handling raw material for Pakistan steel. It is the first integrated port of Pakistan that combined the function of multipurpose deep seaport and a designated industrial zone. It also offers transport and other infrastructural facilities for industrial development. Main categories of cargo handled at Port Bin Qasim include iron ore, coal, grain, furnace oil, edible oil, rice, LPG containers, Jute and fertilizer.

Importance of Gwadar Port

It has a large sea port, It facilitates exports of fruit crop, For minerals output the Chinese are provided financial assistance. It can serve as an entrepôt-port for Central Asia if Afghanistan allows Central Asian goods to pass through its territory. This can generate revenue by collecting large sums as transit fee. Gwadar can be a substitute



port. Warehouses could also open for exports of goods and for storing imported goods. It is a flourishing port. It also has a fish labour.

Reasons for Development of Gwadar Port

For a substitute to release pressure on Port Qasim and Karachi port. Afghanistan could store its goods for exports. If Keamari and Bin Qasim are affected by a strike or by a natural calamity then Gwadar can become a substitute port. It can be short route for Central Asian land-locked states to do trade.

Pakistan National Shipping Corporation was established in 1979.

Functions

Its purpose is to maintain and develop maritime shipping and transport service. It serves as an operating link between major trading partners of the country.

Communication

Telecommunication

The devices which provide rapid long distance communication, convert sound and images into signals which are then transmitted along wires or radio waves in worldwide system are telecommunication.

Importance of Internet

It is the recent development. It is worldwide network that links computer by telephone and satellite. It allows people to send and receive emails, create websites of information on networks called worldwide web (www). Internet helps to obtain information from websites anywhere in the world.

Institutions for Development of Telecommunications

- ❖ Pakistan telecommunication Company Limited (PTCL)
- ❖ Pakistan Telecommunication Authority (PTA)
- ❖ National Telecommunication Corporation (NTC)
- ❖ Pakistan Broadcasting Corporation (PBC)
- ❖ Pakistan Television Corporation (PTC)
- ❖ Pakistan Post Office (PPO)



Devices for Telecommunication

- (a) Telephone (b) Radio (c) Wireless
(d) Mobiles (e) Television (f) Computers
(g) Internet (h) Fax machines

Promotion of Education through Telecommunication

Education can be promoted by making distant learning possible and by bringing more information into classroom to enrich pupils learning. Videos, computer programmers, multimedia are used to teach a topic easily in classroom. It helps to prepare projects they encourage the young generation to create things which benefit the country's development.

Role of Tele-Communication in Industry

It helps to learn new technology to improve the quality of products. There is advertisement through media about companies and products. Marketing opportunities and trends are known. Electrical modes of trading (E-Commerce) are helpful for online banking. The international companies manage their branches in other parts of the world from their head offices e.g., in Europe etc. Websites of companies help to market their products and services.

Government Plans

- ❖ To provide infrastructure needed for telecommunication.
- ❖ Encourage educational channels on TV and Radio
- ❖ Encouragement of software industry in Urdu and regional language.
- ❖ Encouragement of the use of software in education and business.
- ❖ Investment in education and training of IT professionals.
- ❖ Government is keen to encourage the software industry

Importance Communication

- ❖ E-commerce (online sales purchase and payments)
- ❖ The maintenance of law and order because of quick information.
- ❖ It creates awareness of political situations among people.
- ❖ It results in increased cultural interaction which helps to learn about diverse cultures.
- ❖ It also promotes distant education.



Photograph A for Question 4



Photograph B for Question 4



Study Fig. 10 which gives information about the number of internet users and telephone lines in Pakistan.

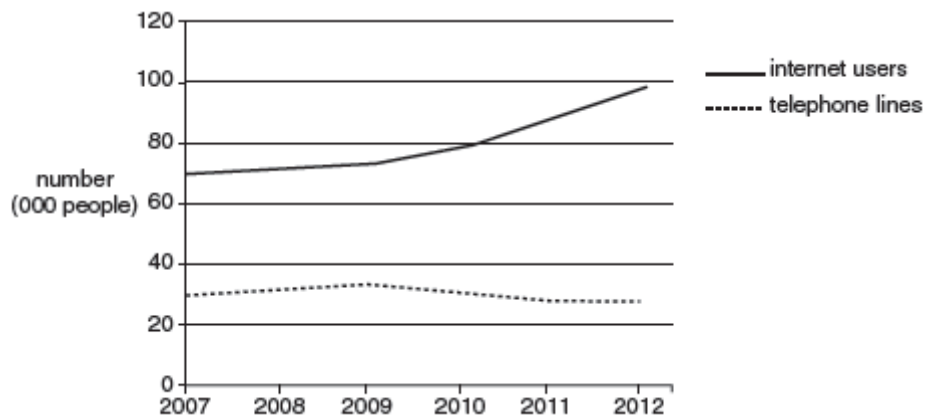
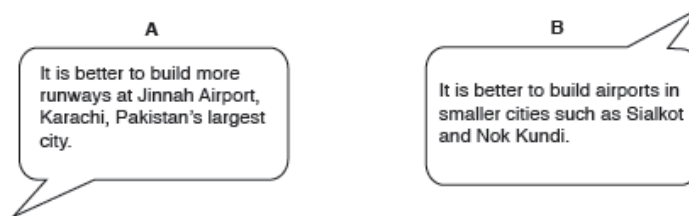


Fig. 10

Read the following two views about possibilities for air transport improvements in Pakistan:



Which view do you agree with more? Give reasons to support your answer. You should consider View A and View B in your answer.

Answer:

VIEW B

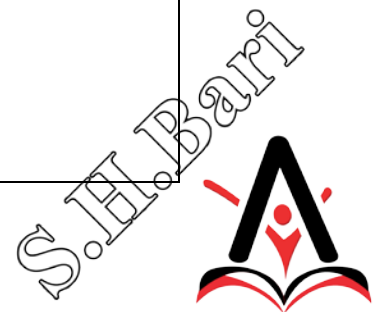
For

- Shorter distance to travel to Iran / Afghanistan / China / India;
- Demand for trade in local manufactured items;
- Even development of air transport infrastructure over country;
- Bring in tourists / income from tourism.

Against

- Cost of providing air facilities at new sites;
- Serve smaller populations – less likely to generate large numbers of passengers / profit;
- Nok Kundi very remote;
- Difficulties in building in Balochistan due to climate.

ETC.



Study Fig. 4.1, a map showing the rail network in Pakistan.

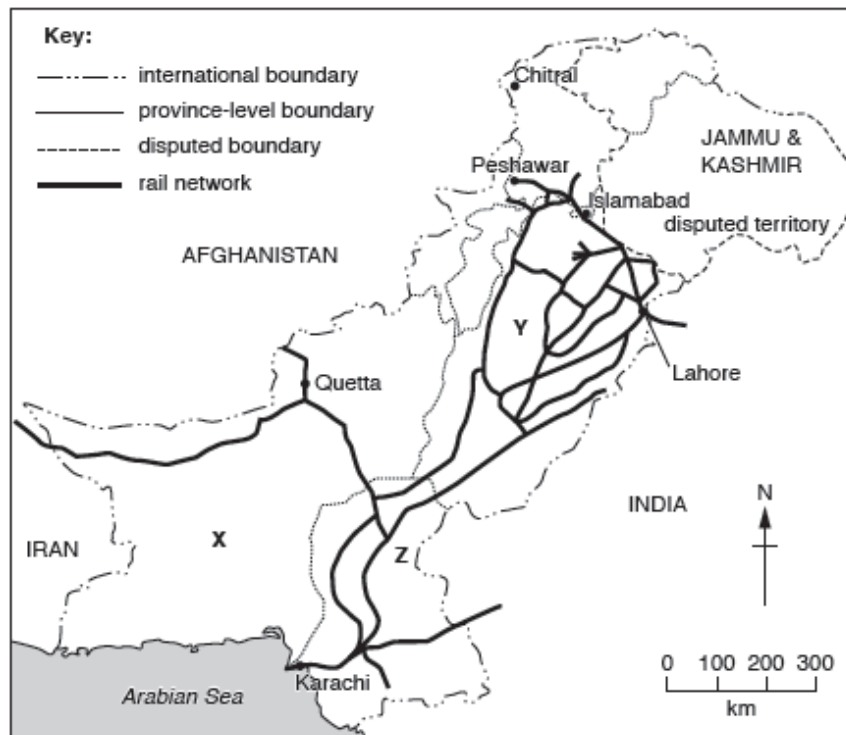


Fig. 4.1

Name the three provinces X, Y and Z.

Answer:

X = Balochistan
Y = Punjab
Z = Sindh

3 @ 1 mark

Using Fig. 4.1 only, identify two countries that have international rail links with Pakistan.

Answer:

Iran
India

2 @ 1 mark

Study Fig. 4.1, which is a map of Pakistan's road network.

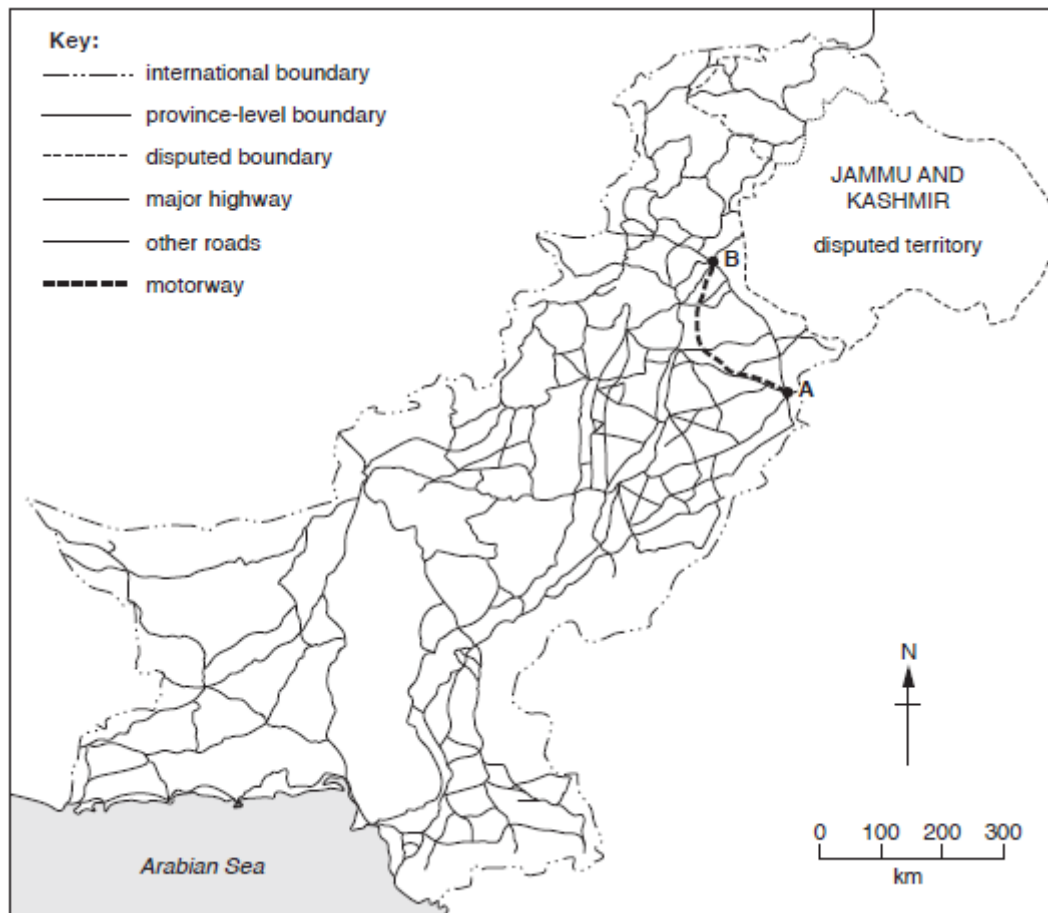


Fig. 4.1

Name the two cities labelled A and B on Fig. 4.1.

Answer:

A= Lahore

B= Rawalpindi / Islamabad

2 @ 1 mark

Name the road on Fig. 4.1 that crosses an international boundary.

Answer:

Karakoram Highway / KKH

1 @ 1 mark

Chapter 12: POPULATION

Population

The number of people living in an area is its population.

Over Population

When the population exceeds its resources, it is termed as over population.

Density of Population

The no. of people living per unit area is the density of population.

Density of Population = Total Population / Total Area

Birth Rate The number of people born on 1000 people

Death Rate The number of people dies on 1000 people.

Rate of Natural Increase: Birth rate – Death rate

Reasons for High Population

- ❖ Desire for more sons to work and earn
- ❖ More hands needed to work in farms especially in rural areas.
- ❖ Religious View: Muslims have a firm belief that Allah gives Rizq to everyone
- ❖ Family Planning Program receive strong resistance on religious grounds
- ❖ No family planning due to lack of awareness
- ❖ Illiteracy among the people
- ❖ Easily marriages increase the span of reproductivity
- ❖ During 1980's many Afghan refugees (3.7 million) migrated to Pakistan as refugees. We have their generations as well of in Pakistan
- ❖ Polygamy leads to more population
- ❖ Low death rate due to proper medication

Problems created by High Population

- ❖ Malnutrition
- ❖ Lack of health facilities
- ❖ Lack of educational facilities
- ❖ Poor living standards
- ❖ Unemployment
- ❖ Lack of accomodation
- ❖ Child Labour



How to Control?

- ❖ We shall think that boys and girls have equal significance
- ❖ Increase in literacy rate
- ❖ In urban areas the need to improve standard of living has given awareness to have small families.
- ❖ Services of Ulemas
- ❖ Education for women
- ❖ Stop early marriages it will reduce the reproductivity period
- ❖ Awareness among people about the benefits of small family size.
- ❖ NGO's should provide assistance and awareness to have small families e.g., Sabz Sitara, Chabi ka Nishan etc.

Different Sectors of Employment**Primary**

It is related to all such occupations that extract raw material as fishing, agriculture, mining and forestry.

Secondary

It deals with the manufacturing of products.

Tertiary

It comprises of all those occupations which provide services as medicines, banking, teaching, management, marketing, transportation etc.

Unemployment

When people do not get jobs this is called as unemployment.

Reasons for Unemployment

- ❖ Overpopulation
- ❖ Lack of political stability
- ❖ Unstable government policies
- ❖ No development works or projects
- ❖ Lack of industrialization
- ❖ Illiteracy
- ❖ Mechanization
- ❖ Rural Urban migration, burden on the urban areas

Effects of Unemployment

- ❖ Lack of modern facilities
- ❖ Malnutrition



- ❖ Lack of health facilities
- ❖ Lack of educational facilities
- ❖ Poor living standards
- ❖ Unemployment
- ❖ Unskilled Labour>>>>low wages
- ❖ Child Labour
- ❖ Diseases

How to Control

- ❖ Increasing industrialization
- ❖ Developing Technical education to develop skilled labour
- ❖ Population control
- ❖ Literacy
- ❖ Political stability
- ❖ Long term government policies
- ❖ Development projects

Migration: Rural Urban Migration

To move from one place to another is called migration.

Rural Push Factors	Urban Pull Factors
Loss of land because of water logging and salinity	Employment opportunities
Low share in inheritance results in less income	Better educational institutes
Strong control of landlords in villages	Hospitals and doctors
Natural calamity such as floods results in destruction	Entertainment facilitates i.e. cinemas, parks etc
No infrastructure available for their progress and development	Better infrastructure such as roads etc
Introduction of machinery in agriculture means less jobs in villages	Better standard of living because of electricity water supply sewerage gas transport

Emigration

To migrate to the other countries due to better living standard, education and medical facilities, a healthy environment and political stability.

Immigration

Incoming of people to Pakistan is called as immigration i.e. to invest their money in some business, for marriages for their children etc.



Problems created by Rural – Urban Migration

- ❖ Burden on the resources in urban areas
- ❖ Unemployment increases
- ❖ Burden on the infrastructure e.g., leads to load shedding
- ❖ Lack of housing people are forced to live in slums
- ❖ Problems of sanitation
- ❖ Lack of institutions as schools and colleges
- ❖ Lack of medicines and hospitals
- ❖ Unplanned growth of cities
- ❖ Living standard deteriorates
- ❖ Poverty and injustice increases leading to high crime rate and terrorist activities
- ❖ People who are unemployed can be a cause of theft, murder, smuggling they can be forced to become drug addicts

How to Control?

- ❖ Providing modern facilities in rural areas
- ❖ Better health facilities as hospitals
- ❖ Better educational facilities as schools and colleges
- ❖ Developing the infrastructure i.e. roads, electricity etc
- ❖ Providing job opportunities by equal distribution of resources and making policies
- ❖ Industrialization away from urban areas

Density

The no. of people living in per square kilometer area is called as density of population.

Factors for Density of Population

- ❖ Relief features (plain, rugged, badland)
- ❖ Drainage features (river, sea, lakes).
- ❖ Infrastructure is available or not.
- ❖ Transportation facilities(land, sea, road, rail)
- ❖ Climate (Extreme, mild, moderate).
- ❖ Natural resources (forests, fish, soil, minerals etc.)

Census

Census means to count the total population in a country generally after every 10 years. It includes data about birth rate, death rate, literacy rate, no. of males, females children and age groups etc.



Population Pyramid It indicates the birth rate, death rate, age groups and gender	
Why Death Rates were Higher During 1905–35?	There were many diseases which could not be cured as cholera, malaria etc. There was no proper sewage system which resulted in many diseases. There was little development in medical science, no antibiotics or life saving drugs was present.
Why was there High Birth Rate During 1905–35?	Large families were considered a pride, there was no family planning or birth control. More children and family members when needed to work in farms.
Why did the Death Rate Drop in 1935 – 70?	It dropped as proper sanitation was available; life saving drugs, antibiotics was also present. Proper vaccination facilities, hospitals, doctors and nurses were there. There was a decrease in infant mortality rate. There was better transportation system to move doctors, food, medicines etc. to the places where they were needed. There was better nutrition as variety of food and increased food production.
Why was there a Fall in Birth and Death Rate During 1970-05?	It was due to family planning programs by the government and NGO's like Sabz Sitara, Chabi ka Nishan etc. Increased industrialization and mechanized farming needing fewer workers. Greater number of women in Urban Areas who have started to work. There is improvement in living standards with a desire for more material possessions rather than children. Women marry at a later age in Urban Areas which leads to low birth rate and death rate has also decreased as new medical facilities as hospitals, clinics and doctors.

Study Fig. 7 which gives information about urban and rural population in Pakistan in two different years.

(b) Study Fig. 7 which gives information about urban and rural population in Pakistan in two different years.

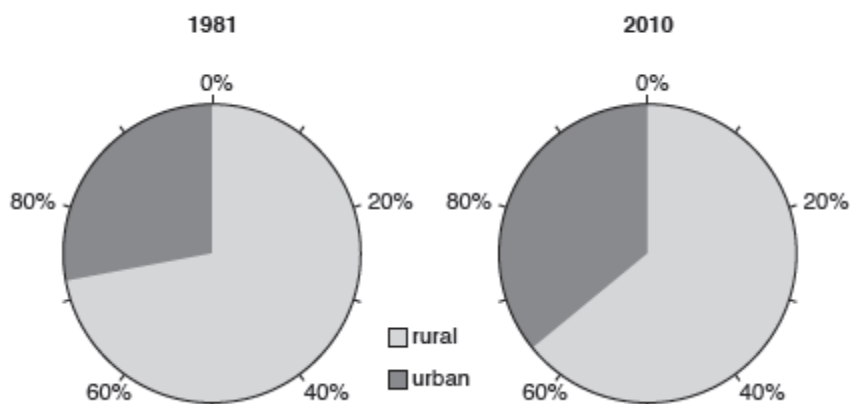


Fig. 7

By how much has the rural population decreased between 1981 and 2010?

Answer:

Study Fig.7 which gives information about urban and rural population in Pakistan in two different years.

By how much has the rural population decreased between 1981 and 2010?

[1]

8% (tolerance 7–8%) Also accept alternative interpretation 11%

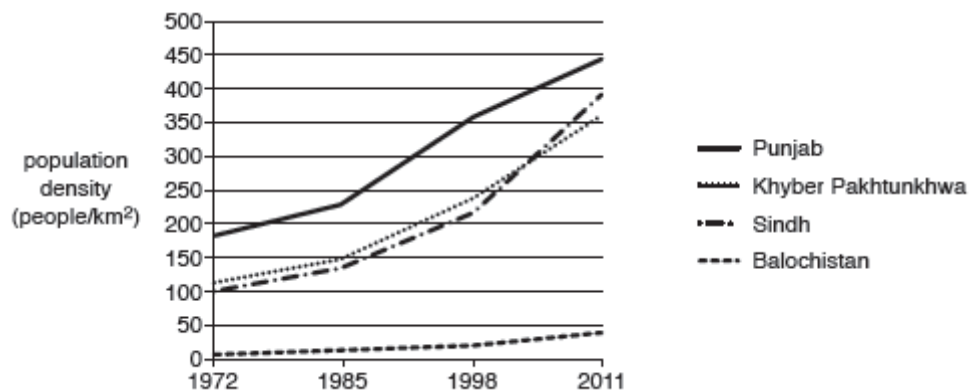


Fig. 8

Describe two of the main changes over time that can be seen in this graph.

- 1
- 2[2]

Answer:

Study Fig. 8 which is a graph showing the changes in population density for different provinces between 1972 and 2011. Describe two of the main changes over time that can be seen in this graph.

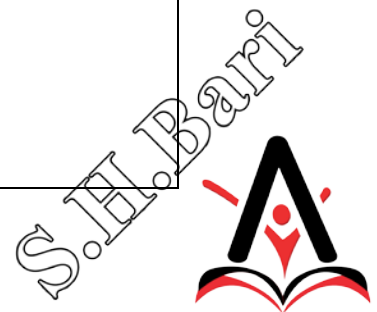
All four increased

P, KPK, and S all increased at same rate

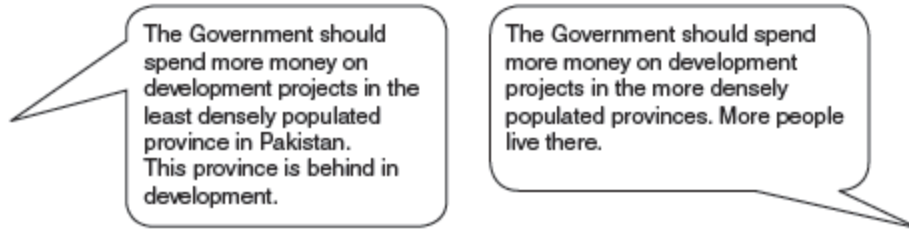
B increased at a slow rate

P/KPK/S increased at a fast rate

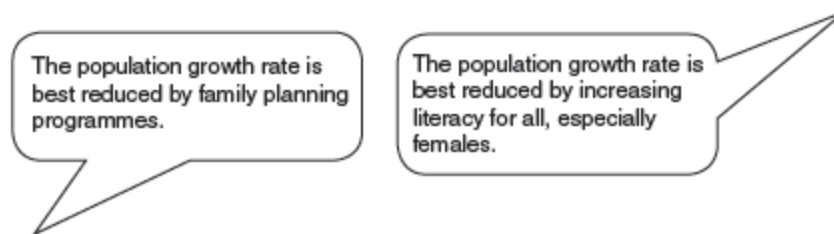
S became more densely populated than KPK over the time period



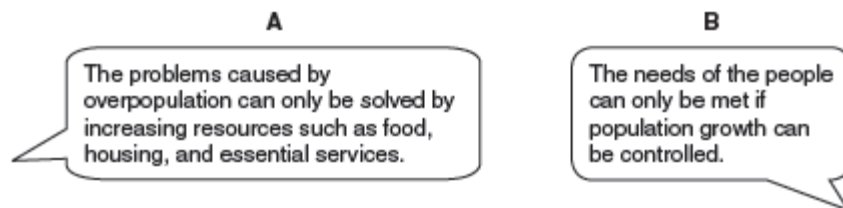
Read the following two views:



Read the following two views about reducing the population growth rate in Pakistan.



Read the following two views about overpopulation in Pakistan:



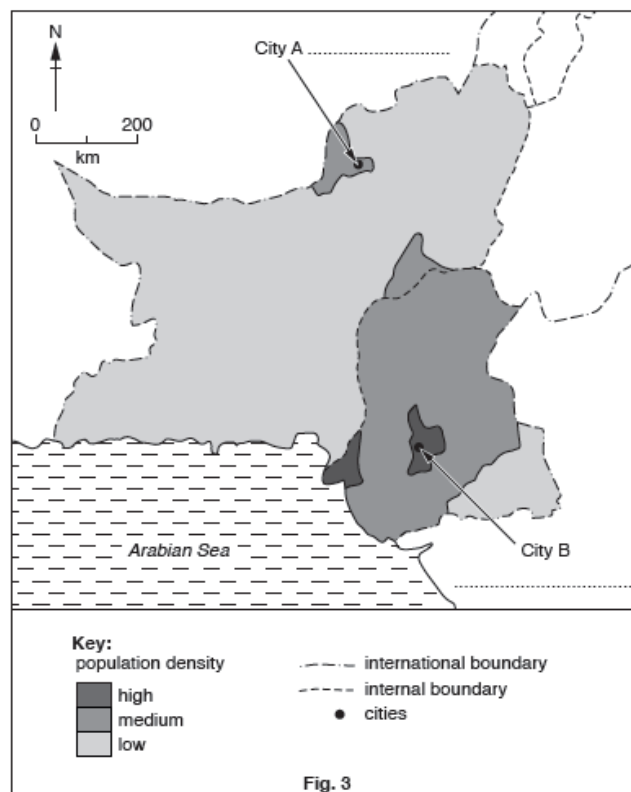
Which view do you agree with more? Give reasons to support your answer and refer to places or examples you have studied.

Answer:

Photograph A for Question 1



Study Fig. 3, a map which shows the distribution of population in southern Pakistan.



(i) On the map name the following: City A; City B. [2]

Answer:

A = Quetta
B = Hyderabad

2 @ 1 mark

(ii) Name one area of low population density shown on Fig. 3.

Answer:

- EITHER Balochistan / Tharparkar / Thar Desert / Kharan Desert / Zhob Desert / Cholistan Desert / Chagi Desert / Makran Desert;
- OR a named district in Balochistan: Awaran / Barkhan / Bolan / Chagi / Dera Bugti / Gwadar / JhalMagsi / Kachi / Kalat / Kech / Kharan / Khuzdar / Kohla / Lasbela / Loralai / Mastung / Musa Khel Bazar / Nushki / Panjgur / Piskin / Sherani / Qila A Saifullah / Sibi / Wazuk / Ziarat / Zhob;
- OR a named mountain range: Suleiman Range / Chaghi Range / Central Brahui Range / Toba Kakar Range / Makran Range / Kharan Range / Pab Range / Kirthan Range.

1 @ 1 mark

Study Fig. 5.1, a population pyramid for Pakistan in 2014.

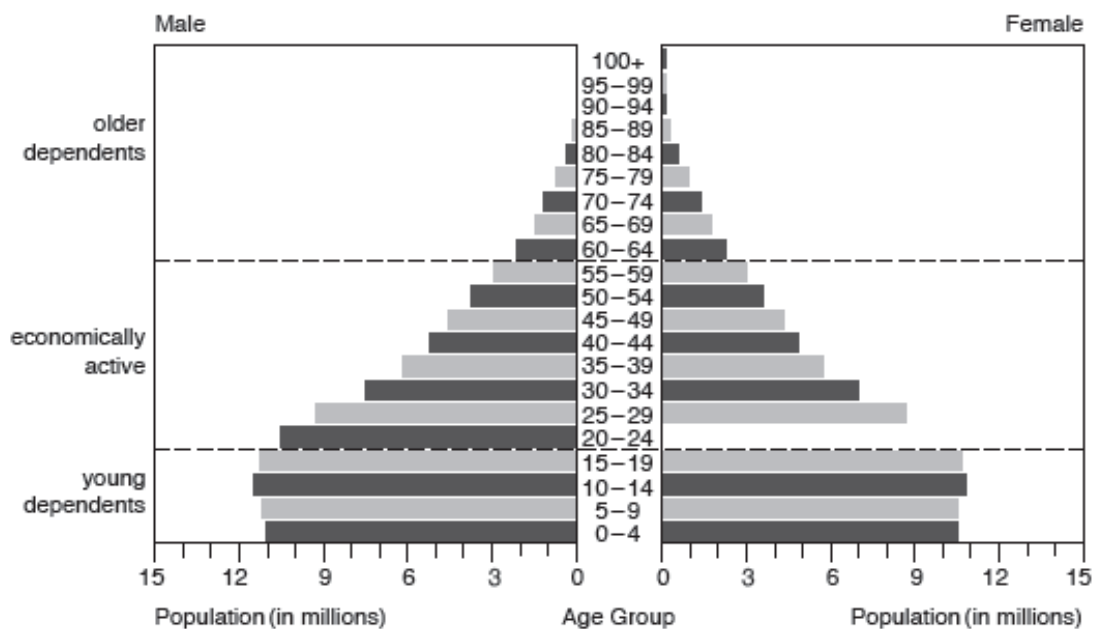


Fig. 5.1

Study Fig. 5.2 showing selected population information about Pakistan.

Total population	187 million
Urban population	70.87 million
Rural population	116.13 million
Population growth rate	2.0% per annum
Total fertility rate	3.3 per thousand
Birth rate	26.8 per thousand
Death rate	7.0 per thousand

Fig. 5.2

Using data from Fig. 5.2 calculate the natural population increase for Pakistan.

Show your working in the box below.

Answer:

Natural Increase = birth rate – death rate
 $26.80 - 7.00 = 19.8$ (per thousand)

2 @ 1 mark

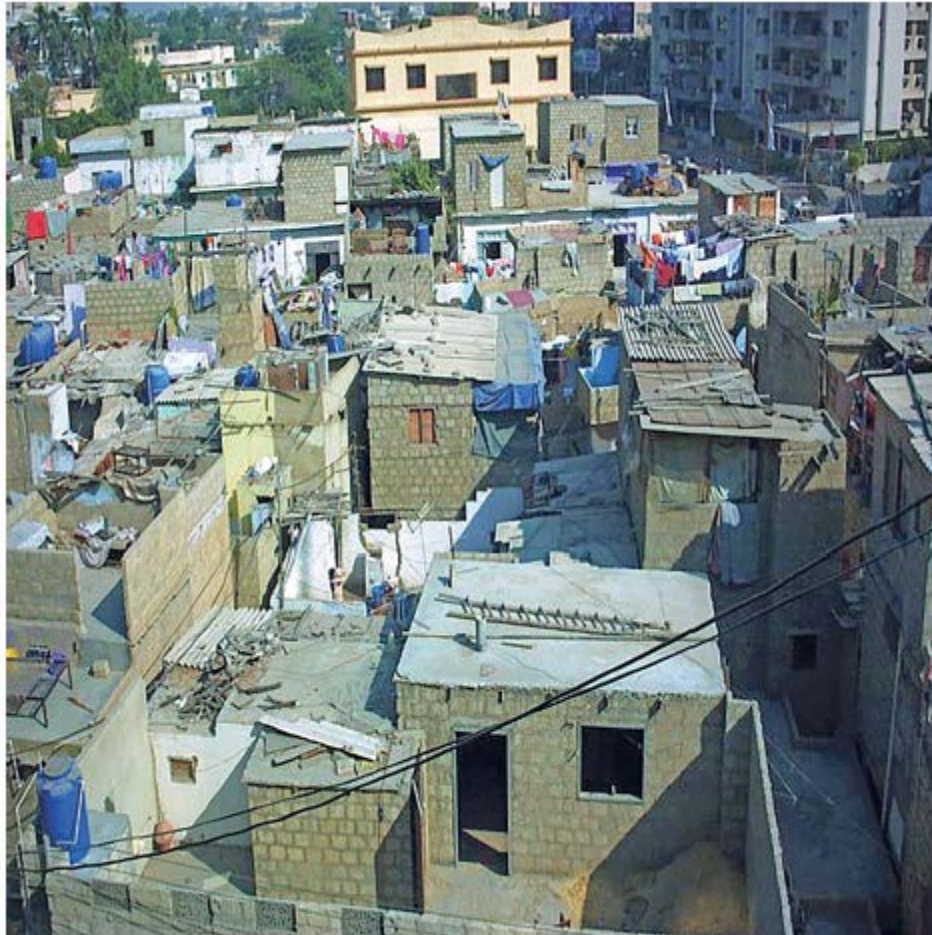
Describe the Figures

Fig. 5.2 for Question 5



Describe the Figures

Fig. 5.3 for Question 5



Paper 2 –The environment of Pakistan

Question 1

Example candidate response – high

- 1 (a) (i) On the outline map of Pakistan Fig. 1 mark and shade **two** areas which experience low annual rainfall (125 mm or less). [2]

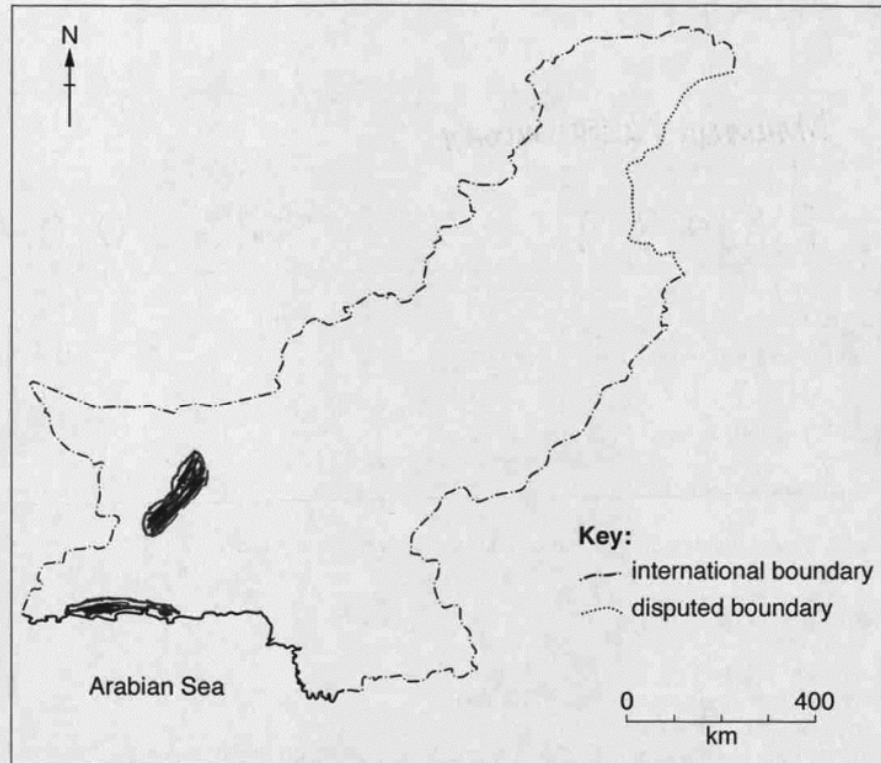


Fig. 1

- (ii) Name the crop which is mainly grown in these areas of low annual rainfall.

Dates

[1]

- (iii) Explain the difficulties for people living in areas of low rainfall.

The agricultural activities are difficult to take place due to less water for crops. Also people must then face extreme heat. Droughts may also occur which add to the difficulties. Such areas may not provide hydel HEP and irrigation in such areas is difficult.

[3]

Example candidate response – high, continued

(b) (i) Study Fig. 2 which shows rainfall data for two cities on the River Indus.

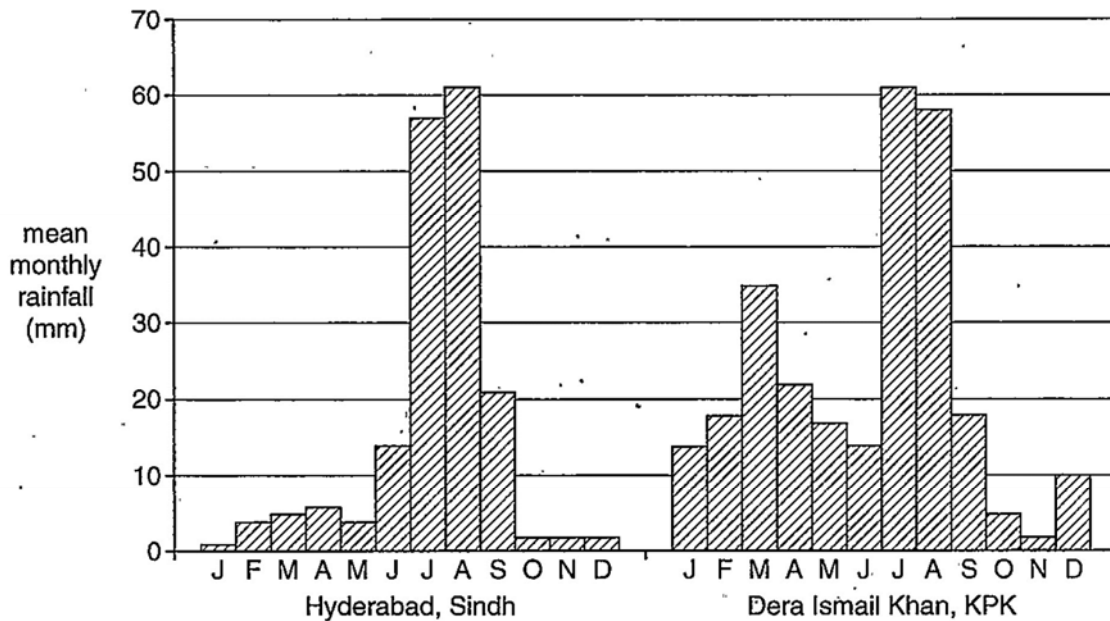


Fig. 2

A. Compare the amount and pattern of monthly rainfall in Hyderabad with that of Dera Ismail Khan.

From January to May the rainfall in Hyderabad is less not more than 5mm while more than 12mm in D.I Khan. In June the rainfall is about 12mm for both. July and August are very wet with wettest months for over 60mm (July for D.I Khan and August for Hyderabad). Then rainfall decreases in September. October and November are dry. December is wetter in D.I Khan. [3]

B. Give three reasons for any similarities or differences in the two patterns of rainfall.

1. From December to May, rainfall is more in D.I Khan due to western depression.
2. In July and August, it is monsoon that causes a high rainfall in both of the cities.
3. As D.I Khan does not have an inversion layer, it enjoys convectional currents in April and May but not so in Hyderabad. [3]

Example candidate response – high, continued

- (ii) Explain the effect of flooding on the local economy and transport links in communities along the River Indus.

Flooding causes the destruction of crops thereby the local economy dependant on farming is hurt. Moreover livestock is also damaged thereby hurting the economy. Also flooding causes blocked roads, and the water is needed to be drained or else it may damage road. More Floods also help increase fertility thereby leading to better yields and recharge groundwater supply which may be used later. [4]

- (c) Give an example of a dry port and explain why it is located where it is.

Example Lahore Dry Port

Explanation Lahore is an extremely big city with a lot of industries. It manufactures a lot of things which are needed to be exported. As it is away from sea, dry ports are needed. This would also help ~~we~~ investors to invest more. Besides, Lahore is the provincial capital of Punjab. [3]

Example candidate response – high, continued

- (d) It has been proposed that a new motorway should be built in Sindh from Hyderabad via Thatta to Keti Bandar on the Indus Delta, a town with a population of less than 25 000.

Explain why this motorway might be needed. To what extent would this proposal be possible?

This proposal is, without any doubt, advantageous. Doing so will help Keti Bandar's local industries to flourish as a new market may be explored. Also the unemployed people may move to Hyderabad for jobs so unemployment decreases. Also this will help encourage government to develop Keti Bandar. Adjoining areas would also be benefitted. Raw materials from Keti Bandar will be used by industries in Hyderabad. Also the population of Keti Bandar will increase. But this proposal faces a problem of lack of investment as Pakistan is poor. This would also result in loss of agricultural land. Also the lack of skill makes the project difficult. It will increase internal migration.^[6]

[Total: 25]



Examiner comment – high

(1a)(i) In this question the candidate followed the instructions correctly, marking and shading two areas on the map. There are three distinct regions where rainfall is below 125 mm. A mark was awarded here for correctly identifying an area in one of the regions. A further mark would have been awarded if the candidate had shaded an area of a different region.

This candidate correctly identified an area within Balochistan. Fewer candidates correctly shaded areas in the other two regions of the map. A number of candidates confused named deserts with areas that experience rainfall of less than 125 mm or less. For example, the Thal desert was often shaded.

Mark awarded = 1 out of 2

(1a)(ii) Most candidates referred to secondary crops such as wheat. Here the candidate carefully reads the question and identifies the main crop as dates. This question tests knowledge of the maps in the course atlas that show crops, crops and cultivated areas, and fruit grown in Pakistan.

Mark awarded = 1 out of 1

(1a)(iii) In this question the candidate correctly identifies that facing extreme heat is a difficulty for people living in an area of low annual rainfall, and also correctly identifies that drought may occur. However, both these points are listed on the same line of the mark scheme as they are related ideas so only one mark can be awarded here.

Further marks could have been awarded if the candidate had given more detail and used more precise subject-specific language. The candidate needed to mention the lack of water for irrigation or that the type of crops that could be grown are limited because of this. For example it would not be possible to grow crops like rice.

Mark awarded = 1 out of 3

(1b)(i)(A) This response shows the candidate looked at the amount and pattern of monthly rainfall in both of the cities as asked. They start by correctly comparing one period, from January to May, in both cities, followed by the statistic for June, correctly telling us that the rainfall is the same in both cities in that month. When comparing graphs it is important to make clear comparisons (candidates should imagine that the person who is being told the information cannot see the graph) to compare like with like (in this case the same month with the same month) and to give accurate and precise information. To be awarded the full three marks the candidate needed to tell us that both cities receive high or the highest amounts of rain in July and August, rather than comparing July in Hyderabad with August in Dera Ismail Khan.

Marks are only awarded for comparisons and not for general description or explanation of the graphs. Generally, marks are awarded for one illustration of one idea. For example, one mark is awarded for 'the amount of rainfall in June is identical in both cities', but saying 'the amount of rainfall in November is identical in both cities' would not be credited as well, although this is correct information, because it is the same kind of comparison.

Mark awarded = 2 out of 3

(1b)(i)(B), In responding to this question the candidate gains the full three marks by demonstrating a good knowledge of the reasons. Candidates in general gave more accurate responses to this part of the question. Most at least mentioned that both cities experienced monsoon rainfall.

Mark awarded = 3 out of 3



(1b)(ii) The candidate correctly identifies that flooding causes the destruction of crops and livestock. These are similar ideas, and so one mark is awarded. The candidate also mentions roads being blocked and damaged by floods (again a similar idea). To be awarded further marks, the candidate needed to develop the good ideas they included in a more precise way. For example, that the farmer would lose money because of the crops/livestock being destroyed, or that the blocked roads meant that journeys would take longer. Another very good idea that could have been credited if it had been clearer and more detailed, was the increase in the soil fertility after a flood where alluvium has been deposited.

Some candidates moved away from the focus of the question and wrote about the effects on the national economy or the social effects on the local area, for example, the effect of flooding on housing. However, a number of candidates referred to the disruption of the electricity supply, telecommunications and the inability to send emails, phone customers, etc.

Mark awarded = 2 out of 4

(1c) The candidate's response here gained the full three marks for correctly identifying a dry port and giving the locational factors, as requested in the question.

Mark awarded = 3 out of 3

(1d) The candidate has kept their answer closely focused on this question and has included examples. There are a number of very good points about why the motorway might be needed, and a number of very good points about why the proposal might not be possible. The candidate is awarded Level 2, three marks. To gain higher marks, the candidate needed to develop more of the points made, and also needed to include arguments as to why the proposal may be possible. The response needs to clearly identify the reasons why the motorway is needed, then go on to say how far the proposal is possible (rather than raising these points together). The candidate also needed to develop their reasons for why the proposal may not be possible, and end with a clear conclusion.

Mark awarded = 3 out of 6

Total marks awarded = 16 out of 25



Question 2

Example candidate response – high

- 2 (a) Study Figs 3 and 4 which give information about the extraction of three metallic minerals in Pakistan in 2010–11.

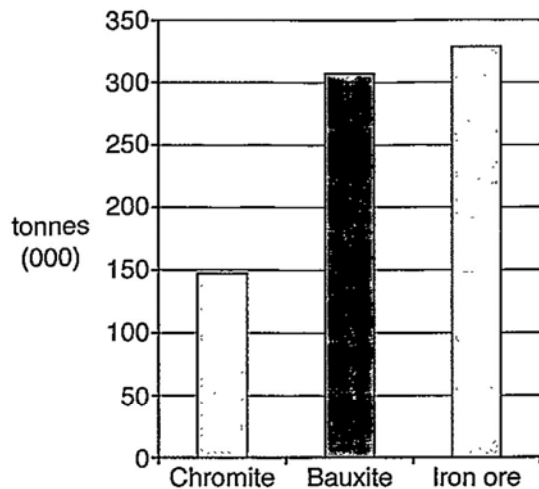


Fig. 3

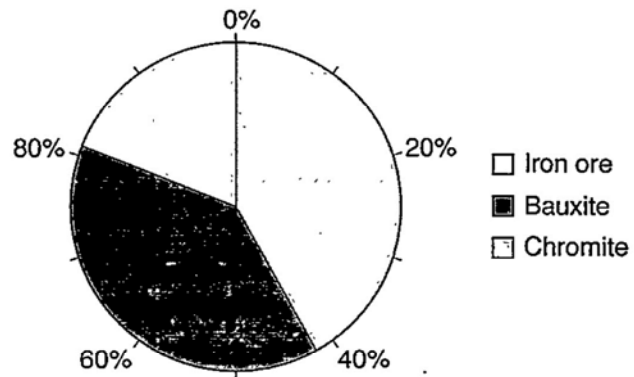


Fig. 4

- (i) How much iron ore was extracted in 2010–11?
~~225 tonnes~~ 330 tonnes[1]
- (ii) State the difference between the type of information being provided in Fig. 3 compared to that in Fig. 4.
 Fig. 3 provides the exact amount whereas Fig. 4 provides the percentage share.[1]
- (b) (i) Give **one** use for the mineral chromite and name **one** area where it is extracted in Pakistan.
 Use Used as an alloy to make stainless steel.
 Area Balochistan[2]

Example candidate response – high, continued

- (ii) What are the benefits of extracting mineral resources for local people and the national economy?

Extracting minerals would require labour force and thus provide employment. Mineral resources are of high demand so it will boost economy, e.g. gold and copper. If Pakistan extract copper itself, it won't need to import it thus reducing ~~pressure~~^{burden} on foreign exchange. Pakistan can export these resources. As economy will boost, more development projects can be run helping local people. [4]

- (iii) Explain the effects of mineral extraction on the natural environment.

Mineral extraction have some negative impacts on environment. The machinery used may produce loud noise thus happens noise pollution. Deforestation might take place to reach mines. There is land degradation due to digging and land might collapse too. The machinery and waste might be thrown into rivers and streams resulting in polluting them. [4]

Example candidate response – high, continued

(c) (i) Fig. 5 is a diagram of a thermal power station.

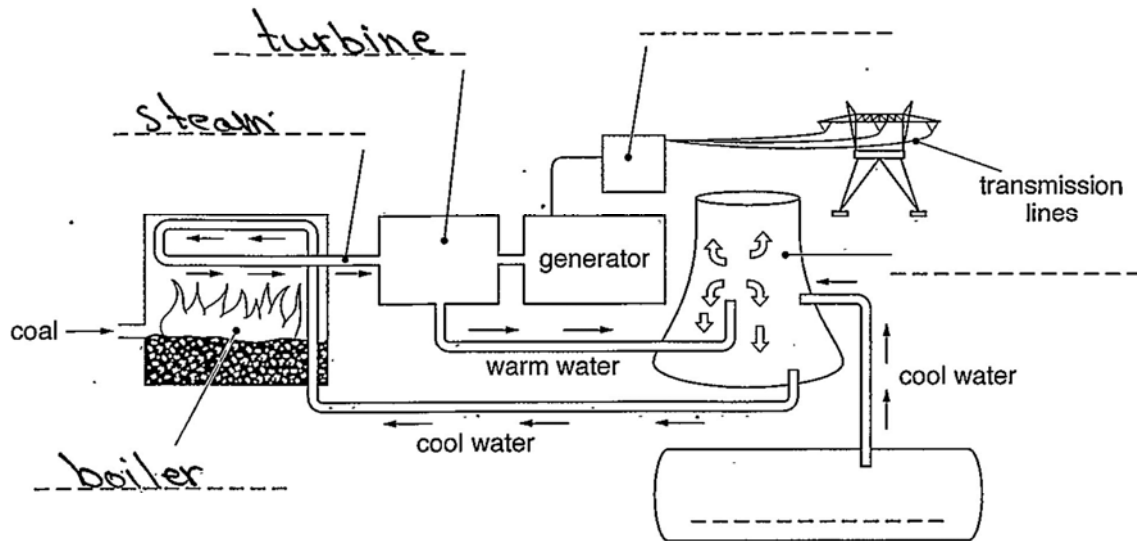


Fig. 5

Choose **three** terms from the list below and use them to label the diagram in **three** of the spaces provided.

reservoir transformer boiler turbine cooling tower steam [3]

(ii) Explain why burning fossil fuels in power stations is unsustainable.

It is unsustainable because fossil fuels are not renewable. They will run out at some stage. They have deposits in earth but someday these deposits will be exhausted. Then the fossil fuel that is burnt is gone, turned into smoke. We can not make fossil fuels as they are natural.

[4]

Example candidate response – high, continued

(d)

Pakistan Railways now carry less than 10 percent of Pakistan's passenger traffic and 5 percent of its freight.

To what extent is it possible to develop railways further in Pakistan? Support your answer by using examples you have studied.

Government must replace the old and useless engines with new ones. Electric trains can be introduced which are faster so people will prefer them. The train cabins must have facilities and be more comfortable. But for all this, huge investment is required. Government is already short of capital and needs to allocate budget in other projects. The tracks are to be maintained or at some places replaced as well. They are old and out dated. Government can try to reduce ticket prices to make railways cheaper. Developing railways is possible to some extent. New routes should be made to make railways more efficient. [6]

[Total: 25]



Examiner comment – high

(2a)(i) Although the candidate looked at the correct chart and the correct area of the bar chart, for the mark to be awarded they needed to give the correct numerical value of 330 000 tonnes. A significant minority of candidates omitted the tonnes or the 000 tonnes in their response. It is essential that all areas of the graph are read carefully, including titles, axes labels and keys, as well as the numerical data represented.

Mark awarded = 0 out of 1

(2a)(ii) The correct answer was given in the form of 'amount' and 'percentage share'. A variety of responses were given credit, in order to allow for the different words that candidates used to describe the same concepts.

Mark awarded = 1 out of 1

(2b)(i) The mark was awarded for 'stainless steel'. Goods rather than processes gained the mark. Very few candidates were able to identify one of the correct areas for the extraction of chromite, i.e. 'Muslimbagh/Zhob (Valley)/Wad'. Although these areas are in Balochistan, this is a province, and mining for chromite does not take place over the entire province. Candidates generally need to be more specific and accurate when naming the different areas of Pakistan where different activities take place.

Mark awarded = 1 out of 2

(2b)(ii) This response was awarded three marks for 'provide employment' for a benefit to local people, 'gold and copper' as the named raw materials and 'won't need to import it (copper)' for the idea of reducing imports. A further mark could have been awarded if the candidate had been more specific. For example, with regard to 'export these resources', they could have included the idea of exports increasing, or that export earnings contribute to the balance of payments because the export of minerals is a source of foreign exchange. The candidate could have gained a mark if the idea of 'more development projects can be run helping local people' had been expanded to cite a specific project, for example road improvement.

Mark awarded = 3 out of 4

(2b)(iii) This was very well answered, gaining full marks for four developed points. The candidate gave the source of noise pollution (loud noise from machinery), the reason why deforestation occurred (to reach the mines), that land might collapse too due to digging and that dumping waste in rivers would pollute them. It is important that candidates develop the points they make when answering questions that ask them to 'explain'.

Mark awarded = 4 out of 4

(2c)(i) The candidate correctly identified three terms on the diagram.

Mark awarded = 3 out of 3

(2c)(ii) One mark was awarded here, because, although the candidate made a number of points, they repeated the same point – that fossil fuels are non-renewable. The candidate needed to address other aspects of the unsustainability of fossil fuels. They should have considered not just the fuels themselves, but also the source, extraction and effects of using such fuels. The candidate could have gained another mark if they had expanded on the CO₂ emissions produced by fossil fuels which in turn contribute to global warming.

Mark awarded = 1 out of 4



Examiner comment – high, continued

(2d) This type of question requires the candidate to present developed arguments for and against a proposal or possibility and make a clear evaluation of those arguments. There is enough development here to award this Level 2, four marks. The argument presented about why the government should reduce ticket prices to make the railways cheaper (an argument for the possibility) as well as the point about the challenge of other budget priorities (an argument against the possibility) are the elements that gain credit. However, although the candidate states 'Developing railways is possible to some extent' the argument against the possibility of development is much more persuasive, i.e. there's no money to do it.

There were a number of points addressing the problems with the railways and what the government should do to develop them, but little which evaluated the extent to which development of the railways is possible (as asked by the question). This response is a very good example of a candidate who has presented points on both sides of an argument but failed to base their conclusion on clear evidence. More careful reading of the question, and more careful planning and organisation of points for and against the feasibility of development would have helped gain credit at Level 3.

Mark awarded = 4 out of 6

Total marks awarded = 17 out of 25



Question 3

Example candidate response – high

- 3 (a) Study Fig. 6 which gives information about the area of Pakistan under cultivation.

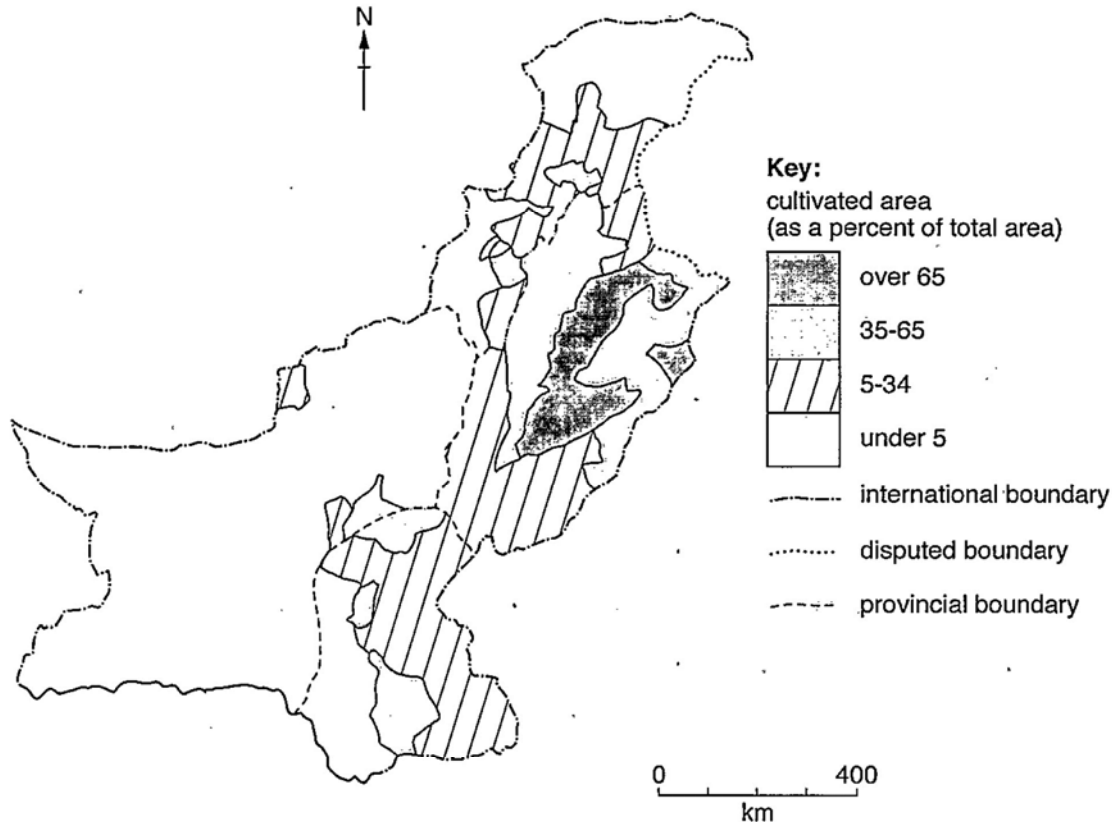


Fig. 6

- (i) A. Name a district that has a cultivated area of over 65%.

Lahore

- B. How much of Faisalabad District is cultivated?

35-65 %

[2]

- (ii) Suggest reasons why so many districts of Pakistan have a cultivated area of less than 5%.

Settlements away from rivers have a very low population. There are either mountains or deserts that support very ~~scare~~ scarce population. Many areas have no proper irrigation systems as there is not much need for water. These areas also do not support agriculture thus water is only given to ^{those areas} 5.



Example candidate response – high, continued

- (iii) Explain why agricultural land is no longer producing crops in many regions of Pakistan.

Land that is assigned for agriculture is to be used strictly for agriculture only. Growing crops on the same land again and again without breaks rids the soil of its natural minerals. Thus crops are ^{not} either grown properly. Farmers are mostly uneducated and do not know how much fertilizer to use and how to properly grow crops. They are also poor and can not afford good fertilizers. [4]

- (b) Study Fig. 7 which gives information about urban and rural population in Pakistan in two different years.

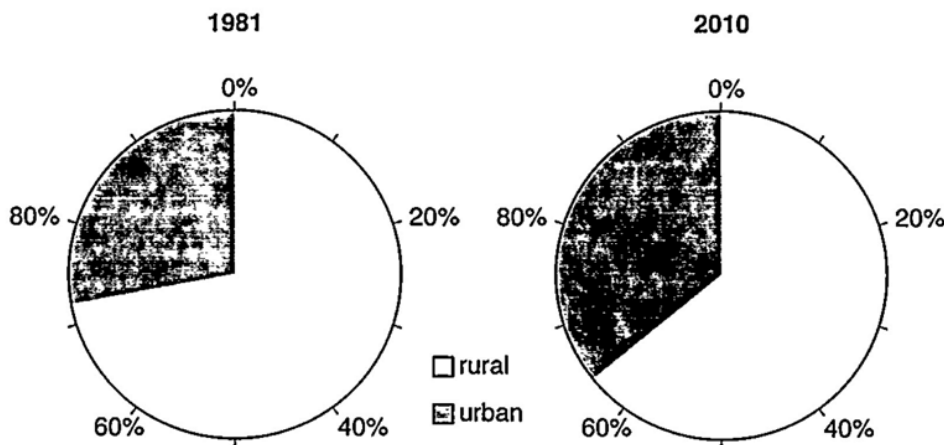


Fig. 7

- (i) By how much has the rural population decreased between 1981 and 2010?

Rural population has decreased by around 8%. [1]

- (ii) Loss of agricultural land is one reason for the rural population decreasing. Describe three other push factors that are causing people to move to urban areas.

1. Rural areas do not have / provide good services such as hospitals, education etc.
2. There is usually only one kind of job (e.g. farming) with low income.
3. Rural areas usually do not have good infrastructure. Water, electricity & gas are ^{mostly} unavailable. [3]

Example candidate response – high, continued

- (iii) Explain the effects on rural areas when large amounts of people migrate to urban areas.

Rural areas become less populated. Usually, it is men who migrate to cities. They leave their family behind that is left to take care of themselves without much money. However, if the immigrant has found a good job and sends enough money to his family, they can get better living conditions and the children can be educated. Economic situation of village improves.

- (c) Study Fig. 8 which is a graph showing the changes in population density for different provinces between 1972 and 2011.

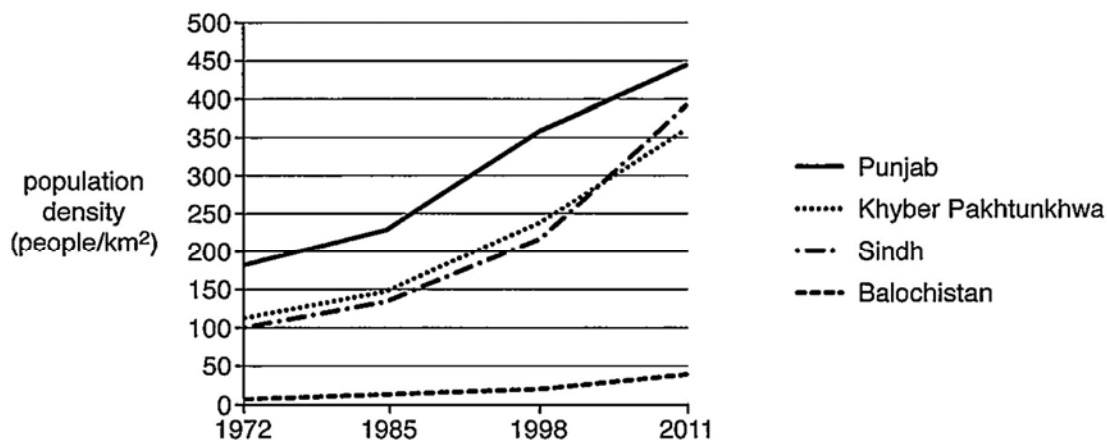


Fig. 8

Describe **two** of the main changes over time that can be seen in this graph.

1. Since 1972 to 1985, 1998, population has grown extremely rapidly in Punjab. Slowed down after 1998.
2. Till 1998, population grew in Punjab. After 1998, rate of growth of Sindh got more than that of Punjab.

Example candidate response – high, continued

(d) Read the following two views:

The Government should spend more money on development projects in the least densely populated province in Pakistan. This province is behind in development.

The Government should spend more money on development projects in the more densely populated provinces. More people live there.

Which view do you agree with more? Give reasons and refer to places or examples you have studied to support your answer.

The government should definitely concentrate on areas with less population. It is true that these areas are still not developed properly. People living in scarcely populated areas also deserve good living standards. Other than that, if the area is developed, there will be less rate of immigration and population will become stable. Along with development come hospitals, schools and good infrastructure. This creates more jobs for skilled and unskilled labourers. Roads will encourage trade and the economic situation will improve. I think that the whole country should be equally developed so there is no extreme poverty or ^{extreme wealth}. [6]

[Total: 25]

Examiner comment – high

(3a)(i)(A) Lahore was the most popular answer, but unfortunately it's not correct. A few candidates were able to identify, from the map, a district that has a cultivated area of over 65%, and most of them gave Gujranwala or Multan. This suggests that candidates need to become more familiar with the location of places within Pakistan.

(3a)(i)(B) The candidate correctly identified Faisalabad District on the map and noted from the key that the cultivated area was 35–65%, gaining one mark for this. When candidates did not give the percentage unit they were not awarded the mark, because the question was '*How much of*' so the answer needed to be in terms of a percentage, a share or a proportion.

Mark awarded = 1 out of 2

(3a)(ii) The candidate gained two marks for correctly identifying 'mountains' and 'deserts'. However, the answer focused on settlements away from rivers, rather than the reason why so many districts of Pakistan have a cultivated area of less than 5%. The thrust of the answer was that these areas have little water, and if the candidate had developed the idea of places being away from rivers, a further mark could have been gained. More careful reading of the question would have led to a more directed response, gaining the full marks available.

It would not have been correct to give lack of irrigation, waterlogging and salinity as the reasons why so many districts of Pakistan have cultivated areas of less than 5%. Equally, talking about 'less rainfall' without including something against which 'less' could be compared could not be credited as we have to ask the question, 'less than what'?

Mark awarded = 2 out of 3

(3a)(iii) The candidate was credited for correctly talking about 'overcultivation', that the growing of crops on the same land again and again without a break depletes the minerals in soil. There was potentially another mark in the first sentence about land use, where the candidate could have stated that land formerly used for agriculture is now used for housing or industry. None of the ideas about farmers in the last three and a half lines are creditable.

Many candidates gave good developed responses to this question, particularly around waterlogging and salinity. Candidates in the high range also talked about alternative uses for land, the migration of the workforce and problems with landlords

Mark awarded = 1 out of 4

(3b)(i) In common with the vast majority of candidates, the correct answer was given and gained one mark. Very few candidates omitted the unit (%).

Mark awarded = 1 out of 1

(3b)(ii) This was very well answered, gaining maximum marks. Three push factors, those that force people to move to urban regions, were given: 'do not have/provide good services such as hospitals', 'low income', 'water, electricity and gas are mostly unavailable'.

Some candidates focused on pull factors (reasons why people are attracted to something) and so couldn't be credited. Some candidates needed to be more careful about the qualifying words they used; for example they wrote about 'no schools' in rural areas, rather than about the quality of the schools, and so couldn't be credited.

Mark awarded = 3 out of 3



Examiner comment – high, continued

(3b)(iii) This question was answered well and was awarded three marks. The points were explained and developed, giving a good illustration of what happens when large numbers of people migrate to urban areas.

This candidate also mentioned the positive effect on rural areas of remittances being sent back by the migrants who find work. Many candidates needed to think beyond 'decrease in population' (which wasn't credited) and reduction in agricultural activity/output. Some candidates responded by talking about the effects on urban areas, which, after analysis of both this question and question **(b)(ii)**, suggests that candidates need to have a clearer idea about the definitions of *urban*, *rural* and *push and pull factors*.

Mark awarded = 3 out of 4

(3c) This question asks the candidate to describe two of the main changes over time, so the examiner is looking for major trends and patterns that span the entire period of the graph, from 1972 to 2011. The candidate breaks down the time period into phases, which does not illustrate a main change, and repeats ideas. This response couldn't be credited. The examiner was looking for 'All four have increased', 'Punjab KPK and Sindh all increased at a fast rate/the same rate' and 'Balochistan increased at a slow rate.'

Mark awarded = 0 out of 2

(3d) This type of question requires the candidate to provide a developed response that addresses both points of view and makes a clear evaluation. This response had the potential to reach Level 3 as the points made were clear, well developed and well organised. However, only one view was addressed – the first view, so the marks awarded were Level 2, three marks, which is the maximum that can be awarded when only one view is addressed. With an equally well-developed opposing view, along with evaluation and examples, this response would have reached the top of Level 3. This was a similar situation for many other candidates.

Mark awarded = 3 out of 6

Total marks awarded = 14 out of 25



Question 4

Example candidate response – high

- 4 (a) (i) Study Fig. 9 which shows formal and informal sector employment in Pakistan over three years.

Sector	2008–09	2009–10	2010–11
Formal	26.7	26.7	26.2
Informal	73.3	73.3	73.8

Fig. 9

Which sector of employment makes up the largest share of the labour force in Pakistan over these years?

..... Informal sector employment [1]

- (ii) Tourism is a service industry. Name **two** other service industries.

1 Insurance
 2 Banking [2]

- (iii) Study Figs 10 and 11 which give information about visitors into Pakistan in 1999.

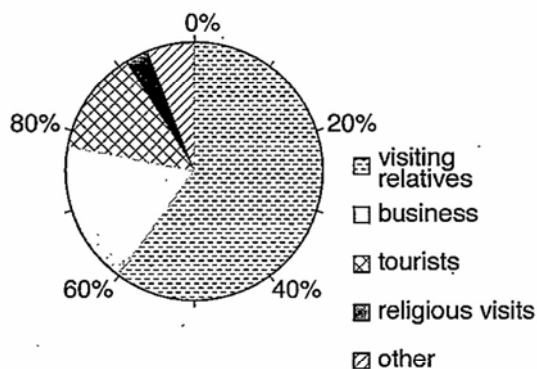


Fig. 10

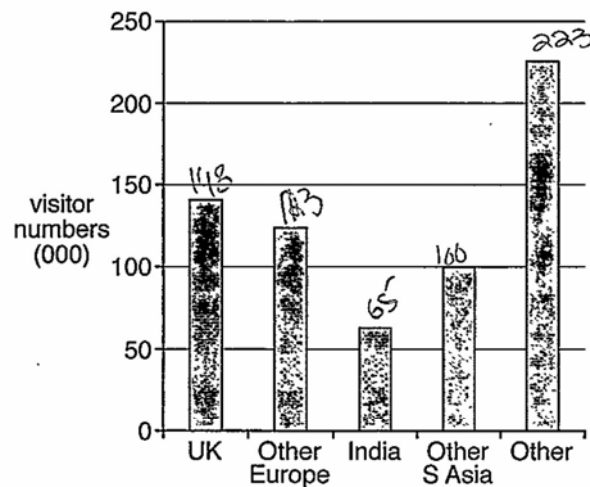


Fig. 11

- A. What percentage of visitors into Pakistan was classed as tourists?

..... 13% [1]

- B. What was the total number of visitors into Pakistan?

..... 659,000 [2]

- C. Suggest **one** reason why more people visited relatives compared to visiting Pakistan as tourists.

..... less people visit as tourists due to increased terrorism which is a threat to the lives of foreigners [3]

Example candidate response – high, continued

- (b) (i) Study Photographs A and B (Insert) which show airports in Gilgit and Chitral.

Using the photographs and your own knowledge describe the problems in providing air transport in the northern areas of Pakistan.

Dev. Developing air transport in northern areas is difficult due to rugged landscape and unsuitable topography. In mountainous areas, it is expensive to develop air transport. Also, demand is not high for air transport as people earn low incomes and have a low standard of living making air transport inconvenient in northern areas. [3]

- (ii) Explain the advantages to Pakistan as a developing country of providing more air transport routes.

Providing more air transport routes enables stimulates trade resulting in better balance of payments position. Gross Domestic Product will also increase as well. Also, industries will open up which will result in more jobs being created. More businessmen could travel which might attract investment. Tourism might increase as well resulting in more income for the locals. This will then result in a higher standard of living. Tourism could also result in more foreign exchange for the country. [4]

- (c) (i) Name or describe a border crossing by road between Pakistan and a neighbouring country. Which country is linked to Pakistan by this road?

Border crossing ... Karakoram Highway

Country ... China [2]



Example candidate response – high, continued

- (ii) How useful is the border crossing you have named or described in (i) for trade? Give reasons for your answer.

~~Since Pakistan~~ Border crossing is useful as this increases trade with China. Trading by road is cheaper and is a flexible form of transport. This will be beneficial for ~~the~~ both the countries as standard of living will increase for the citizens and more could be exported contributing to the GNP of the countries. ~~Industri~~ Diplomatic relations could be improved. China could invest in Pakistan which results in jobs being created hence resulting in low crime rates. [4]

- (d) There are advantages and disadvantages to Pakistan of trading with different countries or groups of countries.

Read the following two views:

Pakistan would benefit from stronger trade links with China.

There are more advantages to Pakistan in maintaining trade with EU (European Union) countries.

Which view do you agree with more? Give reasons and refer to places or examples you have studied to support your answer.

There are more advantages to Pakistan in maintaining trade with EU countries as this helps Pakistan getting access to wider markets. Balance of payments position is improved of the country. Also, free trade could be done with less sanctions hence increasing international competitiveness of the countries goods and services. Chances of war is then reduced with the ^{European Union} ~~western~~ countries. However, Pakistan would benefit from stronger trade links with China as China is our neighbouring country and it is cheaper and convenient to develop trade links with China. I agree ^{more} with the statement that says "there are more advantages to Pakistan in maintaining trade with European Union countries as dependence on other countries is reduced for Pakistan." [6]

[Total: 25]

Examiner comment – high

(4a)(i) In common with the vast majority of the candidates the answer given was correct, and awarded one mark.

Mark awarded = 1 out of 1

(4a)(ii) Two correct answers were given for two marks. A wide range of service industries were named by candidates, demonstrating there was good understanding of what constitute services and the service sector.

Mark awarded = 2 out of 2

(4a)(iii)(A) This was clearly answered, with the correct figure and units given.

(4a)(iii)(B) The correct number of 000s was given to the basic calculation of 659 to give a total of 659 000. This demonstrates that the candidate had read the axis label carefully.

(4a)(iii)(C) Again, a clear, correct response here gives the candidate the maximum number of three marks overall.

Generally, candidates answered parts **(A)** and **(C)** well, giving the correct figure within tolerance and units at **(A)** and a variety of valid reasons, such as 'people work abroad and return to Pakistan to visit relatives.' Fewer candidates gave a correct answer for **(B)**, with many candidates omitting the 000s from the visitor numbers, which meant they didn't get the mark.

Mark awarded = 3 out of 3

(4b)(i) The candidate identified that it is difficult to develop air transport in northern areas because of the mountains and was awarded a mark for this. More focus on what the specific problems were in providing air transport in the northern areas of Pakistan was needed here to gain the other two marks. The wording in the question, in conjunction with Photographs A and B in the Insert, should have directed the candidate towards problems caused by the physical geography and logistical issues, rather than general economic problems (although lack of funds for specified air transport improvements or new technology would have gained a mark).

Mark awarded = 1 out of 3

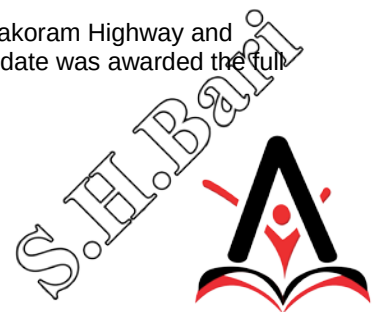
(4b)(ii) This was well answered and provides a very good example of the level of ability of those candidates in the high range. A key word in this question was *more* air transport routes, so ideas around 'increasing' should feature here. The candidate talks about 'stimulating trade', 'more businessmen' taking trips, 'attract investment' 'tourism might increase', thereby earning the full four marks available for this question. Other good points here were 'more income for the locals' and 'more foreign exchange'. The possibility that the provision of more air transport routes might affect the balance of payments and the Gross Domestic Product is more complex, and a development too far in this instance, so these points were not credited on the mark scheme for this question.

Most candidates gave precise and directed responses to this question, often talking about the types of products that could be exported, and candidates in the high range covered a wide range of points that were creditable on the mark scheme.

Mark awarded = 4 out of 4

(4c)(i) The border crossings were generally well known by candidates, with the Karakoram Highway and China the response most often given. Most candidates did well here, and this candidate was awarded the full two marks for their response.

Mark awarded = 2 out of 2



Examiner comment – high, continued

(4c)(ii) The candidate needed to evaluate how useful the crossing was to gain a reserved mark, so this could not be credited. The candidate did explain the other areas of the response, noting that trading by road is 'cheaper', that trade would 'increase' and that diplomatic relations would 'improve', thus gaining three marks.

Most of the candidates were positive about the benefits of the border crossings, particularly those that named the Karakoram Highway and China. Few candidates mentioned in connection with this crossing that the road was blocked/closed in winter.

Mark awarded = 3 out of 4

(4d) The candidate gave a developed argument that addressed the advantages to Pakistan of maintaining trade with China. They were able to offer some evaluation: 'there are more advantages to Pakistan in maintaining trade with European countries'. If there had been a developed argument that addressed the second statement about the benefits to Pakistan of having stronger trade links with the EU, rather than a series of undeveloped ideas, or ideas which were not completely valid 'free trade could be done with less sanctions', then the response would have been awarded Level 3. As it stands, the maximum mark that can be achieved is Level 2, three marks, because of the lack of developed argument addressing the second point of view. Development could have taken the form of 'the balance of payments is improved because Pakistan is able to export a number of different goods, such as sports goods and cotton textiles in large quantities, to a very large European market, which also means it does not depend on one country for its export sales.'.

Mark awarded = 3 out of 6

Total marks awarded = 19 out of 25



Question 5

Example candidate response – high

5 (a) Study Fig. 12 which gives information on birth rates and death rates in Pakistan.

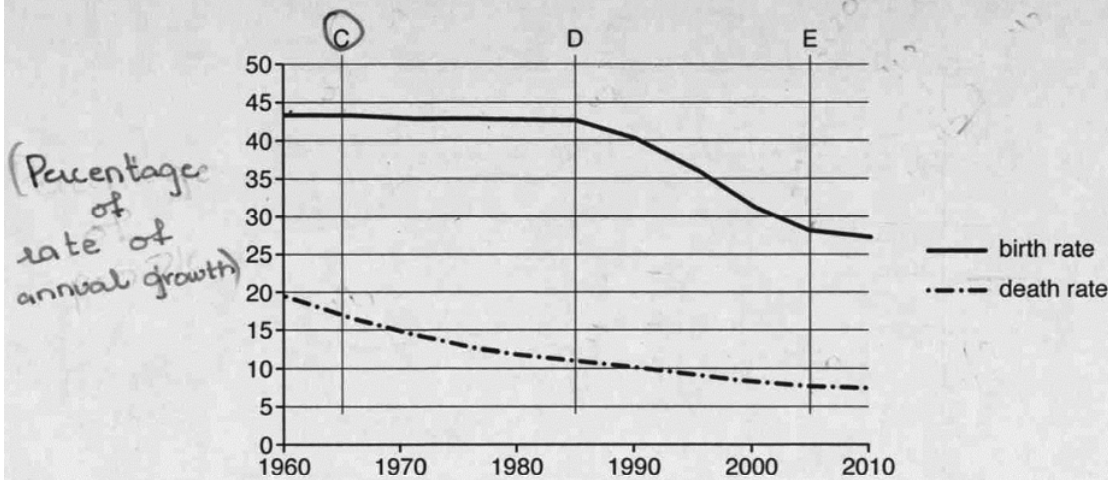


Fig. 12

(i) On Fig. 12:

- Add a suitable label for the y-axis (vertical axis).
- Circle one of the letters C, D, or E on the graph to show the point where natural increase is highest. [2]

(ii) Give **two** reasons why Pakistan continues to have a high birth rate.

- The rate of illiteracy is still high and people are not aware of problems due to high birth.
- Lack of implementation on family planning programme. [2]

(iii) The death rate in Pakistan has been decreasing every year since 1960. Suggest reasons for this.

The reason for reduction is better medical facilities and cure of diseases like cholera and malaria. Moreover, due to the improvement in sanitation facilities leading to less diseases being born. Better transport facility for transportation of doctors and nurses. [3]

Example candidate response – high, continued

- (b) Study Fig. 13 which shows changes in population growth rates for two countries between 2000 and 2012.

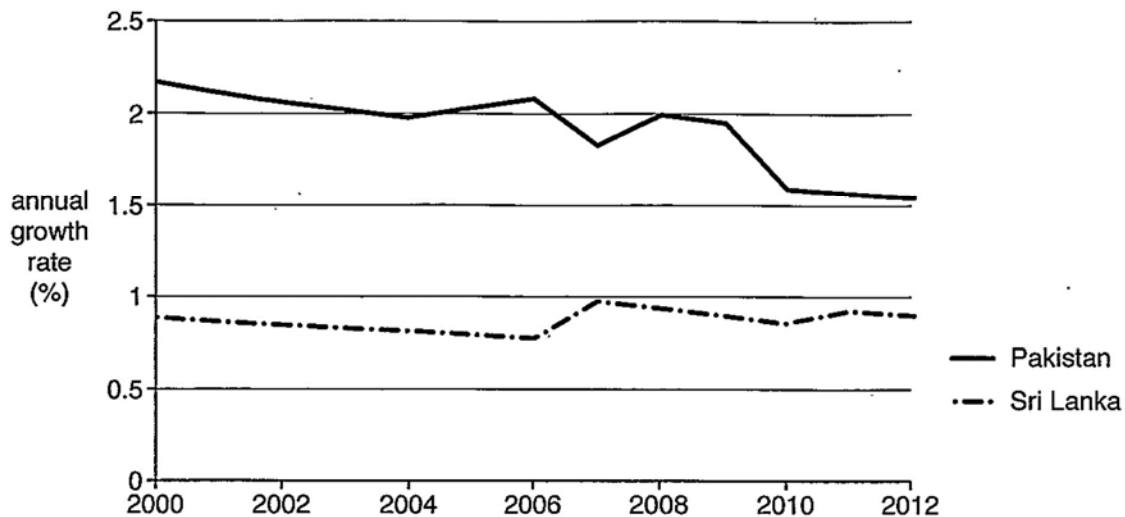


Fig. 13

- (i) Compare the main changes in the population growth rate of Pakistan with that of Sri Lanka between these two dates.

Since 2000 Pakistan's growth rate had been 2.3% way more than 0.9% of Sri Lanka; due to lack of literacy but in 2006-2008 there has been a decrease in Pakistan growth rate which dropped to 1.7%; however in 2006-2008 Sri Lanka's growth rate has increased to 1.0% [3]

Example candidate response – high, continued

- (ii) One of the factors affecting population growth is international migration. In recent years, Pakistan has experienced more emigration than immigration.

A. What is meant by the term 'immigration'?

The ~~movement~~ ^{movement} of people out of the country.

B. Explain what has caused high levels of emigration and the effects of this on the economy of Pakistan.

Emigration is the movement of people into Pakistan. The Afghan refugees plus the emigration of the rural people, in search of job opportunities or to meet their relatives. However, this causing housing problems, leading to more squatter settlements. Moreover, it leads to pollution and this leads to drop in the economy as more funds are applied in these problems so less funds are available for industry. [5]

- (c) (i) What is meant by the term 'sustainable population growth'?

Sustainable population growth means the growth should be in such a way which does not ^{create problem in} ~~disturb~~ the environment. [1]

- (ii) Suggest three problems that might be caused by unsustainable population growth.

- 1 Housing problems, leading to more katchi abadi and squatter settlements.
- 2 Cutting of trees and clearance of forest to make roads and houses.
- 3 Pollution and overcrowding and less resources being available and increase rate of crimes. [3]

Example candidate response – high, continued

(d) Read the following two views about reducing the population growth rate in Pakistan.

The population growth rate is best reduced by family planning programmes:

The population growth rate is best reduced by increasing literacy for all, especially females.

Which view do you agree with more? Give reasons and refer to examples you have studied from Pakistan to support your answer.

I agree that by increasing literacy for everyone will be the best idea. As literacy creates awareness among people that material possession are more important and they will be aware of the problems caused due to high birth rate. Family planning programmes such as "subsitua" and "chabi ka nishan" have been made but were not benefitted as people of rural areas do not understand these things. Therefore literacy especially for females will make them aware of the consequences of high birth rate, which leads to unemployment and rural, urban migration.

[6]

[Total: 25]

Examiner comment – high

(5a)(i) The candidate was not able to correctly label the y-axis, with 'per 1000 population (per year)' or 'average births and deaths per 1000 people'. They also did not gain the mark for identifying 'D' as the point where the natural increase is the highest.

Mark awarded = 0 out of 2

(5a)(ii) Two marks were given here for clear explanations as to why Pakistan continues to have a high birth rate. Overall, candidates demonstrated a good knowledge of the various reasons for this.

Mark awarded = 2 out of 2

(5a)(iii) Full marks were given here because the candidate gave four reasons why the death rate has been decreasing since 1960, for example: 'better medical facilities', 'cure of diseases like cholera' and 'improvement in sanitation facilities'. The majority of candidates commented on improvements in health care and improvements in food and hygiene, as well as the reduction in infant mortality, demonstrating a thorough knowledge of this area.

Mark awarded = 3 out of 3

(5b)(i) The candidate should have given more accurate figures for the start and end population growth rates for Pakistan and Sri Lanka to gain a mark. The question asks for a comparison of the main changes between the two dates. Candidates did not need to mention other dates within the overall range of 2000 to 2012, or break the overall range into shorter time periods, or give a description of every change or trend within those shorter time periods. Better data interpretation skills would result in an improvement in the responses to questions that address Assessment Objective 4 of the syllabus.

Mark awarded = 0 out of 3

(5b)(ii) This is an example of a good candidate who has confused 'immigration' (the movement of people into a country) and 'emigration' (the movement of people out of a country). Unfortunately, it is therefore not possible to credit any points in the response, although the candidate shows a good understanding of the problems involved when people move from one country/area to another.

Mark awarded = 0 out of 5

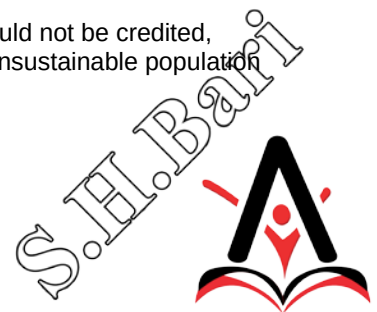
(5c)(i) This response could have earned a mark had it contained more development and thought. The candidate clearly understood that sustainable population growth was concerned with the relationship between population growth and the environment. More specific mention of issues such as use of resources could have been made; those candidates that referred to resources in their answer invariably gained the mark. Common errors such as 'population remains the same', 'population doesn't overburden the economy' or 'population can keep growing' demonstrate that improve knowledge of the importance of sustainable development (which is a theme across Paper 2) is needed.

Mark awarded = 0 out of 1

(5c)(ii) This was a very good response with attention being paid to the idea that social, economic and environmental problems increase: 'more katchi abadi and squatter settlements', 'less resources available', an 'increase in the rate of crime' and 'clearance of forest to make roads and homes'. All three marks were awarded.

Candidates who just mentioned 'unemployment' or 'crime' or 'housing problems' could not be credited, because these are all problems which exist in a society which is not experiencing unsustainable population growth.

Mark awarded = 3 out of 3



Examiner comment – high, continued

(5d) This is a good example of a response awarded Level 3, six marks. There are developed statements addressing both points of view. There is clear evaluation: 'I agree that by increasing literacy for everyone will be the best idea', and the content of the response supports this evaluation. There are also examples given of family planning programmes. The argument is presented clearly in an organised way.

Of all the part (d) questions, this question elicited the best answers. More candidates addressed both points of view with developed ideas than for other similar questions in the rest of the paper, and good examples were included. This meant that candidates were able to access the higher marks through the evaluation of two developed points of view.

Mark awarded = 6 out of 6

Total marks awarded = 14 out of 25



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